

A REVISIT OF PATCH SOLUTIONS FOR THE 2D LOGLOG-EULER TYPE EQUATION

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ABSTRACT. In this paper, we revisit the patch solutions for a class of inviscid whole-space active scalar equations that interpolate between the 2D Euler equation and the α -SQG equation. Compared with the 2D Euler equation in vorticity form, there is an additional Fourier multiplier $m(\Lambda)$ ($\Lambda = (-\Delta)^{1/2}$) in the Biot-Savart law. If the symbol m satisfies the Osgood-type condition

$$\int_2^{+\infty} \frac{1}{r(\log r)m(r)} dr = +\infty$$

and certain mild assumptions, the system is referred to as the 2D Loglog-Euler type equation.

First, we prove a Yudovich-type theorem establishing the existence and uniqueness of a global weak solution for the Loglog-Euler type equation associated with bounded and integrable initial data. This result directly applies to patch solutions, which are weak solutions corresponding to patch initial data given by characteristic functions of disjoint, regular, bounded domains.

Next, we revisit the seminal result by Elgindi [44] and provide a different proof under explicit assumptions on m , showing that for the 2D Loglog-Euler type equation with $C^{1,\mu}$ ($0 < \mu < 1$) single-patch initial data, the evolved patch boundary globally preserves the $C^{1,\mu-\varepsilon}$ regularity for any $\varepsilon \in (0, \mu)$. In contrast to the frequency-space argument in [44], we develop an entirely physical-space-based approach that avoids the Littlewood-Paley theory and offers advantages for potential extensions to the half-plane or bounded smooth domains.

Furthermore, we investigate the global propagation of higher-order $C^{n,\mu}$ boundary regularity for patch solutions with any $n \in \mathbb{N}^*$, and analyze the evolution of multiple patches.

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1. INTRODUCTION

In this paper, we study a family of two-dimensional (2D) active scalar equations

$$\begin{cases} \partial_t \omega + u \cdot \nabla \omega = 0, & (x, t) \in \mathbb{R}^2 \times \mathbb{R}_+, \\ u = \nabla^\perp (-\Delta)^{-1} m(\Lambda) \omega, & (x, t) \in \mathbb{R}^2 \times \mathbb{R}_+, \\ \omega|_{t=0}(x) = \omega_0(x), & x \in \mathbb{R}^2, \end{cases} \quad (1)$$

where $\nabla^\perp \triangleq (\partial_{x_2}, -\partial_{x_1})$ and $\Lambda \triangleq (-\Delta)^{\frac{1}{2}}$, the vector $u = (u_1, u_2)$ denotes the velocity field, while the scalar field ω represents the vorticity (or, depending on the model, the density or temperature) of the fluid. The operator $m(\Lambda)$ is the Fourier multiplier with radial symbol $m(\xi) = m(|\xi|)$ satisfying the following assumptions:

(H1) $m \in C^{n+4}(\mathbb{R}_+)$, $n \in \mathbb{N}^*$ and

$$\forall r > 0, \quad m(r) > 0, \quad m'(r) \geq 0; \quad \text{and} \quad \lim_{r \rightarrow 0^+} r m'(r) \text{ exists.}$$

Moreover, m' satisfies the Mikhlin-Hörmander condition: there exists a constant $C > 0$ such that

$$\left| \frac{d^k}{dr^k} m'(r) \right| \leq C r^{-k} m'(r), \quad \forall k = 1, \dots, n+3, \quad \forall r > 0.$$

(H2) Set $\tilde{m}(r) \triangleq m(e^r)$. One of the following conditions holds:

(H2a) there exist constants $\beta \in [0, +\infty]$, $\beta_1 \in [0, +\infty)$, and $\beta_2 \in (-2, +\infty)$ such that the following limits exist:

$$\lim_{r \rightarrow +\infty} m(r) = +\infty, \quad \lim_{r \rightarrow +\infty} \frac{r(\log r) \tilde{m}'(r)}{\tilde{m}(r)} = \beta, \quad \lim_{r \rightarrow +\infty} \frac{r \tilde{m}'(r)}{\tilde{m}(r)} = \beta_1, \quad \lim_{r \rightarrow +\infty} \frac{r \tilde{m}''(r)}{\tilde{m}'(r)} = \beta_2;$$

(H2b) there exists a constant $\alpha \in (0, 2)$ such that

$$\lim_{r \rightarrow +\infty} \frac{r m'(r)}{m(r)} = \alpha;$$

(H2c) there exists a constant $C > 0$ such that

$$\lim_{r \rightarrow +\infty} m(r) = C < +\infty.$$

We emphasize that the technical assumptions in (H2) impose a non-oscillatory condition on m near infinity. The three cases correspond to different asymptotic growth regimes: bounded m in (H2c), power-law growth in (H2b), and intermediate growth in (H2a). In particular, the existence of the limits in (H2) rules out highly oscillatory behavior near infinity, such as, $m(r) = \log \log(e^2 + 2r + \sin r)$. To keep (H2a) as general as possible, we allow β to take the value $+\infty$.

Active scalar equations (1) under assumptions (H1)-(H2) arise naturally in fluid-dynamical models and have attracted considerable attention. Typical examples include the following.

- *2D Euler equation:* $m(r) \equiv 1$. In this case, $m(\Lambda) \equiv \text{Id}$, and equation (1) reduces to the two-dimensional Euler equation in vorticity form. This equation describes the motion of an inviscid incompressible fluid in two dimensions and is one of the fundamental models in fluid dynamics. It is a prototypical example satisfying (H2c).
- *Inviscid α -SQG equation:* $m(r) = r^\alpha$, $\alpha \in (0, 2)$. Here,

$$m(\Lambda) = (-\Delta)^{\frac{\alpha}{2}} = \Lambda^\alpha.$$

For $\alpha = 1$, equation (1) becomes the well-known SQG equation, which serves as a simplified model for atmospheric circulation near the tropopause [58] and for ocean dynamics in the upper layers [76]. The regime $0 < \alpha < 1$ was introduced by Córdoba, Fontelos, Mancho, and Rodrigo [26] as a class of models interpolating between the 2D Euler and SQG equations. This family of equations satisfies (H2b).

- *2D Loglog-Euler equation:* $m(r) = \log^\beta(1 + \log(1 + r^2))$, $\beta \in (0, 1]$. In this case, the multiplier is

$$m(\Lambda) = \log^\beta(1 + \log(1 - \Delta)). \quad (2)$$

This equation was introduced by Chae, Constantin, and Wu [15] as part of a broader class of models interpolating the 2D Euler and α -SQG equations. It was subsequently studied independently by Dabkowski, Kiselev, Silvestre, and Vicol [31], and by Elgindi [44].

The 2D Loglog-Euler equation is a prototypical example satisfying (H2a). Further examples obeying (H2a) include

$$m(r) = \log^{\beta_1}(1 + r) \quad \text{and} \quad m(r) = \log(1 + \log(1 + (\log(1 + r))))).$$

We refer to models satisfying this intermediate-growth condition collectively as the *2D Loglog-Euler type equations*; they are the primary focus of this paper.

For comparison, we briefly mention a related but distinct family of models, commonly referred to as logarithmically modified 2D Euler equations, in which the velocity is given by $u = \nabla^\perp(-\Delta)^{-1}T_\gamma(\Lambda)\omega$, with typical choices

$$T_\gamma(\Lambda) = \log^{-\gamma}(e + \Lambda) \quad \text{or} \quad T_\gamma(\Lambda) = \log^{-\gamma}(e + \Lambda^2).$$

Chae and Wu [17] established local well-posedness in the borderline Sobolev setting for $\gamma > 1/2$; Dong and Li [39] proved global well-posedness for $\gamma \geq 3/2$; while Kwon [74] showed strong ill-posedness for the remaining range $0 < \gamma \leq 1/2$. These logarithmically modified models are fundamentally different from the equations considered in this paper. Their multipliers $m(|\xi|) = T_\gamma(|\xi|)$ decay at high frequencies and therefore introduce a logarithmic smoothing effect compared with the 2D Euler equation. In contrast, the multipliers considered under assumption (H2) do not decay at high frequencies: they may remain bounded or grow without bound. The latter, covered by (H2a) and (H2b), leads to velocity fields that are (slightly) more singular than the Euler velocity.

The well-posedness theory for the 2D Euler equation dates back to the classical works of Wolibner [96] and Hölder [59] in the 1930s, who established global well-posedness for smooth solutions with Hölder-continuous vorticity. A major advance was made by Yudovich [98], who proved the global existence and uniqueness of weak solutions with bounded and decaying vorticity; see also [4, 19,

[30, 81, 79] for other accessible proofs. The velocity field associated with a Yudovich solution is generally not Lipschitz continuous in space, but is instead log-Lipschitz in the sense that

$$|u(x, t) - u(y, t)| \leq C|x - y| \left(1 + \log \frac{1}{|x - y|} \right), \quad \forall t > 0, \quad 0 < |x - y| \leq 1. \quad (3)$$

As observed by Bahouri and Chemin [3], such a log-Lipschitz velocity field may lead to a deterioration of the regularity of passively transported quantities; see [34, 11, 62, 35] for recent developments. Global existence of weak solutions, possibly without uniqueness, was established for vorticity in the larger class $L_t^\infty L_x^p$, $1 < p < \infty$, in [98, 38]. The uniqueness theory was subsequently extended beyond bounded vorticity. Yudovich [99] proved uniqueness for $\omega \in \cap_{p_0 \leq p < \infty} L_t^\infty L_x^p$ provided that $\|\omega(t)\|_{L^p} \leq C\theta(p)$, where $\theta(p)$ grows sufficiently slow in p (e.g. $\theta(p) = \log p$); Vishik [92] introduced a different uniqueness class $\omega \in L_t^\infty(L^{p_0} \cap B_\Pi)$, $p_0 \in (1, 2)$, where $\int_1^\infty \Pi(\tau)^{-1} d\tau = \infty$ and $B_\Pi \triangleq \{f \in \mathcal{S}'(\mathbb{R}^2) : \sum_{j=-1}^N \|\Delta_j f\|_{L^\infty} = O(\Pi(N))\}$; Taniuchi [91] further extended the result of [99] by proving uniqueness for $\omega \in L_t^\infty Y_\Theta$, where $\Theta : [1, \infty) \rightarrow [1, \infty)$ is increasing and satisfies the Osgood type condition $\int_2^\infty \frac{dp}{p\Theta(p)} = \infty$ and $Y_\Theta \triangleq \{f \in \mathcal{S}'(\mathbb{R}^2) : \sup_{p \in [1, \infty)} \frac{\|f\|_{L^p}}{\Theta(p)} < \infty\}$; see [6, 5] for related refinements. Elgindi et al. [49] developed a novel class of solutions beyond the Yudovich class for which local well-posedness and finite-time singularity formation can be established. Concerning non-uniqueness for the 2D Euler equation, with or without forcing, and vorticity in $L_t^\infty L_x^p$ ($p < \infty$), see [93, 94, 1, 13, 9, 82, 10] for recent developments. Beyond the Yudovich theory, the 2D Euler equation has also been extensively studied from several other perspectives, including ill-posedness in critical and supercritical function spaces [8, 45, 48, 63, 29], well-posedness in singular planar domains [54, 55, 56], (optimal) bounds on the growth of vorticity gradient [36, 72, 100, 41, 102, 66].

For the α -SQG equation, the local well-posedness and ill-posedness theories in Sobolev and Hölder spaces are now well understood; see, e.g., [95, 16, 23] and [27, 28, 64, 21], respectively. The global existence of weak solutions was established in [89, 80, 16, 24, 84, 77], while non-uniqueness results for the (forced) α -SQG equation was obtained in [12, 20, 60, 32, 14] and references therein. He and Kiselev [57] constructed solutions of α -SQG equation for every $\alpha \in (0, 2)$ that exhibit either infinite-in-time growth of derivatives or finite-time singularity formation. Despite these advances, the global well-posedness of smooth solutions to the α -SQG equation for any $\alpha \in (0, 2)$ on either $\mathbb{R}^2$ or the torus $\mathbb{T}^2$ remains a remarkable open problem. This problem has been resolved only in certain domains with boundary. Indeed, for the α -SQG equation with $\alpha \in (0, \frac{1}{2}]$ in the half-plane, Zlatoš [101] established local well-posedness in the anisotropic Lipschitz-type spaces and proved the finite-time singularity formation for smooth initial data (see also [65]), where the rigid boundary plays an essential role in the singularity mechanism.

Based on the study of the 2D Euler and α -SQG equations, an intriguing question arises:

How far can we move from the 2D Euler equation towards the α -SQG equation while still maintaining global well-posedness of smooth (or non-smooth) solutions?

Motivated by this question, Chae, Constantin, and Wu [15] introduced the Loglog-Euler equation (1) with multiplier (2), and proved the global well-posedness of smooth solutions. Later, the Osgood-type condition

$$\int_2^{+\infty} \frac{1}{r(\log r)m(r)} dr = \int_{\log 2}^{+\infty} \frac{dr}{r\tilde{m}(r)} = +\infty. \quad (\text{Osg})$$

was proposed as a possible threshold for global well-posedness. Elgindi [44] and Dabkowski et al. [31] independently established the global regularity of smooth solutions to (1) under the Osgood condition (Osg), together with several mild assumptions on the multiplier. Whether (Osg) is the critical threshold for global regularity of smooth solutions to (1) remains an open problem.

We note that if m satisfies (H2c) (e.g., the 2D Euler case), (Osg) holds automatically, whereas if m satisfies (H2b) (e.g., the α -SQG equation), (Osg) fails. Our main interest lies in the intermediate case (H2a), where (Osg) requires $\beta \leq 1$. See Lemma 2.1 for more details.

In this paper, we focus primarily on patch (or vortex patch) solutions of the active scalar equation (1), which are weak solutions associated with patch-like initial data consisting of either a single patch ($N = 1$) or multiple patches ($N > 1$):

$$\omega_0(x) = \sum_{j=1}^N a_j \mathbf{1}_{D_j(0)}(x), \quad a_j \in \mathbb{R}, \quad \mathbf{1}_{D_j(0)}(x) \triangleq \begin{cases} 1, & x \in D_j(0), \\ 0, & x \in \mathbb{R}^2 \setminus D_j(0), \end{cases} \quad (4)$$

where $D_j(0) \subset \mathbb{R}^2$, $j = 1, \dots, N$ are simply connected disjoint bounded domains with regular boundaries $\partial D_j(0)$. Patch solutions provide a natural mathematical model for the evolution of bounded domains with concentrated scalar fields in incompressible fluids, particularly excelling in capturing sharp interfacial dynamics. For the 2D Euler equation, the celebrated theorem of Yudovich [98] establishes global existence and uniqueness of patch solutions with initial data of the form (4), while preserving the patch structure. However, the flow map (defined by (5)) provided in the Yudovich theory is a homeomorphism of Hölder class $C^{\exp(-Ct)}$, so the patch boundary is only known *a priori* to belong to $C^{\exp(-Ct)}$. This naturally led to the classical *vortex patch problem* of 2D Euler equation, posed in the 1980s, which asks whether the initial smoothness of the patch boundaries is preserved for all time. This question was answered affirmatively by Chemin [18], who proved global persistence of $C^{k,\gamma}$ -regularity with $k \in \mathbb{N}^*$ and $\gamma \in (0, 1)$ for the vortex patch boundaries. Bertozzi and Constantin [7] subsequently gave a more elementary geometric proof. Alternative proofs were later obtained in [88, 86]. This $C^{1,\gamma}$ theory for vortex patches was extended to the 2D Euler equation in the half-plane [71] and smooth bounded domains [67], where patch boundaries are allowed to touch the rigid domain boundary (see [37, 43] for earlier results). Recently, Kiselev and Luo [69] established the strong ill-posedness of C^2 vortex patches. We also refer to [33, 22, 47, 46] for results on vortex patches with boundary singularities (e.g. corners).

For the α -SQG equation, there is no analogue of Yudovich's theory that directly yields the existence and uniqueness of patch solutions. Nevertheless, the evolution of patch boundaries can still be studied through the contour dynamics formulation. Rodrigo [87] first proved the local existence and uniqueness of C^∞ patch solutions for the α -SQG equation with $\alpha \in (0, 1)$. By exploiting the cancellation structure of the contour equation, local well-posedness of α -SQG patch solutions in L^2 -based Sobolev spaces has been established in a series of works [50, 16, 73, 25, 52, 51]: H^n ($n \geq 2$) for $\alpha \in (0, 1)$, H^{2+s} ($s > 0$) for $\alpha = 1$, and H^n ($n \geq 3$) for $\alpha \in (1, 2)$. Kiselev and Luo [70] further demonstrated strong ill-posedness for α -SQG patches with $\alpha \in (0, 1)$ in the Hölder space $C^{2,\gamma}$ ($\gamma \in (0, 1)$) and in the Sobolev space $W^{2,p}$, $p \neq 2$. Splash singularities for α -patches with $\alpha \in (0, 2)$ were ruled out in [53, 68, 61]. For the α -SQG equation in half-plane with a rigid boundary, Kiselev, Ryzhik, Yao, and Zlatoš [71, 73] showed the local well-posedness of multiple patch solutions and constructed patch-like initial data leading to finite-time singularity formation for $\alpha \in (0, \frac{1}{12})$. Together with the global well-posedness result for multiple vortex patches of the 2D Euler equation [71], this dichotomy highlights the critical transition at $\alpha = 0$ in the behavior of patch solution. The regime of finite-time singularity formation for α -patches was subsequently extended to $\alpha \in (0, \frac{1}{3})$ by Gancedo and Patel [52]; see also [101] for a further improvement.

For the active scalar equation (1) with m satisfying (H2a), the associated vortex-patch dynamics is slightly more singular than in the bounded- m case (including the 2D Euler equation). Indeed, even for a circular patch, the induced velocity field is not Lipschitz continuous near the patch boundary; see part (ii) of Proposition 4.3. This suggests that some loss of boundary regularity of the vortex patch may occur during evolution. When the Osgood-type condition (Osg) holds, together with other mild assumptions, Elgindi [44] identified new cancellation mechanisms and, by applying

the losing-estimates method [3], established global $C^{1,\gamma-\varepsilon}$ ($0 < \varepsilon < \gamma < 1$) regularity for patch boundaries of the whole-space 2D Loglog-Euler type equation (1), starting from $C^{1,\gamma}$ initial patch boundaries. In contrast, when the Osgood-type condition (Osg) fails (i.e., $\int_2^{+\infty} \frac{1}{r(\log r)m(r)} dr < +\infty$), the authors and Miao [83] proved finite-time singularity formation for patch solutions of (1) in the half-plane with a rigid boundary, thereby extending the singularity results of [71, 52] for the α -SQG equations. These complementary results suggest that the Osgood condition (Osg) may serve as the sharp threshold separating global regularity from finite-time singularity formation for patch solutions of (1).

In this paper, we revisit the seminal work of Elgindi [44] on patch solutions to the 2D Loglog-Euler type equation (1), and provide an alternative proof under the explicit assumptions (H1)-(H2a)-(Osg). Unlike the frequency-space-based argument involving losing estimates in [44], we develop an entirely physical-space-based approach that avoids the use of Littlewood-Paley theory. This formulation offers several advantages, particularly for extensions to the half-plane setting (see Remark 3) or to bounded smooth domains. We further investigate the global propagation of $C^{n,\gamma}$ boundary regularity for patch solutions with arbitrary $n \in \mathbb{N}^*$, and extend the analysis to the multi-patch setting.

Our first result establishes a Yudovich-type well-posedness theorem for the 2D Loglog-Euler type equation (1), which provides the foundation for the study of patch solutions.

Theorem 1.1. *Assume that $m(\xi) = m(|\xi|)$ is a radial function satisfying (H1)-(H2a)-(Osg). Let $\omega_0 \in L^1 \cap L^\infty(\mathbb{R}^2)$, then the 2D Loglog-Euler type equation (1) admits a unique global weak solution (in the sense of Definition 3.1)*

$$\omega \in L^\infty([0, +\infty); L^1 \cap L^\infty(\mathbb{R}^2)), \quad \omega(x, t) = \omega_0(\Phi_t^{-1}(x)),$$

where $\Phi_t \triangleq \Phi_{t,0}$, $\Phi_t^{-1} \triangleq \Phi_{0,t}$, and the two-parameter flow map $\Phi_{t,s} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is uniquely defined by

$$\frac{d\Phi_{t,s}(x)}{dt} = u(\Phi_{t,s}(x), t), \quad \Phi_{t,s}(x)|_{t=s} = x. \quad (5)$$

Moreover, the flow map Φ_t satisfies

$$\frac{1}{H^{-1}(H(|x-y|^{-1}) + Ct)} \leq |\Phi_t^{\pm 1}(x) - \Phi_t^{\pm 1}(y)| \leq \frac{1}{H^{-1}(H(|x-y|^{-1}) - Ct)}, \quad (6)$$

where $C > 0$ depends only on $\|\omega_0\|_{L^1 \cap L^\infty}$ and the map $H : (0, +\infty) \rightarrow \mathbb{R}$ is defined by

$$H(r) \triangleq \begin{cases} \int_2^r \frac{1}{\tilde{r}(\log \tilde{r})m(\tilde{r})} d\tilde{r}, & \text{for } r \geq 2, \\ \frac{1}{(\log 2)m(2)} \log \frac{r}{2}, & \text{for } r \in (0, 2), \end{cases} \quad (7)$$

and $H^{-1} : \mathbb{R} \rightarrow (0, +\infty)$ is its inverse function.

Remark 1. For the Euler equation ($m(r) \equiv 1$), the definition (7) gives $H(r) \approx \log \log r$ as $r \rightarrow +\infty$. Consequently, the inverse function H^{-1} exhibits double-exponential growth:

$$H^{-1}(y) \approx \exp(C \exp(y)), \quad \text{as } y \rightarrow +\infty.$$

When $\lim_{r \rightarrow +\infty} m(r) = +\infty$, $H(r)$ grows more slowly than $\log \log r$ as $r \rightarrow +\infty$, and hence H^{-1} grows faster than the double-exponential scale. The Osgood condition (Osg) guarantees H maps $(0, +\infty)$ onto $\mathbb{R}$, so that H^{-1} is globally defined on $\mathbb{R}$. In general, H^{-1} may exhibit arbitrarily fast growth.

A vector field u is called an Osgood vector field (see [2, 75]) if $|u(x) - u(y)| \leq \Psi(|x - y|)$ for every $x, y \in \mathbb{R}^d$, where $\Psi : (0, +\infty) \rightarrow (0, +\infty)$ is increasing, satisfies $\lim_{r \downarrow 0} \Psi(r) = 0$ and $\int_0^1 \frac{dr}{\Psi(r)} = +\infty$. Ambrosio and Bernard [2] established uniqueness for bounded signed measure-valued solutions of

the transport equation driven by a divergence-free Osgood vector field; see La [75] and Drivas et al. [42] for the propagation of regularity and singularity for such transport equations. The velocity field u in Theorem 1.1 is an Osgood vector field by virtue of the continuity estimate (36). Thus, our result places the Loglog-Euler type equation within the framework of the Osgood transport theory.

Finally, we note that Theorem 1.1 is analogous to the results in [99, 92, 91, 6, 5] on slightly generalized Yudovich classes.

Theorem 1.1 ensures the global well-posedness of patch solutions for the 2D Loglog-Euler type equation with initial data as in (4), and shows that the patch structure is globally preserved:

$$\omega(x, t) = \sum_{j=1}^N a_j \mathbf{1}_{D_j(t)}(x), \quad D_j(t) = \Phi_t(D_j(0)), \quad (8)$$

where $\Phi_t = \Phi_{t,0}$ denotes the flow map defined by (5).

We now study the vortex patch problem and investigate the global regularity of the evolving patch boundaries. More precisely, we aim to address the following question:

If the initial patch boundaries $\partial D_j(0)$ in (4) belong to the Hölder class $C^{n,\mu}$ with $n \in \mathbb{N}^$ and $\mu \in (0, 1)$ for all $j \in 1, 2, \dots, N$, what is the optimal regularity of $\partial D_j(t)$ for every $t > 0$ and $j \in 1, 2, \dots, N$?*

We first focus on the evolution of a single patch, namely the case $N = 1$ with $a_1 = 1$:

$$\omega(x, t) = \mathbf{1}_{D(t)}(x), \quad D(t) = \Phi_t(D_0). \quad (9)$$

Our main result on the propagation of boundary regularity is the following.

Theorem 1.2. *Let $\mu \in (0, 1)$ and $n \in \mathbb{N}^*$. Assume that $m(\xi) = m(|\xi|)$ is a radial function satisfying (H1)-(H2a)-(Osg). Let $\omega_0 = \mathbf{1}_{D_0}$, where $D_0 \subset \mathbb{R}^2$ is a bounded simply connected domain with $\partial D_0 \in C^{n,\mu}$. Let ω be the corresponding unique global patch solution of the 2D Loglog-Euler equation (1). Then, for any $t > 0$ and every $\varepsilon \in (0, \mu)$, the patch boundary $\partial D(t) = \Phi_t(\partial D_0)$ satisfies $\partial D(t) \in C^{n,\mu-\varepsilon}$.*

More precisely, let $z(\cdot, t)$ be a parameterization of $\partial D(t)$ (see (46) and (52)). Then, for every $\varepsilon > 0$ and every $\epsilon > 0$, there exists a constant $C = C(\mu, \varepsilon, \epsilon, \|z_0\|_{C^{n,\mu}}) > 0$ such that

$$\|z(t)\|_{C^{n,\mu-\varepsilon}} \leq C \left(H^{-1} \left(C(1+t) \log^{\beta+\epsilon}(e+t) \right) \right)^{C \log^{\beta+\epsilon}(e+t)}, \quad (10)$$

where the map H is defined by (7).

Theorem 1.2 shows that the initial $C^{n,\mu}$ boundary regularity persists in time, up to the loss of an arbitrarily small exponent. However, the estimate (10) indicates that the corresponding bound may grow rapidly in time, up to $H^{-1}(t)$, with additional logarithmic factors. According to Remark 1, this growth can be double-exponential in time, or even faster. This is consistent with the results for the two-dimensional Euler case (see Remark 2).

The next theorem extends the preceding boundary regularity result to the multi-patch setting.

Theorem 1.3. *Let $\mu \in (0, 1)$ and $n \in \mathbb{N}^*$. Assume that $m(\xi) = m(|\xi|)$ is a radial function satisfying (H1)-(H2a)-(Osg). Let ω_0 be the multi-patch initial data given by (4), where $D_j(0) \subset \mathbb{R}^2$ are bounded, simply connected, and pairwise disjoint domains with $\partial D_j(0) \in C^{n,\mu}$ for $j = 1, \dots, N$. Let ω be the corresponding unique global patch solution of the 2D Loglog-Euler equation (1). Then, for any $t > 0$, the patches $D_j(t)$ remain pairwise disjoint. Moreover, for every $\varepsilon \in (0, \mu)$,*

$$\partial D_j(t) \in C^{n,\mu-\varepsilon}, \quad j \in \{1, \dots, N\}.$$

A more detailed version of Theorem 1.3 is given in Theorem 6.1, which provides a more quantitative bound in (120). Compared to the bound (10) for the single-patch case, the growth behaves like $\exp(H^{-1}(t))$, which is triple-exponential in time or even faster. The additional exponential factor arises from the fact that the distance between patches may decrease over time, thereby influencing the boundary regularity. See Remark 7 for further discussion.

In what follows, we outline the main ideas underlying the proofs of our main theorems, emphasizing the key analytical mechanisms and novelties involved.

For the proof of Theorem 1.1, we first establish the crucial modulus-of-continuity estimate (42) for the velocity field and derive the corresponding estimate (6) for the flow map. Combined with an approximation argument based on kernel regularization, these *a priori* estimates yield the global existence of Yudovich-type solutions.

For uniqueness, the classical proof for the 2D Euler equation (see [18, 79]) relies on the harmonic-analysis estimate

$$\|\nabla^\perp \nabla(-\Delta)^{-1} \omega\|_{L^p} \leq C \frac{p^2}{p-1} \|\omega\|_{L^p}, \quad 1 < p < \infty,$$

where $C > 0$ is universal. Obtaining an analogous estimate for $\nabla^\perp \nabla(-\Delta)^{-1} m(\Lambda) \omega$ with an unbounded multiplier m appears difficult. Instead, we adopt the approach of [81, Sec 2.3], which is based on the analysis of the flow map. More precisely, we introduce the quantity $\delta(t)$ defined in (44), analogous to [81, Eq. (3.30)], but formulated on the whole space rather than a bounded smooth domain. Combining the modulus-of-continuity estimate with the flow map bounds, we derive an Osgood-type inequality that closes the uniqueness argument.

Next, we outline the proof of Theorem 1.2. Following the level-set formulation introduced in Section 4.1, we represent the patch boundary $\partial D(t)$ by a level-set function φ , namely $\varphi(x, t) = 0$ for all $x \in \partial D(t)$ (see Definition 4.1). Denote by $W \triangleq \nabla^\perp \varphi$ the tangential vector field along $\partial D(t)$. We also introduce the tangential differential operator

$$\partial_W \triangleq W \cdot \nabla, \quad \text{and} \quad \partial_W^k \triangleq \underbrace{\partial_W \cdots \partial_W}_{k \text{ times}}.$$

As shown in Section 4.1, for every $t > 0$ and $n \in \mathbb{N}^*$, $\mu \in (0, 1)$,

$$\|\partial_W^{n-1} W(\cdot, t)\|_{C^\mu} < +\infty \implies \partial D(t) \in C^{n, \mu},$$

so the propagation of boundary regularity reduces to estimating tangential derivatives of W .

For the case $n = 1$, Elgindi [44] established the global propagation of $C^{1, \mu-\varepsilon}$ boundary regularity by combining frequency-space techniques with the method of losing estimates [3, 4], simultaneously controlling $\|W(t)\|_{\dot{C}^{\mu-\varepsilon}}$ and $|W(t)|_{\inf} \triangleq \inf_{x \in \partial D(t)} |W(x, t)|$.

In contrast, we develop a new approach that is entirely physical-space-based. Compared with the frequency-space framework, our method is better suited to extensions to domains with boundaries.

Our approach relies on three key ingredients.

- A near-Lipschitz estimate for the velocity field $u = \nabla^\perp(-\Delta)^{-1} m(\Lambda)(\mathbf{1}_{D(t)})$.

In Lemma 4.1, we show that, in the single-patch setting, u admits a modulus of continuity of the form $\rho \mapsto \rho(m(\rho^{-1}) + 1)$ up to the logarithmic factor

$$f(t) \triangleq \log \left(1 + \frac{\|W(t)\|_{\dot{C}^\gamma}}{|W(t)|_{\inf}} \right), \quad \gamma \in (0, \mu].$$

For general vorticity $\omega \in L^1 \cap L^\infty$, the corresponding modulus is $\rho \mapsto \rho(\log \rho^{-1} m(\rho^{-1}) + 1)$; see (37). Thus, the patch geometry yields an improvement by a logarithmic factor, which is crucial for implementing the method of losing estimates.

- A key integral representation formula and refined Hölder-type estimate of $\partial_W u$.

In Lemma 4.4, we derive a singular integral representation for $\partial_W u$ that generalizes [7, Proposition 2] from the 2D Euler setting. The justification is more delicate here because ∇u is generally unbounded and the multiplier $m(\Lambda)$ introduces an additional singularity; see, for example, [44, p. 984]. To overcome these difficulties, we adopt a more general approach to justify the integral representation, partially inspired by [86].

The refined Hölder-type estimate in Lemma 4.5 further extends the following bound from [7, Corollary 1], which holds for the 2D Euler case:

$$\|\partial_W u\|_{\dot{C}^\mu} \leq C \|\nabla u\|_{L^\infty} (\|W\|_{\dot{C}^\mu} + 1).$$

To account for the multiplier $m(\Lambda)$, we introduce the m -adapted Hölder space $(C_{\mathbf{m}}^\mu, \|\cdot\|_{C_{\mathbf{m}}^\mu})$ defined in (59). Combining this framework with the near-Lipschitz estimate, we control $\|\partial_W u\|_{C_{\mathbf{m}}^\mu}$ in terms of $\|W\|_{\dot{C}^\mu}$, up to the logarithmic factor f .

- *A refined estimate for the flow map $\Phi_{t,s}$.*

The near-Lipschitz regularity of the velocity yields the sharper continuity estimate for the flow map $\Phi_{t,s}$ defined in (5). This significantly improves upon the Yudovich-type estimate (6) and allows us to estimate $W(x, t)$ directly along the flow map.

Combining these ingredients with the method of losing estimates in the physical-space framework, we obtain the propagation estimate for $\|W(\cdot, t)\|_{C^{\mu-\varepsilon}}$ (see (94)).

To control the logarithmic factor f , we further derive a lower bound for $|W(t)|_{\inf}$ in (81), following the strategy of [44]. The key ingredient is the pointwise estimate on $\nabla u \mathbf{w} \cdot \mathbf{w}$ established in Lemma 4.6, where $\mathbf{w} = \frac{W}{|W|}$.

Combining the estimates for $\|W(\cdot, t)\|_{\dot{C}^{\mu-\varepsilon}}$ and $|W(t)|_{\inf}$, we show that the logarithmic quantity f satisfies the Osgood-type inequality (100). This yields

$$|W(t)|_{\inf} \geq |W_0|_{\inf} \left(H^{-1}(C(1+t) \log^{\beta+\varepsilon}(e+t)) \right)^{-C}, \quad (11)$$

$$\|W(t)\|_{C^{\mu-\varepsilon}(\mathbb{R}^2)} \leq C \|W_0\|_{C^\mu} \left(H^{-1}(C(1+t) \log^{\beta+\varepsilon}(e+t)) \right)^{C \log^{\beta+\varepsilon}(e+t)}, \quad (12)$$

which immediately implies the boundary estimate (10) with $n = 1$.

For general $n \in \mathbb{N}^*$, the proof of Theorem 1.2 follows the same strategy as in the case $n = 1$. The main new ingredient is a higher-order analogue of the key integral representation (108) for $\partial_W^k u$. Based on this formula, we establish the Hölder-type estimate for $\partial_W^k u$ in Lemma 5.4, treating the leading term involving $\partial_W^{k-1} W$ as in the $n = 1$ case, and estimating the lower-order striated terms separately. An induction argument combined with the method of losing estimates then yields

$$\|\partial_W^{n-1} W(t)\|_{C^{\mu-\varepsilon}(\mathbb{R}^2)} \leq C \|\varphi_0\|_{C^{n,\mu}} \left(H^{-1}(C(1+t) \log^{\beta+\varepsilon}(e+t)) \right)^{C \log^{\beta+\varepsilon}(e+t)}, \quad (13)$$

which completes the proof of Theorem 1.2.

For the multi-patch problem, we first use Theorem 1.1 to obtain a uniform positive lower bound on the distance between distinct patches throughout the evolution. Consequently, the interaction terms remain regular, while the only singular contribution comes from the patch under consideration. The single-patch estimates therefore apply to each patch separately, leading to the proof of Theorem 1.3.

Finally, we present some remarks as follows.

Remark 2 (Bounded multiplier case). Theorems 1.1, 1.2, and 1.3 remain valid under assumptions (H1)-(H2c), with $\beta = \varepsilon = \epsilon = 0$ in the corresponding statements. Consequently, these results apply to the quasi-geostrophic shallow water equation [40] (i.e. $m(r) = \frac{r^2}{r^2 + \lambda^2}$, $\lambda > 0$); see also the recent preprint [78] on the same model. We do not attempt to formulate the most general assumptions

under which our arguments remain valid. For example, they also apply to the Euler- λ equation [90] (i.e. $m(r) = \frac{1}{1+\lambda^2 r^2}$, $\lambda > 0$) and to the logarithmically modified 2D Euler equations [17, 39, 74], although m is not monotone increasing. In particular, the bounds (11)-(13) for the single-patch case and (119)-(120) for the multi-patch case coincide with the corresponding bounds for the 2D Euler equation, as given in [7, Eqs. (2.13), (2.11)] and [71, p. 930], respectively.

Indeed, when m is bounded, the m -adapted space C_m^μ coincides with the standard Hölder space C^μ . Moreover, the proofs of the main estimates (see Lemmas 4.1, 4.5, 4.6, and 5.4) carry over with only minor modifications. Since the velocity field is Lipschitz continuous (Lemma 4.1), the losing-estimates method is no longer needed. Consequently, one recovers the global persistence of $C^{n,\mu}$ boundary regularity for patch solutions.

Remark 3 (Half-plane case). Consider the vortex patch problem for the 2D Loglog-Euler type equation (1) in the half-space $\mathbb{R}_+^2$ with a no-flow (rigid) boundary condition. If the initial patch boundaries are disjoint from the rigid boundary $\partial\mathbb{R}_+^2$, then one may perform an odd extension across the boundary and apply the proof of Theorem 1.3 to obtain global persistence of patch boundary regularity in the half-space, exactly as the whole-space case.

If the patch boundaries touch the rigid boundary, as in the singularity scenario studied in [71, 52, 83], the situation becomes more delicate. It is natural to combine the techniques of [71] for the 2D Euler equation with the arguments developed here to establish global persistence result in the presence of boundary contact. Most of the estimates used in [71] can be extended without substantial difficulty. The main obstacle is an analogue of [71, Eq. (3.29)] (corresponding to $m \equiv 1$ and $\gamma = 0$ in (14)):

Given $\tilde{B}_x \triangleq B(O_x, r_x)$ and $u_{\tilde{B}_x}(z) = \nabla^\perp(-\Delta)^{-1}m(\Lambda)(\mathbf{1}_{\tilde{B}_x})(z)$, does it hold that for all $|z - O_x| = r_x + d(x)$ and $d(x) \in (0, \frac{1}{4}r_x]$ and for some $\gamma \in [0, 1)$,

$$|\nabla^2 u_{\tilde{B}_x}(z)| \leq C(1 + m(d(x)^{-1}))d(x)^{-\gamma}r_x^{-1+\gamma} \quad (14)$$

However, part (iii) of Proposition 4.3 shows that this estimate (14) already fails for the unit disc patch. Therefore, the argument of [71] cannot be extended directly to the Loglog-Euler type setting. New ideas will be required to resolve this boundary-touching vortex patch problem.

The remainder of this paper is organized as follows. In Section 2, we collect several useful properties of $m(r)$ and of the kernel function $G(\rho)$, and present the expression formula for the velocity $u = \nabla^\perp(-\Delta)^{-1}m(\Lambda)\omega$. In Section 3, we prove Theorem 1.1, establishing the Yudovich-type result for the 2D Loglog-Euler type equation. Section 4 is devoted to the proof of Theorem 1.2 with $n = 1$, concerning the single-patch case, while the general case $n \in \mathbb{N}^*$ of Theorem 1.2 is proved in Section 5. Finally, the proof of Theorem 1.3 on multiple patches is presented in Section 6. The appendices contain the proof of the kernel estimates, the circular-patch analysis underlying Proposition 4.3, and the regularization and compactness arguments deferred from Sections 3–6.

Notations. We use C to denote a generic positive constant, which may vary from line to line. Its dependence on parameters will be specified when necessary. We write $A \approx_p B$ if there exists a constant $C > 0$, depending only on p , such that $C^{-1}A \leq B \leq CA$.

2. PRELIMINARIES

In this section, we shall deduce some useful properties of $m(r)$, write out the expression formula of the velocity field $u(x)$ in convolution form, and recall various estimates of the kernel function $G(\rho)$ determined by the multiplier m .

2.1. Properties of $m(r)$. We collect below several useful consequences of assumptions (H1)-(H2a). We first note that the limits in (H2a) can equivalently be written as:

$$\begin{aligned} \lim_{r \rightarrow +\infty} \frac{r(\log r)(\log \log r)m'(r)}{m(r)} &= \lim_{r \rightarrow +\infty} \frac{r(\log r)\tilde{m}'(r)}{\tilde{m}(r)} = \beta, \\ \lim_{r \rightarrow +\infty} \frac{r(\log r)m'(r)}{m(r)} &= \lim_{r \rightarrow +\infty} \frac{r\tilde{m}'(r)}{\tilde{m}(r)} = \beta_1, \end{aligned} \quad (15)$$

$$\lim_{r \rightarrow +\infty} \frac{(\log r)(rm''(r) + m'(r))}{m'(r)} = \lim_{r \rightarrow +\infty} \frac{r\tilde{m}''(r)}{\tilde{m}'(r)} = \beta_2, \quad (16)$$

with $\beta \in [0, +\infty]$, $\beta_1 \in [0, +\infty)$, $\beta_2 \in (-2, +\infty)$.

- Under assumption (H1), for every $\lambda > 0$,

$$m(\lambda r) \approx_\lambda m(r), \quad \forall r > 0. \quad (17)$$

Moreover, there exists a constant $C > 0$ such that

$$m'(r) \leq Cr^{-1}m(r). \quad (18)$$

See, e.g., [83, Eq. (43)] for (17). To prove (18), we first use (H1) to obtain

$$\frac{d}{dr}(rm'(r)) = m'(r) + rm''(r) \leq Cm'(r).$$

Integrating over $(0, r)$ gives

$$rm'(r) \leq \lim_{r \rightarrow 0^+} rm'(r) + C(m(r) - m(0^+)) \leq Cm(r),$$

where we have used the fact that $m(0^+) \geq 0$, and

$$\lim_{r \rightarrow 0^+} rm'(r) = 0. \quad (19)$$

Indeed, if this limit were equal to some $c > 0$, then $m'(r) > \frac{c}{2r}$ for all sufficiently small r , contradicting the integrability of m' near the origin.

- Under assumption (H2a), if $\beta < +\infty$, for every $\varepsilon > 0$, there exists a large constant $r_\varepsilon \geq 2$ such that

$$r \mapsto (\log r)^{-\beta-\varepsilon} \tilde{m}(r) \text{ is decreasing on } [r_\varepsilon, \infty). \quad (20)$$

Indeed,

$$\frac{d}{dr}(\log r)^{-\beta-\varepsilon} \tilde{m}(r) = \frac{\tilde{m}(r)}{r(\log r)^{\beta+1+\varepsilon}} \left(\frac{r(\log r)\tilde{m}'(r)}{\tilde{m}(r)} - \beta - \varepsilon \right);$$

By (H2a), the expression in parentheses converges to $-\varepsilon$ as $r \rightarrow +\infty$ and is therefore negative for all sufficiently large r . This proves (20).

If $\beta = +\infty$, the same argument applied to (15) yields the following weaker conclusion: for every $\varepsilon > 0$, there exists a large constant $r_\varepsilon \geq 2$ such that

$$r \mapsto (\log r)^{-\beta_1-\varepsilon} m(r) \text{ is decreasing on } [r_\varepsilon, \infty).$$

Consequently,

$$m(r) \leq C_\varepsilon \max \{1, (\log_+ r)^{\beta_1+\varepsilon}\}, \quad \forall r > 0. \quad (21)$$

where $C_\varepsilon > 0$ depending only on ε and the multiplier m .

- Under assumption (H2a), there exists a sufficiently large constant $r_0 \geq 2$ such that

$$r \mapsto r \tilde{m}(r) \text{ is increasing and convex in } [r_0, +\infty). \quad (22)$$

Indeed,

$$\frac{d}{dr}(r\tilde{m}(r)) = \tilde{m}(r)\left(1 + \frac{r\tilde{m}'(r)}{\tilde{m}(r)}\right), \quad \text{and} \quad \frac{d^2}{dr^2}(r\tilde{m}(r)) = \tilde{m}'(r)\left(2 + \frac{r\tilde{m}''(r)}{\tilde{m}'(r)}\right),$$

and

$$1 + \frac{r\tilde{m}'(r)}{\tilde{m}(r)} \xrightarrow{r \rightarrow +\infty} 1 + \beta_1 > 0, \quad 2 + \frac{r\tilde{m}''(r)}{\tilde{m}'(r)} \xrightarrow{r \rightarrow +\infty} 2 + \beta_2 > 0.$$

Since $\tilde{m}(r) > 0$ and $\tilde{m}'(r) \geq 0$ for all $r > 0$, the desired monotonicity and convexity follow.

- Suppose $\beta < +\infty$. For any $\kappa \geq 1$, there exists a constant $C = C(\kappa, \beta) > 0$ such that

$$\tilde{m}(r^\kappa) \leq C\tilde{m}(r), \quad \forall r > 0. \quad (23)$$

Indeed, the inequality follows from (20) with $\varepsilon = 1$. For $r \leq r_1$, we apply the monotonicity of $\tilde{m}$ to get $\tilde{m}(r^\kappa) \leq \tilde{m}(r_1^\kappa) \leq C_{r_1}\tilde{m}(0) \leq C_{r_1}\tilde{m}(r)$. For $r \geq r_1$, it follows from (20) that

$$\tilde{m}(r^\kappa) \leq \left(\frac{\log r^\kappa}{\log r}\right)^{\beta+1} \tilde{m}(r) = \kappa^{\beta+1} \tilde{m}(r).$$

Applying the same argument to m , with β replaced by β_1 , yields

$$m(r^\kappa) \leq C m(r), \quad \forall r > 0, \quad (24)$$

where $C > 0$ depends on κ and β_1 . In obtaining (24), for $r \leq r_1$, we additionally use (17) to estimate $m(r^\kappa) \leq m(r_1^{\kappa-1}r) \leq C_{r_1}m(r)$. Although (24) is weaker than (23), it remains valid when $\beta = +\infty$.

As a consequence of (24), for every $\lambda > 0$,

$$\tilde{m}(\lambda r) \approx_\lambda \tilde{m}(r), \quad \forall r > 0. \quad (25)$$

Indeed, $\tilde{m}(\lambda r) = m((e^r)^\lambda)$, and the conclusion follows immediately from (24) together with the monotonicity of m .

- For any $\mu \in [\mu_1, \mu_2] \subset (0, 1)$, there exists $C_{\mu_1} > 0$ such that

$$\rho_1^\mu m(\rho_1^{-1}) \leq C_{\mu_1} \rho_2^\mu (m(\rho_2^{-1}) + 1), \quad \forall \rho_2 \geq \rho_1 > 0. \quad (26)$$

Indeed, there exists some $c_{\mu_1} > 0$ such that for every $\rho \in (0, c_{\mu_1})$,

$$\frac{d(\rho^\mu m(\rho^{-1}))}{d\rho} = \rho^{\mu-1} \left(\mu m(\rho^{-1}) - \rho^{-1} m'(\rho^{-1}) \right) \geq \rho^{\mu-1} m(\rho^{-1}) \left(\mu_1 - \frac{\rho^{-1} m'(\rho^{-1})}{m(\rho^{-1})} \right) > 0,$$

since from (H2a), we have

$$\lim_{\rho \rightarrow 0} \frac{\rho^{-1} m'(\rho^{-1})}{m(\rho^{-1})} = \lim_{r \rightarrow +\infty} \frac{r m'(r)}{m(r)} = \lim_{r \rightarrow +\infty} \frac{\beta_1}{\log r} = 0.$$

Hence, (26) holds with $C_{\mu_1} = 1$ for the case $\rho_2 \leq c_{\mu_1}$. When $\rho_2 > c_{\mu_1}$, we have

$$\text{For } \rho_1 \leq c_{\mu_1} : \quad \rho_1^\mu m(\rho_1^{-1}) \leq c_{\mu_1}^\mu m(c_{\mu_1}^{-1}) \leq \rho_2^\mu m(c_{\mu_1}^{-1}),$$

$$\text{For } \rho_1 > c_{\mu_1} : \quad \rho_1^\mu m(\rho_1^{-1}) \leq \rho_2^\mu m(c_{\mu_1}^{-1}),$$

and (26) holds with $C_{\mu_1} = m(c_{\mu_1}^{-1})$.

When the Osgood condition (Osg) holds, the limits in (H2) admit the following characterization.

Lemma 2.1. *Suppose that $m(r)$ satisfies (H1)-(H2). Then:*

- If (H2a) holds, then (Osg) imposes $\beta \leq 1$, $\beta_1 = 0$, and $\beta_2 = -1$.
- If (H2b) holds, then (Osg) must fail.
- If (H2c) holds, then (Osg) must hold.

Proof. We first show that under assumptions (H2a) and (Osg), we have $\beta \leq 1$. Suppose $\beta > 1$. Through a similar argument as in (20), we get

$$\forall \varepsilon > 0, \quad r \mapsto (\log r)^{-\beta+\varepsilon} \tilde{m}(r) \text{ is increasing on } [r_\varepsilon, \infty),$$

for a large constant r_ε . Taking $\varepsilon = \frac{\beta-1}{2} > 0$, we have

$$\tilde{m}(r) = (\log r)^{1+\varepsilon} (\log r)^{-(1+\varepsilon)} \tilde{m}(r) \geq (\log r)^{1+\varepsilon} (\log r_\varepsilon)^{-(1+\varepsilon)} \tilde{m}(r_\varepsilon) = C_{r_\varepsilon} (\log r)^{1+\varepsilon}, \quad \forall r \geq r_\varepsilon.$$

This further implies

$$\int_{r_\varepsilon}^{+\infty} \frac{dr}{r \tilde{m}(r)} \leq \frac{1}{C_{r_\varepsilon}} \int_{r_\varepsilon}^{+\infty} \frac{dr}{r (\log r)^{1+\varepsilon}} = \frac{1}{C_{r_\varepsilon} \varepsilon (\log r_\varepsilon)^\varepsilon} < +\infty,$$

which contradicts the Osgood condition (Osg). Hence, we conclude that $\beta \leq 1$.

For β_1 , we have

$$\beta_1 = \lim_{r \rightarrow +\infty} \frac{r \tilde{m}'(r)}{\tilde{m}(r)} = \lim_{r \rightarrow +\infty} \frac{\beta}{\log r} \leq \lim_{r \rightarrow +\infty} \frac{1}{\log r} = 0.$$

On the other hand, $\beta_1 \geq 0$ due to (H1). Therefore, $\beta_1 = 0$.

Finally, we apply l'Hôpital's rule and get

$$\beta_1 = \lim_{r \rightarrow +\infty} \frac{r \tilde{m}'(r)}{\tilde{m}(r)} = \lim_{r \rightarrow +\infty} \frac{\tilde{m}'(r) + r \tilde{m}''(r)}{\tilde{m}'(r)} = 1 + \beta_2.$$

We conclude with $\beta_2 = \beta_1 - 1 = -1$.

Under assumption (H2b), we have $m(r) \geq Cr^{\alpha/2}$ for $r > r_0$ with a large enough r_0 . Hence,

$$\int_2^{+\infty} \frac{1}{r(\log r)m(r)} dr \leq \int_2^{r_0} \frac{1}{r(\log r)m(r)} dr + \int_{r_0}^{+\infty} \frac{1}{r(\log r)Cr^{\frac{\alpha}{2}}} dr < +\infty.$$

Therefore, the Osgood condition (Osg) fails.

Finally, under assumption (H2c), we have $m(r) \leq C$ for any $r > 0$, and the Osgood condition (Osg) must hold:

$$\int_2^{+\infty} \frac{1}{r(\log r)m(r)} dr \geq \int_2^{+\infty} \frac{1}{Cr(\log r)} dr = +\infty.$$

□

2.2. Expression formula of the velocity $u(x)$. We collect some useful representation formulas for the velocity field $u(x)$.

Lemma 2.2. *Let $u(x) = \nabla^\perp(-\Delta)^{-1}m(\Lambda)\omega(x)$, where $m(\Lambda)$ is a Fourier multiplier with radial symbol $m(\xi) = m(|\xi|)$ satisfying that $m(\xi) \in C^2(\mathbb{R}^2 \setminus \{0\})$ and $\lim_{r \rightarrow 0^+} rm'(r)$ exist, and also*

$$\lim_{r \rightarrow +\infty} r^{-\frac{1}{2}}m(r) = 0, \quad \lim_{r \rightarrow +\infty} r^{\frac{1}{2}}m'(r) = 0.$$

Then the following statements hold.

(i) *For every $x \in \mathbb{R}^2$ we have*

$$u(x) = K * \omega(x) = \int_{\mathbb{R}^2} K(x-y)\omega(y)dy, \quad K(x) \triangleq \frac{x^\perp}{|x|^2}G(|x|), \quad (27)$$

where $x^\perp = (-x_2, x_1)$, and

$$\begin{aligned} G(\rho) &\triangleq \frac{1}{2\pi} m(0^+) + \frac{1}{2\pi} \int_0^\infty J_0(\rho r) m'(r) dr \\ &= \frac{1}{2\pi} m(0^+) + \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} e^{ix \cdot \xi} \frac{m'(|\xi|)}{|\xi|} d\xi, \end{aligned} \quad (28)$$

where $m(0^+) \triangleq \lim_{r \rightarrow 0^+} m(r)$ and J_0 denotes the Bessel function of the first kind of order zero.

(ii) For every $x \in \mathbb{R}^2$, the symmetric part $\mathbf{S}(\nabla u)(x)$ of $\nabla u(x)$ can be expressed as

$$\mathbf{S}(\nabla u)(x) \triangleq \frac{\nabla u(x) + (\nabla u)^T(x)}{2} = \text{p.v.} \int_{\mathbb{R}^2} (\nabla K)_S(x-y) \omega(y) dy, \quad (29)$$

where

$$(\nabla K)_S(x) \triangleq \frac{1}{2} \frac{2G(|x|) - |x|G'(|x|)}{|x|^2} \sigma(x), \quad \sigma(x) \triangleq \begin{pmatrix} \frac{2x_1 x_2}{|x|^2} & \frac{x_2^2 - x_1^2}{|x|^2} \\ \frac{x_2^2 - x_1^2}{|x|^2} & -\frac{2x_1 x_2}{|x|^2} \end{pmatrix}. \quad (30)$$

Remark 4. We remind that, if $\omega(x) = \mathbf{1}_D(x)$ where $D \subset \mathbb{R}^2$ is a simply connected bounded open domain, then $\nabla u(x)$ is well defined for any $x \notin D$ and $\nabla u(x) = \nabla K * \omega(x)$, with $\nabla K(\cdot)$ given by (31). In addition, for $x \in D$, we also see that

$$u(x) = \int_D K(x-y) dy = \int_{D \setminus B(x, d_x)} K(x-y) dy,$$

where $K(x) = \frac{x^\perp}{|x|^2} G(|x|)$ and $d_x \triangleq d(x, \partial D)$. Consequently, $\nabla u(x)$ is bounded outside and inside the domain D , but it may be unbounded on the boundary ∂D .

Proof of Lemma 2.2. The statement (i) has been proved in [83, Lemma 1]. In the following, we only prove statement (ii).

We start with the calculation of ∇u in the distributional sense. For every test function $\tilde{\chi} \in C_c^\infty(\mathbb{R}^2)$, we apply the expression (27) and get

$$\begin{aligned} (\partial_{x_j} u_i, \tilde{\chi}) &= -(u_i, \partial_{x_j} \tilde{\chi}) = - \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} K_i(x-y) \omega(y) dy \partial_{x_j} \tilde{\chi}(x) dx \\ &= - \lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}^2} \int_{|x-y| \geq \varepsilon} K_i(x-y) \partial_{x_j} \tilde{\chi}(x) dx \omega(y) dy \\ &= \lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}^2} \left(\int_{|x-y| \geq \varepsilon} \partial_{x_j} K_i(x-y) \tilde{\chi}(x) dx - \int_{|x-y|=\varepsilon} K_i(x-y) \frac{x_j - y_j}{|x-y|} \tilde{\chi}(x) dS(x) \right) \omega(y) dy. \end{aligned}$$

We focus on the term

$$I_\varepsilon^{i,j}(y) \triangleq \int_{|x-y|=\varepsilon} K_i(x-y) \frac{x_j - y_j}{|x-y|} \tilde{\chi}(x) dS(x).$$

Since the limit $\lim_{\varepsilon \rightarrow 0} I_\varepsilon^{i,j}(y)$ may be unbounded, we need to explore extra cancellations. Define

$$J_\varepsilon^{i,j}(y) \triangleq \tilde{\chi}(y) \int_{|x-y|=\varepsilon} K_i(x-y) \frac{x_j - y_j}{|x-y|} dS(x).$$

Then, we control the difference by

$$\begin{aligned} |I_\varepsilon^{i,j}(y) - J_\varepsilon^{i,j}(y)| &= \left| \int_{|x-y|=\varepsilon} K_i(x-y) \frac{x_j - y_j}{|x-y|} (\tilde{\chi}(x) - \tilde{\chi}(y)) dS(x) \right| \\ &\leq \frac{G(\varepsilon)}{\varepsilon} \cdot 1 \cdot \|\nabla \tilde{\chi}\|_{L^\infty} \varepsilon \cdot 2\pi \varepsilon \leq C \varepsilon m(\varepsilon^{-1}) \xrightarrow{\varepsilon \rightarrow 0} 0, \end{aligned}$$

where we have used (32) (the proof is independent of the statement (ii)) and (21). For $J_\varepsilon^{i,j}(y)$, by symmetry we have

$$\begin{aligned} J_\varepsilon^{1,1}(y) &= -J_\varepsilon^{2,2}(y) = \tilde{\chi}(y) \int_{|x-y|=\varepsilon} \frac{(x_1 - y_1)(x_2 - y_2)}{|x - y|^3} dS(x) = 0, \\ J_\varepsilon^{1,2}(y) + J_\varepsilon^{2,1}(y) &= \tilde{\chi}(y) \int_{|x-y|=\varepsilon} \frac{(x_1 - y_1)^2 - (x_2 - y_2)^2}{|x - y|^3} dS(x) = 0. \end{aligned}$$

Therefore, we conclude with

$$\begin{aligned} \mathbf{S}(\nabla u)(x) &= \lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}^2} \int_{|x-y| \geq \varepsilon} \frac{\nabla K(x-y) + (\nabla K)^T(x-y)}{2} \tilde{\chi}(x) dx \omega(y) dy \\ &\quad + \lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}^2} \left(\frac{I_\varepsilon^{1,1}}{I_\varepsilon^{1,2}(y) + I_\varepsilon^{2,1}(y)} \frac{I_\varepsilon^{1,2}(y) + I_\varepsilon^{2,1}(y)}{2} \right) \omega(y) dy \\ &= \int_{\mathbb{R}^2} \left(\text{p.v.} \int_{\mathbb{R}^2} (\nabla K)_S(x-y) \omega(y) dy \right) \tilde{\chi}(x) dx, \end{aligned}$$

where direct computation yields

$$\nabla K(x) = \frac{1}{2} \frac{2G(|x|) - |x|G'(|x|)}{|x|^2} \sigma(x) - \frac{1}{2} \frac{G'(|x|)}{|x|} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \triangleq (\nabla K)_S(x) + (\nabla K)_A(x), \quad (31)$$

where $\sigma(x)$ defined by (30), also $(\nabla K)_S = \frac{\nabla K + (\nabla K)^T}{2}$ and $(\nabla K)_A = \frac{\nabla K - (\nabla K)^T}{2}$ are the symmetric and antisymmetric parts of ∇K , respectively. This finishes the proof of (29). $\square$

2.3. Properties of $G(\rho)$. We establish several useful estimates for $G(\rho)$ defined by (28) under suitable assumptions on the multiplier m .

Lemma 2.3. *Assume that $m(r) = m(|\xi|)$ satisfies (H1)-(H2a) with $n \in \mathbb{N}^*$. Then there exist constants $\bar{c}_0 > 0$ and $C > 0$ such that $G(\rho) = G(|x|)$ defined by (28) verifies the following statements:*

$$\text{for } \rho \in (0, \bar{c}_0), \quad G(\rho) \approx m(\rho^{-1}), \quad (32)$$

and

$$\text{for } \rho \in (0, \bar{c}_0), \quad |G^{(l)}(\rho)| \leq \frac{Cm(\rho^{-1})}{\rho^l}, \quad \forall l \in \{1, 2, \dots, n+1\}, \quad (33)$$

and

$$\text{for } \rho \in [\bar{c}_0, +\infty), \quad |G^{(l)}(\rho)| \leq \frac{C}{\rho^l}, \quad \forall l \in \{1, 2, \dots, n+1\}. \quad (34)$$

Lemma 2.3 follows from the explicit representation (28) of G . See, for example, [83, Lemma 2] for a proof of (32) and (33) with $l = 1, 2$, noting that assumptions (H1)-(H2a) required there are readily verified in the present setting. For the reader's convenience, we include a proof of (33) (for general l) and (34) in Appendix A.

As a direct consequence, we obtain the following estimate on the kernel K .

Corollary 2.4. *Assume that $m(r) = m(|\xi|)$ satisfies (H1)-(H2a) with $n \in \mathbb{N}^*$, and K is defined in (27). Then*

$$|\nabla^l K(x)| \leq C|x|^{-(l+1)}(m(|x|^{-1}) + 1), \quad \forall x \in \mathbb{R}^2, \quad \forall l \in \{0, 1, \dots, n+1\}. \quad (35)$$

Proof. From the definition of K in (27), applying the Leibniz rule and Lemma 2.3, we compute

$$|\nabla^l K(x)| \leq \sum_{j=0}^l \frac{l!}{j!} \frac{|G^{(j)}(|x|)|}{|x|^{l-j+1}} \leq \sum_{j=0}^l \frac{l!}{j!} \frac{C(m(|x|^{-1}) + 1)}{|x|^j \cdot |x|^{l-j+1}} \leq \frac{C(m(|x|^{-1}) + 1)}{|x|^{l+1}},$$

for any $l \in \{0, 1, \dots, n+1\}$. □

3. YUDOVICH-TYPE THEOREM FOR THE 2D LOGLOG-EULER TYPE EQUATION

In this section, we establish the Yudovich-type theorem for the 2D Loglog-Euler equation (1) associated with initial data $\omega_0 \in L^1 \cap L^\infty(\mathbb{R}^2)$.

We first introduce the definition of weak solutions for the equation (1).

Definition 3.1 (Weak solutions). Let $\omega_0 \in L^1 \cap L^\infty(\mathbb{R}^2)$. We say that (u, ω) is a weak solution to the 2D Loglog-Euler type equation (1) with initial data ω_0 , provided that for any $T > 0$,

- (i) $\omega \in L^\infty([0, T]; L^1 \cap L^\infty(\mathbb{R}^2))$.
- (ii) $u = \nabla^\perp(-\Delta)^{-1}m(\Lambda)\omega = K * \omega$ with the kernel K given by (27).
- (iii) For every $\tilde{\chi} \in C^1([0, T]; C_0^1(\mathbb{R}^2))$,

$$\int_{\mathbb{R}^2} \omega(x, T) \tilde{\chi}(x, T) dx - \int_{\mathbb{R}^2} \omega_0(x) \tilde{\chi}(x, 0) dx = \int_0^T \int_{\mathbb{R}^2} \omega(x, t) (\partial_t \tilde{\chi} + u \cdot \nabla \tilde{\chi})(x, t) dx dt.$$

The following lemma establishes the modulus-of-continuity estimate for the velocity field associated with bounded and integrable vorticity.

Lemma 3.1. Assume that $m(\xi) = m(|\xi|)$ is a radial function satisfying (H1)-(H2a). For any $\omega \in L^1 \cap L^\infty(\mathbb{R}^2)$, let $u = \nabla^\perp(-\Delta)^{-1}m(\Lambda)(\omega)$. Then

$$\|u\|_{L^\infty(\mathbb{R}^2)} + \sup_{|x-\tilde{x}|>0} \frac{|u(x) - u(\tilde{x})|}{\nu(|x - \tilde{x}|)} \leq C \|\omega\|_{L^1 \cap L^\infty(\mathbb{R}^2)}, \quad (36)$$

where $C > 0$ is a universal constant and the modulus of continuity ν is defined by

$$\nu(\rho) \triangleq \begin{cases} \rho(\log \rho^{-1})m(\rho^{-1}), & \text{for } 0 < \rho \leq \frac{1}{2}, \\ \rho(\log 2)m(2), & \text{for } \rho > \frac{1}{2}. \end{cases} \quad (37)$$

Proof. We start by applying (27) and Lemma 2.3 to deduce that

$$\begin{aligned} |u(x)| &\leq \left| \int_{\{|x-y| \leq \bar{c}_0\}} \frac{(x-y)^\perp}{|x-y|^2} G(|x-y|) \omega(y) dy \right| + \left| \int_{\{|x-y| \geq \bar{c}_0\}} \frac{(x-y)^\perp}{|x-y|^2} G(|x-y|) \omega(y) dy \right| \\ &\leq C \|\omega\|_{L^\infty} \int_0^{\bar{c}_0} m(\rho^{-1}) d\rho + C \int_{\{|x-y| \geq \bar{c}_0\}} |\omega(y)| dy \leq C \|\omega\|_{L^1 \cap L^\infty}. \end{aligned}$$

Here, $\bar{c}_0 > 0$ denotes the constant introduced in Lemma 2.3. Note that from (21), $m(\rho^{-1})$ behaves like $|\log \rho|^{\beta_1+}$ near zero. Hence, the integral $\int_0^{\bar{c}_0} m(\rho^{-1}) d\rho$ is finite. Consequently, $u \in L^\infty(\mathbb{R}^2)$.

Next, we will focus on the continuity property of u . For any $\tilde{x}, x \in \mathbb{R}^2$ such that $|x - \tilde{x}| < \frac{\bar{c}_0}{3}$ (without loss of generality assuming $\bar{c}_0 \leq 1$), we apply (27) and obtain

$$\begin{aligned} |u(x) - u(\tilde{x})| &\leq \int_{\mathbb{R}^2} |K(x-y) - K(\tilde{x}-y)| \cdot |\omega(y)| dy \\ &\leq \int_{\{|x-y| \leq 2|x-\tilde{x}|\}} |K(x-y) - K(\tilde{x}-y)| \cdot |\omega(y)| dy \\ &\quad + \int_{\{|x-y| \geq 2|x-\tilde{x}|\}} |K(x-y) - K(\tilde{x}-y)| \cdot |\omega(y)| dy \\ &\triangleq I + II. \end{aligned}$$

For I , we infer from (32) that

$$I \leq 2\|\omega\|_{L^\infty} \int_{\{|x-y| \leq 3|x-\tilde{x}|\}} \frac{G(|x-y|)}{|x-y|} dy = 4\pi\|\omega\|_{L^\infty} \int_0^{3|x-\tilde{x}|} G(\rho) d\rho \leq C\|\omega\|_{L^\infty} \int_0^{3|x-\tilde{x}|} m(\rho^{-1}) d\rho.$$

It follows from L'Hôpital's rule and (15) in (H2a) that

$$\lim_{\rho \rightarrow 0} \frac{\int_0^\rho m(\tilde{\rho}^{-1}) d\tilde{\rho}}{\rho m(\rho^{-1})} = \lim_{\rho \rightarrow 0} \frac{m(\rho^{-1})}{m(\rho^{-1}) - \rho^{-1} m'(\rho^{-1})} = \lim_{r \rightarrow +\infty} \frac{1}{1 - \frac{r m'(r)}{m(r)}} = 1. \quad (38)$$

Hence, there exists a constant $C > 0$, depending on $\bar{c}_0$, such that for any $\rho < c_0$,

$$\int_0^\rho m(\tilde{\rho}^{-1}) d\tilde{\rho} \leq C \rho m(\rho^{-1}). \quad (39)$$

Therefore, by (39), we deduce

$$I \leq C\|\omega\|_{L^\infty} |x - \tilde{x}| m(|x - \tilde{x}|^{-1}).$$

For II , we apply the mean value theorem and get

$$K(x - y) - K(\tilde{x} - y) = (x - \tilde{x}) \cdot \nabla K(\theta x + (1 - \theta)\tilde{x} - y),$$

for some $\theta \in [0, 1]$. Applying (35) with $l = 1$, we have

$$|\nabla K(x)| \leq C \frac{m(|x|^{-1}) + 1}{|x|^2}, \quad (40)$$

for every $x \neq 0$. Moreover, for $|x - y| \geq 2|x - \tilde{x}|$, we have

$$\frac{1}{2}|x - y| \leq |\theta x + (1 - \theta)\tilde{x} - y| \leq \frac{3}{2}|x - y|,$$

for any $\theta \in [0, 1]$. Therefore, we apply the above estimates and (17) to obtain

$$\begin{aligned} II &\leq C|x - \tilde{x}| \int_{\{|x-y| \geq 2|x-\tilde{x}|\}} \frac{m(|x-y|^{-1}) + 1}{|x-y|^2} |\omega(y)| dy \\ &\leq C\|\omega\|_{L^\infty} |x - \tilde{x}| \int_{\{2|x-\tilde{x}| \leq |x-y| \leq 2\}} \frac{m(|x-y|^{-1}) + 1}{|x-y|^2} dy + C|x - \tilde{x}| \int_{\mathbb{R}^2} |\omega(y)| dy \\ &\leq C\|\omega\|_{L^\infty} |x - \tilde{x}| \left(\int_{\frac{1}{2}}^{\frac{1}{2|x-\tilde{x}|}} \frac{m(r) + 1}{r} dr \right) + C\|\omega\|_{L^1} |x - \tilde{x}| \\ &\leq C\|\omega\|_{L^1 \cap L^\infty} |x - \tilde{x}| m(|x - \tilde{x}|^{-1}) \log |x - \tilde{x}|^{-1} \leq C\|\omega\|_{L^1 \cap L^\infty} \nu(|x - \tilde{x}|), \end{aligned}$$

where we have used the monotonicity of m in the penultimate inequality to get

$$m(r) + 1 \leq m\left(\frac{1}{2|x-\tilde{x}|}\right) + \frac{m\left(\frac{1}{2|x-\tilde{x}|}\right)}{m\left(\frac{1}{2}\right)} \leq \left(1 + \frac{1}{m\left(\frac{1}{2}\right)}\right) m(|x - \tilde{x}|^{-1}).$$

Gathering the above estimates, we conclude with the desired bound

$$|u(x) - u(\tilde{x})| \leq C\|\omega\|_{L^1 \cap L^\infty} \nu(|x - \tilde{x}|).$$

When $|x - \tilde{x}| \geq \frac{\bar{c}_0}{3}$, we have the following direct estimate

$$\frac{|u(x) - u(\tilde{x})|}{\nu(|x - \tilde{x}|)} \leq \frac{2\|u\|_{L^\infty}}{\inf_{\rho \geq \frac{\bar{c}_0}{3}} \nu(\rho)} \leq C\|\omega\|_{L^1 \cap L^\infty}, \quad (41)$$

where we have used the fact that ν is strictly positive away from zero. $\square$

Now we present the proof of Theorem 1.1.

Proof of Theorem 1.1. We first establish the estimates at the *a priori* level. These estimates also hold for the smooth kernel approximations introduced in Appendix C.1, with constants independent of the regularization parameter.

By the transport structure and the incompressibility of the regularized velocity, it follows that

$$\|\omega(t)\|_{L^1 \cap L^\infty} = \|\omega_0\|_{L^1 \cap L^\infty}.$$

Thanks to Lemma 3.1, the velocity field $u = \nabla^\perp(-\Delta)^{-1}m(\Lambda)\omega$ satisfies

$$\|u(t)\|_{L^\infty(\mathbb{R}^2)} + \sup_{|x-y|>0} \frac{|u(x,t) - u(y,t)|}{\nu(|x-y|)} \leq C\|\omega_0\|_{L^1 \cap L^\infty(\mathbb{R}^2)}. \quad (42)$$

The Osgood condition (Osg) translates into

$$\int_0^{\frac{1}{2}} \frac{d\rho}{\nu(\rho)} = \int_0^{\frac{1}{2}} \frac{d\rho}{\rho(\log \rho^{-1})m(\rho^{-1})} = \int_2^{+\infty} \frac{dr}{r(\log r)m(r)} = +\infty.$$

Therefore, the existence and uniqueness theorem for ordinary differential equations with an Osgood modulus of continuity [4, Theorem 3.2] gives a unique local solution to the flow equation (5), starting from every $x \in \mathbb{R}^2$ at every initial time $s \geq 0$. Moreover, estimate (42) gives, for every $\tau, s \geq 0$,

$$|\Phi_{\tau,s}(x) - x| \leq \int_{\min\{\tau,s\}}^{\max\{\tau,s\}} \|u(\cdot, \tilde{s})\|_{L^\infty} d\tilde{s} \leq C\|\omega_0\|_{L^1 \cap L^\infty} |\tau - s|.$$

Thus, no trajectory can escape to infinity in finite time, and the local solution extends globally in both time directions; see also [4, Proposition 3.11]. Consequently, $\Phi_{t,s}$ is well defined for every $t, s \geq 0$.

Next we verify the invertibility of flow map. For any $\tau, s, t \geq 0$, uniqueness of the trajectories implies the two-parameter flow property

$$\Phi_{t,\tau} \circ \Phi_{\tau,s} = \Phi_{t,s}.$$

Indeed, as functions of t , both sides solve the same ordinary differential equation and agree at $t = \tau$. It follows that $\Phi_{s,t} \circ \Phi_{t,s} = \Phi_{s,s} = \text{Id}$ and $\Phi_{t,s} \circ \Phi_{s,t} = \Phi_{t,t} = \text{Id}$. Therefore,

$$\Phi_{t,s}^{-1} = \Phi_{s,t}.$$

In particular, with the notation $\Phi_t = \Phi_{t,0}$ used in Theorem 1.1, we have $\Phi_t^{-1} = \Phi_{0,t}$.

According to (5) and (42), it follows that for every $t, s \in \mathbb{R}_+$,

$$\left| \frac{d(\Phi_{t,s}(x) - \Phi_{t,s}(y))}{dt} \right| \leq |u(\Phi_{t,s}(x), t) - u(\Phi_{t,s}(y), t)| \leq C\|\omega_0\|_{L^1 \cap L^\infty} \nu(|\Phi_{t,s}(x) - \Phi_{t,s}(y)|). \quad (43)$$

The function $H(r)$ defined in (7) is increasing, and from the Osgood condition (Osg),

$$\lim_{r \rightarrow +\infty} H(r) = \int_2^{+\infty} \frac{dr}{r(\log r)m(r)} = +\infty, \quad \text{and} \quad \lim_{r \rightarrow 0} H(r) = -\infty.$$

Hence, $H : (0, +\infty) \rightarrow \mathbb{R}$ is an invertible map. Moreover, H satisfies

$$\frac{d}{d\rho} H(\rho^{-1}) = -\frac{1}{\nu(\rho)}, \quad \forall \rho \in (0, +\infty).$$

Then, from (43) we get

$$\left| H(|\Phi_{t,s}(x) - \Phi_{t,s}(y)|^{-1}) - H(|x - y|^{-1}) \right| = \left| \int_{|x-y|}^{|\Phi_{t,s}(x) - \Phi_{t,s}(y)|} \frac{d\rho}{\nu(\rho)} \right| \leq C\|\omega_0\|_{L^1 \cap L^\infty} |t - s|,$$

and hence

$$|\Phi_{t,s}(x) - \Phi_{t,s}(y)|^{-1} \geq H^{-1} \left(H(|x - y|^{-1}) - C\|\omega_0\|_{L^1 \cap L^\infty} |t - s| \right),$$

$$|\Phi_{t,s}(x) - \Phi_{t,s}(y)|^{-1} \leq H^{-1} \left(H(|x - y|^{-1}) + C \|\omega_0\|_{L^1 \cap L^\infty} |t - s| \right),$$

which directly implies (6).

With the above *a priori* estimates at our disposal, the global existence result of Theorem 1.1 follows from the approximation argument in Appendix C.1.

Next, by adapting the arguments in [81, Sec. 2.3] with suitable modifications for the present setting (see also [92] and [4, Chap. 7] for related approaches based on Littlewood-Paley theory), we prove the uniqueness part of Theorem 1.1. Assume that (ω^1, u^1) and (ω^2, u^2) are two weak solutions on $[0, T]$ of the 2D Loglog-Euler type equation (1) associated with the same initial data $\omega_0 \in L^1 \cap L^\infty(\mathbb{R}^2)$. Let $\Phi_t^i = \Phi_{t,0}^i$ ($i = 1, 2$) denote the flow map generated by the velocity field u^i , satisfying (5). We first justify the Lagrangian representation of the weak solutions. For each $i = 1, 2$, the bound $\omega^i \in L^\infty([0, T]; L^1 \cap L^\infty)$ and Lemma 3.1 imply that u^i is bounded and has an Osgood modulus of continuity in space. Viewing

$$\mu_t^i \triangleq \omega^i(\cdot, t) dx$$

as a finite signed measure, the weak vorticity equation is precisely the continuity equation

$$\partial_t \mu_t^i + \operatorname{div}(u^i \mu_t^i) = 0, \quad \mu_t^i|_{t=0} = \omega_0 dx.$$

The uniqueness theorem for signed measure solutions of the continuity equation driven by an Osgood vector field [2] (see also [3]) therefore gives

$$\mu_t^i = (\Phi_{t,0}^i)_\# (\omega_0 dx).$$

Since u^i is divergence free, its flow map preserves the Lebesgue measure, and hence

$$\omega^i(x, t) = \omega_0(\Phi_{0,t}^i(x)) \quad \text{for a.e. } x \in \mathbb{R}^2.$$

Without loss of generality, we assume $\omega_0 \not\equiv 0$. Set

$$\delta(t) \triangleq \frac{1}{\|\omega_0\|_{L^1}} \int_{\mathbb{R}^2} |\Phi_t^1(x) - \Phi_t^2(x)| \cdot |\omega_0(x)| dx. \quad (44)$$

By (5), it follows that

$$\begin{aligned} \Phi_t^1(x) - \Phi_t^2(x) &= \int_0^t \left(u^1(\Phi_\tau^1(x), \tau) - u^2(\Phi_\tau^2(x), \tau) \right) d\tau \\ &= \int_0^t \left(u^1(\Phi_\tau^1(x), \tau) - u^1(\Phi_\tau^2(x), \tau) \right) d\tau + \int_0^t \left(u^1(\Phi_\tau^2(x), \tau) - u^2(\Phi_\tau^2(x), \tau) \right) d\tau. \end{aligned}$$

For the first term, in view of (42) and (37), there exists some constant $C > 0$ depending only on $\|\omega_0\|_{L^1 \cap L^\infty}$ such that

$$|u^1(\Phi_t^1(x), t) - u^1(\Phi_t^2(x), t)| \leq C\nu(|\Phi_t^1(x) - \Phi_t^2(x)|).$$

For the second term, by the change of variables and using the properties of flow map Φ_t^i , for all $t \in [0, T]$ with $T > 0$, we have

$$\begin{aligned} &\int_{\mathbb{R}^2} |u^1(\Phi_t^2(x), t) - u^2(\Phi_t^2(x), t)| \cdot |\omega_0(x)| dx \\ &= \int_{\mathbb{R}^2} \left| \text{p.v.} \int_{\mathbb{R}^2} K(\Phi_t^2(x) - z) (\omega^1(z, t) - \omega^2(z, t)) dz \right| \cdot |\omega_0(x)| dx \\ &= \int_{\mathbb{R}^2} \left| \int_{\mathbb{R}^2} (K(\Phi_t^2(x) - \Phi_t^1(y)) - K(\Phi_t^2(x) - \Phi_t^2(y))) \omega_0(y) dy \right| \cdot |\omega_0(x)| dx \\ &\leq \int_{\mathbb{R}^2} |\omega_0(y)| \left(\int_{\mathbb{R}^2} |K(\Phi_t^2(x) - \Phi_t^1(y)) - K(\Phi_t^2(x) - \Phi_t^2(y))| \cdot |\omega_0(x)| dx \right) dy \end{aligned}$$

$$\begin{aligned}
&= \int_{\mathbb{R}^2} |\omega_0(y)| \left(\int_{\mathbb{R}^2} |K(x - \Phi_t^1(y)) - K(x - \Phi_t^2(y))| \cdot |\omega^2(x, t)| dx \right) dy \\
&\leq C \|\omega_0\|_{L^1 \cap L^\infty} \int_{\mathbb{R}^2} \nu(|\Phi_t^1(y) - \Phi_t^2(y)|) \cdot |\omega_0(y)| dy,
\end{aligned}$$

where the last inequality follows from the same procedure used to estimate I and II in the proof of Lemma 3.1. Combining the above estimates, we infer that

$$\delta(t) \leq C \int_0^t \int_{\mathbb{R}^2} \nu(|\Phi_\tau^1(x) - \Phi_\tau^2(x)|) \cdot |\omega_0(x)| dx d\tau.$$

If we assume ν to be concave, then by Jensen's inequality we obtain

$$\int_{\mathbb{R}^2} \nu(|\Phi_t^1(x) - \Phi_t^2(x)|) \cdot \frac{|\omega_0(x)|}{\|\omega_0\|_{L^1}} dx \leq \nu \left(\int_{\mathbb{R}^2} |\Phi_t^1(x) - \Phi_t^2(x)| \cdot \frac{|\omega_0(x)|}{\|\omega_0\|_{L^1}} dx \right) = \nu(\delta(t)),$$

which yields

$$\delta(t) \leq C \int_0^t \nu(\delta(\tau)) d\tau. \quad (45)$$

Even if ν is not globally concave, inequality (45) still holds. To see this, we observe that ν is concave near the origin. Indeed,

$$\begin{aligned}
\nu''(0^+) &= \lim_{\rho \rightarrow 0^+} \left(-\rho^{-1} m(\rho^{-1}) + 2\rho^{-2} m'(\rho^{-1}) + \rho^{-3} \log \rho^{-1} m''(\rho^{-1}) \right) \\
&= \lim_{r \rightarrow +\infty} r m(r) \left(-1 + \frac{r m'(r)}{m(r)} \cdot \frac{r \log r m''(r) + 2m'(r)}{m'(r)} \right) = -\infty,
\end{aligned}$$

where we have applied (15)-(16) in (H2a). We then define

$$\nu_1(\rho) \triangleq \begin{cases} \nu(\rho) = \rho(\log \rho^{-1}) m(\rho^{-1}), & \text{if } \rho \in (0, c_1], \\ c_1(\log c_1^{-1}) m(c_1^{-1}) + \nu'(c_1)(\rho - c_1), & \text{if } \rho \in (c_1, +\infty), \end{cases}$$

where the constant $c_1 \in (0, \frac{1}{2})$ is chosen sufficiently small so that ν_1 is concave. Moreover,

$$\nu_1(\rho) \approx_{c_1} \nu(\rho), \quad \forall \rho \in (0, +\infty).$$

Repeating the above estimates with ν_1 in place of ν , we obtain

$$\delta(t) \leq C \int_0^t \int_{\mathbb{R}^2} \nu_1(|\Phi_\tau^1(x) - \Phi_\tau^2(x)|) \cdot |\omega_0(x)| dx d\tau \leq C \int_0^t \nu_1(\delta(\tau)) d\tau \leq C \int_0^t \nu(\delta(\tau)) d\tau.$$

Using the Osgood condition (Osg), we obtain $\delta(t) = 0$ for every $t \in [0, T]$. Hence,

$$|\Phi_t^1(a) - \Phi_t^2(a)| |\omega_0(a)| = 0 \quad \text{for a.e. } a \in \mathbb{R}^2.$$

Therefore, for every $\psi \in C_c(\mathbb{R}^2)$,

$$\int_{\mathbb{R}^2} \psi(x) \omega^1(x, t) dx = \int_{\mathbb{R}^2} \psi(\Phi_t^1(a)) \omega_0(a) da = \int_{\mathbb{R}^2} \psi(\Phi_t^2(a)) \omega_0(a) da = \int_{\mathbb{R}^2} \psi(x) \omega^2(x, t) dx.$$

Thus, $\omega^1(\cdot, t) = \omega^2(\cdot, t)$ almost everywhere. $\square$

4. GLOBAL REGULARITY OF A SINGLE $C^{1,\mu}$ VORTEX PATCH

In this section, we study the global propagation of boundary regularity for a single vortex patch of the 2D Loglog-Euler type equation (1), starting from a $C^{1,\mu}$ patch boundary.

4.1. Mathematical formulation for vortex patch problem. Let $D_0 \subset \mathbb{R}^2$ be a simply connected bounded domain with boundary $\partial D_0 \in C^{n,\mu}$, $n \in \mathbb{N}^*$, $\mu \in (0, 1)$, and let $\omega_0(x) = \mathbf{1}_{D_0}(x) \in L^1 \cap L^\infty$ be the initial data of the 2D Loglog-Euler type equation (1). According to Theorem 1.1, there exists a unique global-in-time patch solution of the form

$$\omega(x, t) = \mathbf{1}_{D(t)}(x), \quad D(t) = \Phi_t(D_0),$$

where the flow map $\Phi_t = \Phi_{t,0}$ is defined by (5). The vortex patch problem is therefore to study the propagation of regularity of $\partial D(t)$ for $t > 0$.

Let $z_0 : \mathbb{T} \rightarrow \partial D_0$ be a $C^{n,\mu}$ -parameterization of the boundary ∂D_0 , where $\mathbb{T}$ denotes the one-dimensional torus. Then

$$z(\xi, t) = \Phi_t(z_0(\xi))$$

defines a parameterization of $\partial D(t)$ and satisfies the following *contour dynamics equation*

$$\partial_t z(\xi, t) = u(z(\xi, t), t), \quad z(\xi, 0) = z_0(\xi). \quad (46)$$

where

$$\begin{aligned} u(z(\xi, t), t) &= - \int_{D(t)} \nabla_y^\perp (\tilde{R}(|z(\xi, t) - y|)) dy = \int_{\partial D(t)} \tilde{R}(|z(\xi, t) - y|) \mathbf{n}^\perp(y, t) dS(y) \\ &= \int_{\mathbb{T}} \tilde{R}(|z(\xi, t) - z(\eta, t)|) \partial_\eta z(\eta, t) d\eta, \end{aligned} \quad (47)$$

with $\tilde{R}(\rho) = \int_\rho^1 \frac{G(r)}{r} dr + C$ and $C \in \mathbb{R}$. This Lagrangian formulation is well suited for establishing local-in-time regularity of $\partial D(t)$. However, even for the 2D Euler equation, a direct proof of global-in-time propagation of boundary regularity from (46)-(47) remains open.

Following [7, 18], we introduce the level-set formulation of the vortex patch problem in Eulerian coordinates.

Definition 4.1 (Level-set characterization of a patch domain). Let D be a simply connected and bounded domain with $\partial D \in C^{k,\mu}$. Let $\varphi \in C^{k,\mu}(\mathbb{R}^2)$ with $k \in \mathbb{N}^*$ and $\mu \in (0, 1)$. We say that φ is a level-set characterization of the domain D if

$$D = \{x \in \mathbb{R}^2 \mid \varphi(x) > 0\}, \quad \inf_{x \in \partial D} |\nabla \varphi(x)| \geq c > 0. \quad (48)$$

A construction of such a level-set function can be found, for example, in [71]. By the definition above, we have

$$\varphi = 0 \quad \text{on } \partial D, \quad \varphi < 0 \quad \text{in } U \cap \overline{D}^c, \quad |\nabla \varphi| \geq c_U > 0 \quad \text{in } U, \quad (49)$$

where $U \subset \mathbb{R}^2$ is a small open neighborhood of ∂D .

Let φ_0 be a $C^{n,\mu}$ level-set characterization of D_0 in the sense of Definition 4.1. For any $x_0 \in \partial D_0$, the solution of the following ordinary differential equation

$$\frac{dz_0(\xi)}{d\xi} = \nabla^\perp \varphi_0(z_0(\xi)) \triangleq W_0(z_0(\xi)), \quad z_0(0) = x_0, \quad (50)$$

yields a $C^{n,\mu}$ -parameterization of the boundary curve ∂D_0 . Let $z(\xi, t)$ be a solution of (46) with z_0 given by (50) and $\varphi(x, t) = \varphi_0(\Phi_t^{-1}(x))$. Then $\varphi(x, t)$ solves

$$\partial_t \varphi + u \cdot \nabla \varphi = 0, \quad u = \nabla^\perp (-\Delta)^{-1} m(\Lambda)(\mathbf{1}_{D(t)}), \quad \varphi|_{t=0} = \varphi_0, \quad (51)$$

with $D(t) = \{x \in \mathbb{R}^2 : \varphi(x, t) > 0\}$. Furthermore, a direct computation shows

$$\partial_\xi z(\xi, t) = W(z(\xi, t), t), \quad \text{with } W(\cdot, t) \triangleq \nabla^\perp \varphi(\cdot, t). \quad (52)$$

Indeed, both $\partial_\xi z$ and $W(z(\xi, t), t)$ are tangent vectors to $\partial D(t)$ and $W \neq 0$ on the boundary. Hence, there exists a scalar function $g(\xi, t)$ such that

$$\partial_\xi z(\xi, t) = g(\xi, t)W(z(\xi, t), t).$$

Differentiating both sides along the flow and using the evolution equations for z and W gives $\partial_t g(\xi, t) = 0$. Since $g(\xi, 0) = 1$ by (50), we have $g(\xi, t) \equiv 1$, which proves (52).

Denoting the striated derivative by $\partial_W = W \cdot \nabla$, we iterate (52) to obtain, for $k \in \mathbb{N}^*$,

$$\partial_\xi^k z(\xi, t) = ((W \cdot \nabla)^{k-1} W)(z(\xi, t), t) = \partial_W^{k-1} W(z(\xi, t), t). \quad (53)$$

Therefore, the regularity of $\partial D(t)$ reduces to estimating the striated derivatives $\partial_W^{k-1} W(\cdot, t)$. Indeed, combining (46), Lemma 3.1 and (53), we have

$$\|z\|_{L^\infty} \leq \|z_0\|_{L^\infty} + \int_0^t \|u(\cdot, s)\|_{L^\infty} ds \leq \|z_0\|_{L^\infty} + C\|\omega_0\|_{L^1 \cap L^\infty} t, \quad (54)$$

and for $k \geq 1$,

$$\|\partial_\xi^k z(\cdot, t)\|_{L^\infty} \leq \|\partial_W^{k-1} W(\cdot, t)\|_{L^\infty}, \quad (55)$$

and

$$\|\partial_\xi^k z(\cdot, t)\|_{\dot{C}^\mu} \leq \|\partial_W^{k-1} W(\cdot, t)\|_{\dot{C}^\mu} \|\partial_\xi z(\cdot, t)\|_{L^\infty}^\mu = \|\partial_W^{k-1} W(\cdot, t)\|_{\dot{C}^\mu} \|W(\cdot, t)\|_{L^\infty}^\mu. \quad (56)$$

Using (51) and the commutator identity $[\partial_W, \partial_t + u \cdot \nabla] = 0$, we infer that for any $k \in \mathbb{N}^*$,

$$(\partial_t + u \cdot \nabla)(\partial_W^{k-1} W) = \partial_W^k u, \quad \partial_W^{k-1} W|_{t=0} = \partial_{W_0}^{k-1} W_0. \quad (57)$$

Hence, at the *a priori* level, the vortex patch problem is reduced to the study of (51) and (57). The justification is provided by the kernel-regularization argument in Appendix C.2.

Before proving the theorem, we introduce the following notation and function spaces:

$$\begin{aligned} |\nabla \varphi(\cdot)|_{\inf} &\triangleq \inf_{x \in \partial D} |\nabla \varphi(x)|, \\ \|\nabla \varphi(\cdot)\|_{\dot{C}^\mu(\mathbb{R}^2)} &\triangleq \sup_{x \neq \tilde{x}} \frac{|\nabla \varphi(x) - \nabla \varphi(\tilde{x})|}{|x - \tilde{x}|^\mu}, \quad \mu \in (0, 1), \\ \|\nabla \varphi(\cdot)\|_{C^\mu(\mathbb{R}^2)} &\triangleq \|\nabla \varphi(\cdot)\|_{L^\infty(\mathbb{R}^2)} + \|\nabla \varphi(\cdot)\|_{\dot{C}^\mu(\mathbb{R}^2)}, \quad \mu \in (0, 1), \end{aligned}$$

and

$$\Delta_\mu \triangleq \frac{\|\nabla \varphi\|_{\dot{C}^\mu(\mathbb{R}^2)}}{|\nabla \varphi|_{\inf}} + 1. \quad (58)$$

The m -adapted Hölder space $C_m^\mu(\mathbb{R}^2)$ ($\mu \in (0, 1]$) is composed of functions f such that

$$\|f\|_{C_m^\mu} = \|f\|_{C_m^\mu(\mathbb{R}^2)} \triangleq \|f\|_{L^\infty(\mathbb{R}^2)} + \sup_{|x-y|>0} \frac{|f(x) - f(y)|}{|x-y|^\mu (m(|x-y|^{-1}) + 1)}. \quad (59)$$

4.2. Near-Lipschitz estimate of u . The following lemma shows that the velocity field retains a near-Lipschitz modulus of continuity under the slowly growing multiplier m .

Lemma 4.1. *Suppose that $m(r)$ satisfies the assumptions (H1)-(H2a) and $D \subset \mathbb{R}^2$ is a bounded domain with $C^{1,\mu}$ ($0 < \mu < 1$)-boundary. Let φ be the level-set characterization of domain D where (48)-(49) hold. Then for any $\gamma \in (0, \mu]$, the velocity $u = \nabla^\perp(-\Delta)^{-1}m(\Lambda)(\mathbf{1}_D)$ satisfies*

$$\|u\|_{C_m^1(\mathbb{R}^2)} \leq C(1 + \log \Delta_\gamma), \quad (60)$$

where the constant $C > 0$ depends only on γ and the domain area $|D|$.

Compared with Lemma 3.1, the key point of Lemma 4.1 is that the cancellation structure of a regular patch improves the velocity estimate by one logarithmic order.

We emphasize that Lemma 4.1 is formulated in the physical space. Its frequency-space counterpart is the following estimate.

Lemma 4.2 ([44, p. 985]). *Let $S_j = \chi(2^{-j}\Lambda)$, $j \in \mathbb{N}$ be the low-frequency cut-off operator in the Littlewood-Paley theory. Under the assumptions of Lemma 4.1, and for every $\gamma \in (0, \mu]$, the following estimate holds:*

$$\|S_j \nabla u\|_{L^\infty(\mathbb{R}^2)} \leq Cm(2^j)(1 + \log \Delta_\gamma),$$

where $C > 0$ depends only on γ and $|D|$ (the area of domain D).

The equivalence between Lemma 4.1 and Lemma 4.2 can be guaranteed by [4, Sec. 2.11].

Proof of Lemma 4.1. Taking advantage of Lemma 3.1, we directly get $\|u\|_{L^\infty} \leq C$. It remains to control the modulus-of-continuity estimate of u in (59).

From the representation (27), for any $x, h \in \mathbb{R}^2$ such that $|h| \leq \frac{\bar{c}_0}{3}$ (recalling that $\bar{c}_0$ is the constant appearing in Lemma 2.3), we write

$$\begin{aligned} u(x) - u(x+h) &= \int_{D \cap \{|x-y| \leq 2|h|\}} K(x-y) dy - \int_{D \cap \{|x-y| \leq 2|h|\}} K(x+h-y) dy \\ &\quad + \int_{D \cap \{|x-y| \geq 2|h|\}} (K(x-y) - K(x+h-y)) dy \\ &\triangleq \mathfrak{N}_1 + \mathfrak{N}_2 + \mathfrak{N}_3. \end{aligned}$$

For $\mathfrak{N}_1$, arguing as the estimation for I in the proof of Lemma 3.1, we obtain

$$|\mathfrak{N}_1| \leq C \int_0^{2|h|} G(\rho) d\rho \leq C \int_0^{2|h|} m(\rho^{-1}) d\rho \leq C|h|m(|h|^{-1}).$$

Similarly, we bound $\mathfrak{N}_2$ by

$$|\mathfrak{N}_2| \leq C \int_0^{3|h|} |G(\rho)| d\rho \leq C|h|m(|h|^{-1}),$$

utilizing the fact $|x+h-y| \leq |x-y| + |h|$.

For $\mathfrak{N}_3$, by mean value theorem, it follows that

$$\begin{aligned} |\mathfrak{N}_3| &= \left| \int_0^1 \int_{D \cap \{|x-y| \geq 2|h|\}} h \cdot \nabla K(x + \theta h - y) dy d\theta \right| \\ &\leq \int_0^1 \left| \int_{D \cap \{|x+\theta h-y| \geq 2|h|\}} h \cdot \nabla K(x + \theta h - y) dy \right| d\theta \\ &\quad + \int_0^1 \int_{\{|x-y| \geq 2|h|\} \triangle \{|x+\theta h-y| \geq 2|h|\}} |h \cdot \nabla K(x + \theta h - y)| dy d\theta \\ &\triangleq \mathfrak{N}_{31} + \mathfrak{N}_{32}, \end{aligned}$$

where the notation $A \triangle B \triangleq (A \setminus B) \cup (B \setminus A)$. For $\mathfrak{N}_{32}$, observe that

$$\{|x-y| \geq 2|h|\} \triangle \{|x+\theta h-y| \geq 2|h|\} \subset \{|h| \leq |x+\theta h-y| \leq 3|h|\}.$$

In view of (40), we infer that

$$\mathfrak{N}_{32} \leq |h| \int_0^1 \int_{\{|h| \leq |x+\theta h-y| \leq 3|h|\}} |\nabla K(x + \theta h - y)| dy d\theta$$

$$\leq C \int_0^1 \int_{|h|}^{3|h|} \frac{m(\rho^{-1})}{\rho^2} \rho d\rho d\theta \leq C|h|m(|h|^{-1}).$$

Note that in the second inequality, we drop the second term in (40), since $|x + \theta h - y| \leq 3|h| \leq \bar{c}_0$.

For the remaining term $\mathfrak{N}_{31}$, we state the following claim:

$$\left| \int_{D \cap \{|x-y| \geq 2|h|\}} \nabla K(x-y) dy \right| \leq C(1 + \log \Delta_\gamma)(m(|h|^{-1}) + 1), \quad \forall \gamma \in (0, 1), \quad (61)$$

where $C > 0$ depends only on γ . Applying (61) and replacing x by $x + \theta h$, we obtain

$$\mathfrak{N}_{31} \leq C|h| \cdot (1 + \log \Delta_\gamma)(m(|h|^{-1}) + 1).$$

Collecting all the estimates, we obtain the desired bound (60). For the case $|h| > \frac{\bar{c}_0}{3}$, we argue analogously to (41) and derive the same bound.

We are left to show the claim (61). Applying (40) and the fact that $m(r)$ is non-decreasing, we obtain

$$\begin{aligned} \left| \int_{D \cap \{|x-y| \geq 2|h|\}} \nabla K(x-y) dy \right| &\leq C \int_{D \cap \{|x-y| \geq 2|h|\}} \frac{m(|x-y|^{-1}) + 1}{|x-y|^2} dy \\ &\leq Cm(|h|^{-1}) \int_{D \cap \{|x-y| \geq 2|h|\}} \frac{dy}{|x-y|^2} + Cm((\frac{\bar{c}_0}{3})^{-1}) \int_{D \cap \{|x-y| \geq 2|h|\}} \frac{dy}{|x-y|^2} \\ &\leq Cm(|h|^{-1}) \int_{2|h|}^L \frac{1}{\rho^2} \rho d\rho \leq Cm(|h|^{-1})(1 + \log |h|^{-1}), \end{aligned} \quad (62)$$

where $L \triangleq \sqrt{|D|/\pi}$ and in the third line we have used the rearrangement inequality.

If $2|h| \geq \delta_\gamma$, with

$$\delta_\gamma \triangleq \Delta_\gamma^{-\frac{1}{\gamma}} = \left(\frac{\|\nabla \varphi\|_{\dot{C}^\gamma}}{|\nabla \varphi|_{\inf}} + 1 \right)^{-\frac{1}{\gamma}},$$

then (61) is a direct consequence of (62).

If $2|h| < \delta_\gamma$, we exploit the symmetry properties of K and the patch structure to establish cancellations, analogous to [7, Geometric lemma]. Denote d_x is the distance between x and ∂D , namely

$$d_x \triangleq \inf_{z \in \partial D} |x - z|. \quad (63)$$

For every $x \in \mathbb{R}^2$, choosing $\tilde{x} \in \partial D$ such that $d_x = |x - \tilde{x}|$, we define the sets

$$\mathcal{S}_\rho(x) \triangleq \{z \mid |z| = 1, x + \rho z \in D\}, \quad \Sigma(x) \triangleq \{z \mid |z| = 1, \nabla_x \varphi(\tilde{x}) \cdot z \geq 0\}, \quad \mathcal{R}_\rho(x) \triangleq \mathcal{S}_\rho(x) \triangle \Sigma(x). \quad (64)$$

Note that

$$D \cap \{|x - y| \geq 2h\} = \{y = x + \rho z : \rho \geq 2h, z \in \mathcal{S}_\rho(x)\},$$

and $\Sigma(x)$ is a semicircle where symmetry can be employed. The geometric lemma in [7] characterizes their difference $\mathcal{R}_\rho(x)$: for all $\rho \geq d_x$, $\gamma \in (0, 1)$ and for each x such that $d_x \leq \delta_\gamma = \Delta_\gamma^{-1/\gamma}$,

$$\mathcal{H}^1(\mathcal{R}_\rho(x)) \leq 2\pi \left((1 + 2^\gamma) \frac{d_x}{\rho} + 2^\gamma \left(\frac{\rho}{\delta_\gamma} \right)^\gamma \right), \quad (65)$$

where $\mathcal{H}^1$ is the Hausdorff measure on the unit circle.

From (31), we decompose ∇K into the symmetric part $(\nabla K)_S$ and the antisymmetric part $(\nabla K)_A$. For the symmetric part, we observe that $\sigma(z)$ has zero mean in unit circle or semicircle:

$$\begin{aligned} \rho < d_x : \quad & \int_{\mathcal{S}_\rho(x)} \sigma(z) dz = 0, \\ \rho \geq d_x : \quad & \left| \int_{\mathcal{S}_\rho(x)} \sigma(z) dz \right| \leq \left| \int_{\Sigma(x)} \sigma(z) dz \right| + \int_{\mathcal{R}_\rho(x)} |\sigma(z)| dz \leq C \mathcal{H}^1(\mathcal{R}_\rho(x)). \end{aligned}$$

Utilizing the above cancellations, and applying (40) and (65), we get

$$\begin{aligned} & \left| \int_{D \cap \{|x-y| \geq 2|h|\}} (\nabla K)_S(x-y) dy \right| = \left| \int_{2|h|}^{\delta_\gamma} \frac{1}{2} \frac{2G(\rho) - \rho G'(\rho)}{\rho^2} \int_{\mathcal{S}_\rho(x)} \sigma(z) dz \rho d\rho \right| \\ & \leq C \int_{\max\{2|h|, d_x\}}^{\delta_\gamma} \frac{m(\rho^{-1}) + 1}{\rho} \mathcal{H}^1(\mathcal{R}_\rho(x)) d\rho \leq C(m(|h|^{-1}) + 1) \int_{d_x}^{\delta_\gamma} \frac{1}{\rho} \left(\frac{d_x}{\rho} + \left(\frac{\rho}{\delta_\gamma} \right)^\gamma \right) d\rho \\ & \leq C(m(|h|^{-1}) + 1). \end{aligned} \tag{66}$$

For the antisymmetric part, we have

$$\int_{D \cap \{|x-y| \geq 2|h|\}} (\nabla K)_A(x-y) dy = \int_{2|h|}^{\delta_\gamma} \frac{G'(\rho)}{2\rho} \mathcal{H}^1(\mathcal{S}_\rho(x)) \rho d\rho \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

For $\mathcal{H}^1(\mathcal{S}_\rho(x))$, we have

$$\begin{aligned} \rho < d_x : \quad & \mathcal{H}^1(\mathcal{S}_\rho(x)) = 2\pi, \\ \rho \geq d_x : \quad & \mathcal{H}^1(\mathcal{S}_\rho(x)) \leq \mathcal{H}^1(\Sigma(x)) + \mathcal{H}^1(\mathcal{R}_\rho(x)) = \pi + \mathcal{H}^1(\mathcal{R}_\rho(x)). \end{aligned}$$

Therefore, by using (32), when $2|h| \geq d_x$, we deduce that

$$\begin{aligned} & \left| \int_{2|h|}^{\delta_\gamma} \frac{G'(\rho)}{2\rho} \mathcal{H}^1(\mathcal{S}_\rho(x)) \rho d\rho \right| \leq \frac{\pi}{2} \left| \int_{2|h|}^{\delta_\gamma} G'(\rho) d\rho \right| + \frac{1}{2} \int_{2|h|}^{\delta_\gamma} |G'(\rho)| \mathcal{H}^1(\mathcal{R}_\rho(x)) d\rho \\ & \leq \frac{\pi}{2} |G(2|h|) - G(\delta_\gamma)| + C \int_{2|h|}^{\delta_\gamma} \frac{m(\rho^{-1}) + 1}{\rho} \mathcal{H}^1(\mathcal{R}_\rho(x)) d\rho \leq C(m(|h|^{-1}) + 1), \end{aligned}$$

and when $2|h| < d_x$, we get

$$\begin{aligned} & \left| \int_{2|h|}^{\delta_\gamma} \frac{G'(\rho)}{2\rho} \mathcal{H}^1(\mathcal{S}_\rho(x)) \rho d\rho \right| \leq \left| \pi \int_{2|h|}^{d_x} G'(\rho) d\rho + \frac{\pi}{2} \int_{d_x}^{\delta_\gamma} G'(\rho) d\rho \right| + \frac{1}{2} \int_{d_x}^{\delta_\gamma} |G'(\rho)| \mathcal{H}^1(\mathcal{R}_\rho(x)) d\rho \\ & \leq \frac{\pi}{2} |2G(2|h|) - G(d_x) - G(\delta_\gamma)| + C(m(d_x^{-1}) + 1) \leq C(m(|h|^{-1}) + 1). \end{aligned}$$

This concludes the proof of (61). $\square$

The following proposition analyzes the velocity induced by a circular patch. In particular, it shows that the dependence on m in the modulus-of-continuity estimate of Lemma 4.1 is sharp. The proof is deferred to Appendix B.

Proposition 4.3 (Boundary asymptotics for a circular patch). *Assume that $m(r)$ satisfies (H1)-(H2a), and let $B = B(0, 1) \subset \mathbb{R}^2$ and $u(x) = \nabla^\perp(-\Delta)^{-1} m(\Lambda) \mathbf{1}_B(x)$.*

(i) *Writing*

$$g(x) = g_0(|x|) = m(\Lambda) \mathbf{1}_B(x),$$

we have

$$g_0(r) = \left(\frac{1}{2} + o(1) \right) \operatorname{sgn}(1-r) m(|1-r|^{-1}), \quad \text{as } r \rightarrow 1, \quad r \neq 1.$$

In particular, $m(\Lambda)\mathbf{1}_B \notin L^\infty(\mathbb{R}^2)$.

- (ii) The velocity u is not Lipschitz continuous in any neighborhood of ∂B . More precisely, for $\nu \in \mathbb{S}^1$,

$$|u((1 \pm h)\nu) - u(\nu)| = \left(\frac{1}{2} + o(1)\right) h m(h^{-1}), \quad \text{as } h \downarrow 0. \quad (67)$$

In addition, u is purely tangential to ∂B and hence

$$u \cdot \mathbf{n} = 0 \quad \text{on } \partial B,$$

with $\mathbf{n}$ the outward unit normal vector of ∂B .

- (iii) The singularity in the second derivatives is anisotropic. Let $\nu \in \mathbb{S}^1$ and set $\tau_\nu \triangleq \nu^\perp$. We have, as $h \downarrow 0$,

$$\nabla^2 u((1 \pm h)\nu) = \left(\frac{1}{2} + o(1)\right) h^{-2} m'(h^{-1}) \tau_\nu \otimes \nu \otimes \nu + O(1 + m(h^{-1})), \quad (68)$$

where the remainder is understood in the norm of vector-valued bilinear forms and is uniform in ν . In particular,

$$|\nabla^2 u((1 \pm h)\nu)| = \left(\frac{1}{2} + o(1)\right) h^{-2} m'(h^{-1}), \quad \text{as } h \downarrow 0.$$

Moreover, for all fixed $\vartheta \in [0, 1)$,

$$\frac{|\nabla^2 u((1 \pm h)\nu)|}{h^{-\vartheta}(1 + m(h^{-1}))} \rightarrow \infty, \quad \text{as } h \downarrow 0.$$

4.3. The bound of $\partial_W u$. Inspired by [7, Proposition 2], we first derive a key integral representation for $\partial_W u$, which fully exploits the cancellation structure of a regular patch.

Lemma 4.4. Let $D \subset \mathbb{R}^2$ be a bounded domain with $C^{1,\mu}$ ($0 < \mu < 1$) boundary, and let φ be a level-set characterization of domain D . Let $W = \nabla^\perp \varphi \in C^\mu(\mathbb{R}^2)$ be the tangential vector field associated with φ . Assume that $u = \nabla^\perp (-\Delta)^{-1} m(\Lambda)(\mathbf{1}_D)$ and m satisfies (H1)-(H2a), then, for every $x \in \mathbb{R}^2$,

$$\partial_W u(x) = W \cdot \nabla u(x) = \text{p.v.} \int_D (W(x) - W(y)) \cdot \nabla K(x - y) dy, \quad (69)$$

where $\nabla K(x)$ is defined in (31).

Proof. Since $W = \nabla^\perp \varphi$ is divergence free, we have

$$\partial_W u_i = \text{div}(W u_i), \quad i = 1, 2.$$

In the following, we will compute $\text{div}(W u_i)$ in the distributional sense. For every $\tilde{\chi} \in C_c^\infty(\mathbb{R}^2)$, we have

$$(\text{div}(W u_i), \tilde{\chi}) = -(W u_i, \nabla \tilde{\chi}) = - \int_{\mathbb{R}^2} u_i(x) W(x) \cdot \nabla \tilde{\chi}(x) dx.$$

In view of the expression formula (27) and Fubini's theorem, it follows that

$$\begin{aligned} (\text{div}(W u_i), \tilde{\chi}) &= - \int_{\mathbb{R}^2} \left(\int_D K_i(x - y) dy \right) W(x) \cdot \nabla \tilde{\chi}(x) dx \\ &= - \int_D \int_{\mathbb{R}^2} K_i(x - y) W(x) \cdot \nabla \tilde{\chi}(x) dx dy \\ &= - \lim_{\varepsilon \rightarrow 0} \int_D \int_{|x-y| \geq \varepsilon} K_i(x - y) W(x) \cdot \nabla \tilde{\chi}(x) dx dy. \end{aligned}$$

Through integration by parts, we find that

$$\int_D \int_{|x-y| \geq \varepsilon} K_i(x-y) W(x) \cdot \nabla \tilde{\chi}(x) dx dy = - \int_D \int_{|x-y| \geq \varepsilon} W(x) \cdot \nabla_x K_i(x-y) \tilde{\chi}(x) dx dy - I_\varepsilon,$$

with

$$I_\varepsilon \triangleq \int_D \int_{|x-y|=\varepsilon} K_i(x-y) W(x) \cdot \frac{(x-y)}{|x-y|} \tilde{\chi}(x) dS(x) dy.$$

Note that $\lim_{\varepsilon \rightarrow 0} I_\varepsilon$ might be infinite. Define

$$J_\varepsilon \triangleq \int_D \int_{|x-y|=\varepsilon} K_i(x-y) W(y) \cdot \frac{(x-y)}{|x-y|} \tilde{\chi}(x) dS(x) dy,$$

and we can take the limit of their difference $R_\varepsilon \triangleq I_\varepsilon - J_\varepsilon$. Indeed, we apply (32) and (21) to obtain

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0} |R_\varepsilon| &= \lim_{\varepsilon \rightarrow 0} \left| \int_D \int_{|x-y|=\varepsilon} K_i(x-y) (W(x) - W(y)) \cdot \frac{(x-y)}{|x-y|} \tilde{\chi}(x) dS(x) dy \right| \\ &\leq \lim_{\varepsilon \rightarrow 0} \int_D \left| \frac{G(\varepsilon)}{\varepsilon} \cdot \|W\|_{\dot{C}^\mu} \varepsilon^\mu \cdot 1 \cdot \|\tilde{\chi}\|_{L^\infty} \cdot 2\pi\varepsilon \right| dy \leq C \lim_{\varepsilon \rightarrow 0} \varepsilon^\mu m(\varepsilon^{-1}) = 0. \end{aligned}$$

For J_ε , using Fubini's theorem and integration by parts, we infer that

$$\begin{aligned} J_\varepsilon &= \int_{\mathbb{R}^2} \int_{D \cap \{|x-y|=\varepsilon\}} K_i(x-y) W(y) \cdot \frac{(x-y)}{|x-y|} dS(y) \tilde{\chi}(x) dx \\ &= \int_{\mathbb{R}^2} \left(\int_{D \cap \{|x-y| \geq \varepsilon\}} W(y) \cdot \nabla_y (K_i(x-y)) dy - \int_{\partial D \cap \{|x-y| \geq \varepsilon\}} K_i(x-y) W(y) \cdot \mathbf{n}(y) dS(y) \right) \tilde{\chi}(x) dx \\ &= \int_{\mathbb{R}^2} \int_{D \cap \{|x-y| \geq \varepsilon\}} W(y) \cdot \nabla_y (K_i(x-y)) dy \tilde{\chi}(x) dx \\ &= - \int_{\mathbb{R}^2} \int_{D \cap \{|x-y| \geq \varepsilon\}} W(y) \cdot \nabla_x K_i(x-y) dy \tilde{\chi}(x) dx, \end{aligned} \tag{70}$$

where in the second equality, we use the fact that $W(y) \cdot \mathbf{n}(y) = 0$ for $y \in \partial D$ with $\mathbf{n}$ the outward normal vector of ∂D . Therefore, based on the above estimates,

$$\begin{aligned} (\operatorname{div}(W u_i), \tilde{\chi}) &= \lim_{\varepsilon \rightarrow 0} \left(\int_{\mathbb{R}^2} \int_{D \cap \{|x-y| \geq \varepsilon\}} (W(x) - W(y)) \cdot \nabla_x K_i(x-y) dy \tilde{\chi}(x) dx + R_\varepsilon \right) \\ &= \int_{\mathbb{R}^2} \left(\text{p.v.} \int_D (W(x) - W(y)) \cdot \nabla_x K_i(x-y) dy \right) \tilde{\chi}(x) dx. \end{aligned}$$

This finishes the proof of (69). $\square$

Using Lemma 4.4, we establish the following Hölder-type estimate for $\partial_W u$.

Lemma 4.5. *Under the assumptions of Lemma 4.4, one has*

$$\|\partial_W u\|_{C_m^\mu(\mathbb{R}^2)} \leq C(1 + \log \Delta_\gamma) \|W\|_{\dot{C}^\mu(\mathbb{R}^2)}, \tag{71}$$

where $C > 0$ depends only on μ , γ and $|D|$.

Proof. By virtue of the expression (69), (40), the rearrangement inequality, and (21), we deduce

$$\begin{aligned} \|\partial_W u\|_{L^\infty} &\leq \int_D |W(x) - W(y)| \cdot |\nabla K(x-y)| dy \\ &\leq C \|W\|_{\dot{C}^\mu(\mathbb{R}^2)} \int_D |x-y|^\mu \frac{m(|x-y|^{-1}) + 1}{|x-y|^2} dy \end{aligned}$$

$$\leq C\|W\|_{\dot{C}^\mu(\mathbb{R}^2)} \int_0^L \frac{m(\rho^{-1}) + 1}{\rho^{1-\mu}} d\rho \leq C\|W\|_{\dot{C}^\mu(\mathbb{R}^2)},$$

with $L = \sqrt{|D|/\pi}$. Choosing $x, h \in \mathbb{R}^2$ such that $|h| \leq \frac{\bar{c}_0}{3}$, we write

$$\partial_W u(x) - \partial_W u(x+h) = \mathfrak{J}_1 + \mathfrak{J}_2 + \mathfrak{J}_3 + \mathfrak{J}_4,$$

where

$$\begin{aligned} \mathfrak{J}_1 &\triangleq \text{p.v.} \int_{D \cap \{|x-y| < 2|h|\}} (W(x) - W(y)) \cdot \nabla K(x-y) dy, \\ \mathfrak{J}_2 &\triangleq -\text{p.v.} \int_{D \cap \{|x-y| < 2|h|\}} (W(x+h) - W(y)) \cdot \nabla K(x+h-y) dy, \\ \mathfrak{J}_3 &\triangleq \int_{D \cap \{|x-y| \geq 2|h|\}} (W(x) - W(x+h)) \cdot \nabla K(x-y) dy, \\ \mathfrak{J}_4 &\triangleq \int_{D \cap \{|x-y| \geq 2|h|\}} (W(x+h) - W(y)) \cdot (\nabla K(x-y) - \nabla K(x+h-y)) dy. \end{aligned}$$

For $\mathfrak{J}_1$, from (40), we get

$$|\mathfrak{J}_1| \leq C\|W\|_{\dot{C}^\mu(\mathbb{R}^2)} \int_{|x-y| \leq 2|h|} |x-y|^\mu \frac{m(|x-y|^{-1})}{|x-y|^2} dy \leq C\|W\|_{\dot{C}^\mu(\mathbb{R}^2)} \int_0^{2|h|} \frac{m(\rho^{-1})}{\rho^{1-\mu}} d\rho.$$

Using (15) in (H2a) and the L'Hôpital rule gives

$$\lim_{\rho \rightarrow 0} \frac{\int_0^\rho \tilde{\rho}^{\mu-1} m(\tilde{\rho}^{-1}) d\tilde{\rho}}{\rho^\mu m(\rho^{-1})} = \lim_{\rho \rightarrow 0} \frac{\rho^{\mu-1} m(\rho^{-1})}{\mu \rho^{\mu-1} m(\rho^{-1}) - \rho^{\mu-1} m'(\rho^{-1}) \rho^{-1}} = \lim_{r \rightarrow +\infty} \frac{1}{\mu - \frac{rm'(r)}{m(r)}} = \frac{1}{\mu}.$$

Then, there exists a constant $C > 0$ depending on μ and $\bar{c}_0$, such that for any $\rho < \bar{c}_0$,

$$\int_0^\rho \tilde{\rho}^{\mu-1} m(\tilde{\rho}^{-1}) d\tilde{\rho} \leq C \rho^\mu m(\rho^{-1}). \quad (72)$$

Together with (17), we find

$$|\mathfrak{J}_1| \leq C\|W\|_{\dot{C}^\mu(\mathbb{R}^2)} |h|^\mu m(|h|^{-1}).$$

Performing the same procedure, we can also show that

$$|\mathfrak{J}_2| \leq C\|W\|_{\dot{C}^\mu(\mathbb{R}^2)} \int_0^{3|h|} \frac{m(\rho^{-1})}{\rho^{1-\mu}} d\rho \leq C\|W\|_{\dot{C}^\mu(\mathbb{R}^2)} |h|^\mu m(|h|^{-1}).$$

For the term $\mathfrak{J}_3$, it follows from (61) that

$$\begin{aligned} |\mathfrak{J}_3| &\leq \|W\|_{\dot{C}^\mu(\mathbb{R}^2)} |h|^\mu \left| \text{p.v.} \int_{D \cap \{|x-y| \geq 2|h|\}} \nabla K(x-y) dy \right| \\ &\leq C(1 + \log \Delta_\gamma) \|W\|_{\dot{C}^\mu(\mathbb{R}^2)} |h|^\mu (m(|h|^{-1}) + 1). \end{aligned}$$

Finally, for the term $\mathfrak{J}_4$, we apply (35) with $l = 2$ and get

$$|\nabla^2 K(x)| \leq C \frac{m(|x|^{-1}) + 1}{|x|^3}.$$

Utilizing the mean value theorem and proceeding as in the treatment of II in Lemma 3.1, we obtain

$$\begin{aligned} |\mathfrak{J}_4| &\leq C\|W\|_{\dot{C}^\mu(\mathbb{R}^2)} \int_{D \cap \{|x-y| \geq 2|h|\}} |x-y|^\mu \frac{m(|x-y|^{-1}) + 1}{|x-y|^3} |h| dy \\ &\leq C\|W\|_{\dot{C}^\mu(\mathbb{R}^2)} |h| \int_{2|h|}^L \frac{m(\rho^{-1}) + 1}{\rho^{3-\mu}} \rho d\rho \leq C\|W\|_{\dot{C}^\mu(\mathbb{R}^2)} |h|^\mu (m(|h|^{-1}) + 1). \end{aligned}$$

Gathering the above estimates leads to the desired result (71). $\square$

Remark 5 (Regularity of the normal velocity). The purpose of this remark is to clarify why the estimate for $\partial_W u$ does not, by itself, imply higher regularity of the intrinsic normal velocity.

Let U be a neighborhood of ∂D on which $|W| \geq c_U > 0$, and set

$$\mathbf{w} = \frac{W}{|W|}, \quad \mathbf{n} = \mathbf{w}^\perp = \frac{W^\perp}{|W|}.$$

At every point where ∇u is defined, since $\text{tr}(\nabla u) = \text{div } u = 0$, a direct computation using the trace-free property of 2×2 matrices yields

$$|(\nabla u)^T \mathbf{n}| = |(\nabla u) \mathbf{w}| = |\partial_{\mathbf{w}} u|, \quad \text{with} \quad \nabla u = \begin{pmatrix} \partial_{x_1} u_1 & \partial_{x_2} u_1 \\ \partial_{x_1} u_2 & \partial_{x_2} u_2 \end{pmatrix}. \quad (73)$$

Thus, the normal projection of the velocity gradient is controlled by the tangential derivative estimate. The only component of ∇u that is not determined by $\partial_W u$ is

$$\mathbf{w} \cdot (\nabla u) \mathbf{n} = (\partial_{\mathbf{n}} u) \cdot \mathbf{w}.$$

This is precisely the component that becomes unbounded in the disc example of Proposition 4.3 (see (138) below). To examine the normal velocity $u \cdot \mathbf{n}$, we have formally

$$\nabla(u \cdot \mathbf{n}) = (\nabla u)^T \mathbf{n} + (\nabla \mathbf{n})^T u.$$

At the basic $C^{1,\mu}$ level of Theorem 1.2, however, the fields W and $\mathbf{n}$ are only C^μ -Hölder continuous in general. Hence, although (73) controls $(\nabla u)^T \mathbf{n}$, it does not imply Lipschitz regularity of the intrinsic normal velocity $u \cdot \mathbf{n}$. Indeed, differentiating this quantity introduces the additional term $(\nabla \mathbf{n})^T u$, which reflects the variation of the moving normal frame. This additional obstruction is geometric: even the projection of a constant vector field onto a merely C^μ unit normal field does not need to be Lipschitz.

Consequently, in studying the regularity of the normal velocity, the estimate for $\partial_W u$ isolates the potentially singular contribution arising from ∇u , while a further loss of regularity may arise from the variation of the moving normal frame. Whether these two contributions cancel is a genuinely nonlinear question. The disc patch provides an extreme example of complete cancellation: although $(\partial_{\mathbf{n}} u) \cdot \mathbf{w}$ becomes unbounded near the boundary, the normal velocity satisfies $u \cdot \mathbf{n} \equiv 0$ on ∂B .

Since the normal component of the velocity determines the evolution of the patch boundary, it would be interesting to construct an explicit single-patch example for which the normal velocity $u \cdot \mathbf{n}$, as a function on the patch boundary, fails to be Lipschitz continuous.

Next, we derive the following refined pointwise estimate for $\partial_W u = W \cdot \nabla u$ in the tangential direction $W = \nabla^\perp \varphi$, which is essentially from [44] but with more technical details here.

Lemma 4.6. *Under the assumptions of Lemma 4.4, we have the following pointwise estimate*

$$|\nabla u \mathbf{w} \cdot \mathbf{w}| \leq C(m(\Delta_\gamma) + 1)(1 + \log \Delta_\gamma), \quad \forall \gamma \in (0, \mu], \quad (74)$$

where $\mathbf{w} \triangleq \frac{W}{|W|} = \frac{\nabla^\perp \varphi}{|\nabla^\perp \varphi|}$, and $C > 0$ depends only on γ and $|D|$.

Proof. Since $\nabla u \mathbf{w} \cdot \mathbf{w} = \mathbf{S}(\nabla u) \mathbf{w} \cdot \mathbf{w}$ with $\mathbf{S}(\nabla u) = \frac{\nabla u + (\nabla u)^T}{2}$ being the symmetric part of ∇u , we will concentrate on the analysis of $\mathbf{S}(\nabla u) \mathbf{w} \cdot \mathbf{w}$. We recall from (29) that

$$\mathbf{S}(\nabla u)(x) = \int_D (\nabla K)_S(x - y) dy = \int_D \frac{1}{2} \frac{2G(|x - y|) - |x - y|G'(|x - y|)}{|x - y|^2} \sigma(x - y) dy,$$

where σ is given by (30). For every $x \in \mathbb{R}^2$, we divide $\mathbf{S}(\nabla u)(x)$ into

$$\begin{aligned} \mathbf{S}(\nabla u)(x) &= \int_{D \cap \{|x-y| \geq \delta_\gamma\}} (\nabla K)_S(x-y) dy + \int_{D \cap \{|x-y| \leq \delta_\gamma\}} (\nabla K)_S(x-y) dy \\ &\triangleq \mathcal{I}_1(x) + \mathcal{I}_2(x), \end{aligned}$$

where $\delta_\gamma \triangleq (\Delta_\gamma)^{-\frac{1}{\gamma}}$.

For $\mathcal{I}_1$, we proceed as (62) and obtain

$$|\mathcal{I}_1(x)| \leq C \int_{D \cap \{|x-y| \geq \delta_\gamma\}} \frac{m(|x-y|^{-1}) + 1}{|x-y|^2} dy \leq Cm(\delta_\gamma^{-1})(1 + \log \delta_\gamma^{-1}),$$

and consequently,

$$|\mathcal{I}_1(x) \mathbf{w}(x) \cdot \mathbf{w}(x)| \leq Cm(\Delta_\gamma)(1 + \log \Delta_\gamma).$$

For $\mathcal{I}_2$, the goal is to exploit additional cancellations in order to sharpen the result of Lemma 4.1. Denote $\tilde{x} \in \partial D$ such that $|x - \tilde{x}| = d_x$, where d_x is defined in (63). If $\delta_\gamma \leq d_x$, then by the mean-zero property of $\sigma(z)$, we have $\mathcal{I}_2(x) = 0$. Thus, we assume that $\delta_\gamma > d_x$ without loss of generality, and

$$\mathcal{I}_2(x) = \int_{D \cap \{d_x \leq |x-y| \leq \delta_\gamma\}} (\nabla K)_S(x-y) dy.$$

Since φ is a level-set characterization of D , we know that $\tilde{x} - x$ is parallel to $\nabla \varphi(\tilde{x})$, and hence orthogonal to $\mathbf{w}(\tilde{x})$. Therefore, we shall seek cancellations in $\mathcal{I}_2(x) \mathbf{w}(\tilde{x}) \cdot \mathbf{w}(\tilde{x})$. Decompose

$$\mathcal{I}_2(x) \mathbf{w}(x) \cdot \mathbf{w}(x) = \mathcal{I}_2(x) \mathbf{w}(\tilde{x}) \cdot \mathbf{w}(\tilde{x}) + \mathcal{I}_2(x) (\mathbf{w}(x) + \mathbf{w}(\tilde{x})) \cdot (\mathbf{w}(x) - \mathbf{w}(\tilde{x})).$$

Let us start with the estimate of the difference

$$|\mathcal{I}_2(x) (\mathbf{w}(x) + \mathbf{w}(\tilde{x})) \cdot (\mathbf{w}(x) - \mathbf{w}(\tilde{x}))| \leq 2|\mathcal{I}_2(x)| \cdot |\mathbf{w}(x) - \mathbf{w}(\tilde{x})|.$$

By virtue of the definitions of $\mathbf{w}(x)$, $\tilde{x}$ and d_x , we have

$$|\mathbf{w}(x) - \mathbf{w}(\tilde{x})| = \frac{|(W(x) - W(\tilde{x})) - \mathbf{w}(\tilde{x})(|W(x)| - |W(\tilde{x})|)|}{|W(x)|} \leq \frac{2\|\nabla \varphi\|_{\dot{C}^\gamma} |x - \tilde{x}|^\gamma}{|\nabla \varphi|_{\inf}} \leq 2\Delta_\gamma (d_x)^\gamma.$$

A similar argument as in (66) yields

$$|\mathcal{I}_2(x)| \leq C \int_{d_x}^{\delta_\gamma} \frac{m(\rho^{-1}) + 1}{\rho} \mathcal{H}^1(\mathcal{R}_\rho(x)) d\rho \leq C(m(d_x^{-1}) + 1).$$

Hence,

$$\begin{aligned} |\mathcal{I}_2(x) (\mathbf{w}(x) + \mathbf{w}(\tilde{x})) \cdot (\mathbf{w}(x) - \mathbf{w}(\tilde{x}))| &\leq C\Delta_\gamma (d_x)^\gamma (m(d_x^{-1}) + 1) \leq C\Delta_\gamma (\delta_\gamma)^\gamma (m(\delta_\gamma^{-1}) + 1) \\ &\leq C(m(\Delta_\gamma^{\frac{1}{\gamma}}) + 1) \leq C(m(\Delta_\gamma) + 1). \end{aligned}$$

where we have used (26) in the second inequality, and (24) in the last inequality.

Next, we focus on the term $\mathcal{I}_2(x) \mathbf{w}(\tilde{x}) \cdot \mathbf{w}(\tilde{x})$. A main observation is that $(\nabla K)_S(x - \cdot) \mathbf{w}(\tilde{x}) \cdot \mathbf{w}(\tilde{x})$ is an odd function with respect to the line

$$\mathbf{L} : s \mapsto x + s\mathbf{w}(\tilde{x})^\perp,$$

which go across x and $\tilde{x}$. To see this, we denote $\bar{y}$ the reflection point of y with respect to $\mathbf{L}$. Then, if we represent $y = x + s_1 \mathbf{w}(\tilde{x})^\perp + s_2 \mathbf{w}(\tilde{x})$, we have $\bar{y} = x + s_1 \mathbf{w}(\tilde{x})^\perp - s_2 \mathbf{w}(\tilde{x})$. This implies

$$|x - \bar{y}| = \sqrt{s_1^2 + s_2^2} = |x - y|.$$

Direct computation shows

$$\sigma(x-y)\mathbf{w}(\tilde{x}) \cdot \mathbf{w}(\tilde{x}) = \frac{2(\mathbf{w}(\tilde{x})^\perp \cdot (x-y))(\mathbf{w}(\tilde{x}) \cdot (x-y))}{|x-y|^2} = \frac{2s_1s_2}{s_1^2 + s_2^2},$$

and

$$\sigma(x-\bar{y})\mathbf{w}(\tilde{x}) \cdot \mathbf{w}(\tilde{x}) = -\frac{2s_1s_2}{s_1^2 + s_2^2} = -\sigma(x-y)\mathbf{w}(\tilde{x}) \cdot \mathbf{w}(\tilde{x}).$$

Hence, we conclude with

$$(\nabla K)_S(x-\bar{y})\mathbf{w}(\tilde{x}) \cdot \mathbf{w}(\tilde{x}) = -(\nabla K)_S(x-y)\mathbf{w}(\tilde{x}) \cdot \mathbf{w}(\tilde{x}).$$

Define the half plane

$$\mathfrak{F} \triangleq \{x \in \mathbb{R}^2 : \nabla\varphi(\tilde{x}) \cdot (x-\tilde{x}) \geq 0\}.$$

Since $\mathfrak{F} \cap \{d_x \leq |x-y| \leq \delta_\gamma\}$ is part of the annulus which is symmetric with respect to $\mathbf{L}$, we deduce that

$$\mathcal{I}_{21}(x)\mathbf{w}(\tilde{x}) \cdot \mathbf{w}(\tilde{x}) = 0, \quad (75)$$

where

$$\mathcal{I}_{21}(x) \triangleq \int_{\mathfrak{F} \cap \{d_x \leq |x-y| \leq \delta_\gamma\}} (\nabla K)_S(x-y) dy.$$

Via (75), it remains to control the difference $\mathcal{I}_{22}(x)\mathbf{w}(\tilde{x}) \cdot \mathbf{w}(\tilde{x})$ where $\mathcal{I}_{22} \triangleq \mathcal{I}_2 - \mathcal{I}_{21}$ satisfies

$$\begin{aligned} |\mathcal{I}_{22}(x)| &\leq \int_{(D \triangle \mathfrak{F}) \cap \{d_x \leq |x-y| \leq \delta_\gamma\}} |(\nabla K)_S(x-y)| dy \leq C \int_{(D \triangle \mathfrak{F}) \cap \{d_x \leq |x-y| \leq \delta_\gamma\}} \frac{m(|x-y|^{-1}) + 1}{|x-y|^2} dy \\ &\leq C \int_{(D \triangle \mathfrak{F}) \cap \{0 \leq |\tilde{x}-y| \leq 2\delta_\gamma\}} \frac{m(|\tilde{x}-y|^{-1}) + 1}{|\tilde{x}-y|^2} dy, \end{aligned}$$

where we apply (40) in the second inequality. For the last inequality, we have used the fact that

$$0 \leq |\tilde{x}-y| \leq |\tilde{x}-x| + |x-y| = d_x + |x-y| \leq 2|x-y|,$$

which implies $\{d_x \leq |x-y| \leq \delta_\gamma\} \subset \{0 \leq |\tilde{x}-y| \leq 2\delta_\gamma\}$. We also make use of the monotonicity of m as well as (17).

Now, we argue that the set $(D \triangle \mathfrak{F}) \cap \{0 \leq |\tilde{x}-y| \leq 2\delta_\gamma\}$ is small. Rewrite the integral in polar coordinates centered in $\tilde{x}$, and borrow the notation in (64). It yields

$$|\mathcal{I}_{22}(x)| \leq C \int_0^{2\delta_\gamma} \frac{m(\rho^{-1}) + 1}{\rho} \mathcal{H}^1(\mathcal{S}_\rho(\tilde{x}) \triangle \Sigma(x)) \rho d\rho.$$

A variation of the geometric lemma (65) (setting $d_x = 0$) reads

$$\mathcal{H}^1(\mathcal{S}_\rho(\tilde{x}) \triangle \Sigma(x)) \leq 2\pi \cdot 2^\gamma \left(\frac{\rho}{\delta_\gamma}\right)^\gamma, \quad \forall \rho > 0.$$

Then, we finish the estimate by

$$\begin{aligned} |\mathcal{I}_{22}(x)| &\leq C \int_0^{2\delta_\gamma} \frac{m(\rho^{-1}) + 1}{\rho} \left(\frac{\rho}{\delta_\gamma}\right)^\gamma d\rho \leq C \left((2\delta_\gamma)^\gamma m((2\delta_\gamma)^{-1}) (\delta_\gamma)^{-\gamma} + \frac{2^\gamma}{\gamma} \right) \\ &\leq C(m(\delta_\gamma^{-1}) + 1) \leq C(m(\Delta_\gamma) + 1), \end{aligned}$$

where we have used (72) in the second inequality, and (24) in the last inequality.

Combining the above estimates, we obtain the desired estimate (74). $\square$

4.4. A refined estimate for the flow map. In this section, we derive a refined estimate for the flow map that improves (6) in Theorem 1.1. It will play a key role in the next section. Define

$$\tilde{\nu}(\rho) \triangleq \rho(m(\rho^{-1}) + 1), \quad \text{and} \quad \tilde{H}(r) \triangleq \int_{\frac{1}{r}}^1 \frac{d\rho}{\tilde{\nu}(\rho)} = \int_{\frac{1}{r}}^1 \frac{d\rho}{\rho(m(\rho^{-1}) + 1)}.$$

Compared with ν and H , one logarithmic factor is removed. As in the case of H , the function $\tilde{H}$ is invertible since $\tilde{H}'(r) = \frac{1}{r^2 \tilde{\nu}(r^{-1})} > 0$. Moreover, the Osgood condition (Osg) implies

$$\lim_{r \rightarrow +\infty} \tilde{H}(r) = \int_0^1 \frac{d\rho}{\rho(m(\rho^{-1}) + 1)} = \int_1^{+\infty} \frac{dr}{r(m(r) + 1)} \geq \frac{\log 2}{1 + \frac{1}{m(1)}} \int_2^{+\infty} \frac{dr}{r(\log r)m(r)} = +\infty.$$

Hence, the inverse function $\tilde{H}^{-1} : \mathbb{R} \rightarrow (0, +\infty)$ is well-defined and increasing.

Lemma 4.7. *Under the assumptions of Theorem 1.1, assume further that $u \in L^1([0, +\infty); C_m^1(\mathbb{R}^2))$. Then, for any $t, s \in \mathbb{R}_+$,*

$$|\Phi_{t,s}(x) - \Phi_{t,s}(y)|^{-1} \geq \tilde{H}^{-1} \left(\tilde{H}(|x - y|^{-1}) - \left| \int_s^t \|u(\tau)\|_{C_m^1(\mathbb{R}^2)} d\tau \right| \right). \quad (76)$$

Moreover, for any $t, s \in \mathbb{R}_+$,

$$|\Phi_{t,s}(x) - \Phi_{t,s}(y)| \leq |x - y| \exp \left((m(|x - y|^{-1}) + 1) \left| \int_s^t \|u(\tau)\|_{C_m^1(\mathbb{R}^2)} d\tau \right| \right). \quad (77)$$

Proof. According to (5) and (59), it follows that for every $t, s \in \mathbb{R}_+$,

$$\left| \frac{d(\Phi_{t,s}(x) - \Phi_{t,s}(y))}{dt} \right| \leq |u(\Phi_{t,s}(x), t) - u(\Phi_{t,s}(y), t)| \leq \|u(t)\|_{C_m^1(\mathbb{R}^2)} \tilde{\nu}(|\Phi_{t,s}(x) - \Phi_{t,s}(y)|).$$

Since $\frac{d}{d\rho} \tilde{H}(\rho^{-1}) = -\frac{1}{\tilde{\nu}(\rho)}$, we get

$$\left| \tilde{H}(|\Phi_{t,s}(x) - \Phi_{t,s}(y)|^{-1}) - \tilde{H}(|x - y|^{-1}) \right| = \left| \int_{|x-y|}^{|\Phi_{t,s}(x) - \Phi_{t,s}(y)|} \frac{d\rho}{\tilde{\nu}(\rho)} \right| \leq \left| \int_s^t \|u(\tau)\|_{C_m^1(\mathbb{R}^2)} d\tau \right|,$$

which directly implies (76).

If $|\Phi_{t,s}(x) - \Phi_{t,s}(y)| \leq |x - y|$, then (77) clearly holds. If $|\Phi_{t,s}(x) - \Phi_{t,s}(y)| > |x - y|$, we have

$$\left| \int_{|x-y|}^{|\Phi_{t,s}(x) - \Phi_{t,s}(y)|} \frac{d\rho}{\tilde{\nu}(\rho)} \right| \geq \frac{1}{m(|x - y|^{-1}) + 1} \int_{|x-y|}^{|\Phi_{t,s}(x) - \Phi_{t,s}(y)|} \frac{d\rho}{\rho} = \frac{\log \frac{|\Phi_{t,s}(x) - \Phi_{t,s}(y)|}{|x-y|}}{m(|x - y|^{-1}) + 1},$$

and consequently,

$$\begin{aligned} |\Phi_{t,s}(x) - \Phi_{t,s}(y)| &\leq |x - y| \exp \left((m(|x - y|^{-1}) + 1) \left| \int_{|x-y|}^{|\Phi_{t,s}(x) - \Phi_{t,s}(y)|} \frac{d\rho}{\tilde{\nu}(\rho)} \right| \right) \\ &\leq |x - y| \exp \left((m(|x - y|^{-1}) + 1) \left| \int_s^t \|u(\tau)\|_{C_m^1(\mathbb{R}^2)} d\tau \right| \right). \end{aligned}$$

This finishes the proof of (77). $\square$

4.5. Closing the estimates. Recall that $\varphi(x, t)$ is a level-set characterization of $D(t)$, satisfying (51), and let $W = \nabla^\perp \varphi$. Then (57) with $k = 1$ reads

$$\partial_t W + (u \cdot \nabla) W = W \cdot \nabla u = \partial_W u, \quad W|_{t=0} = W_0. \quad (78)$$

Integrating along the flow generated by u , we obtain

$$W(x, t) = W_0(\Phi_{0,t}(x)) + \int_0^t (\partial_W u)(s, \Phi_{s,t}(x)) ds. \quad (79)$$

We apply (71) in Lemma 4.5 and obtain the $\|W\|_{L^\infty}$ bound:

$$\|W(t)\|_{L^\infty} \leq \|W_0\|_{L^\infty} + \int_0^t \|\partial_W u(s)\|_{L^\infty} ds \leq \|W_0\|_{L^\infty} + C \int_0^t (1 + \log \Delta_\gamma) \|W(s)\|_{\dot{C}^{\mu(s)}} ds, \quad (80)$$

for any $\mu \in (0, 1)$, $\gamma \in (0, \mu]$ and the mapping $\mu(s) : [0, t] \rightarrow [\frac{\mu}{2}, \mu]$.

Next, we establish a lower bound for $|\nabla \varphi|_{\inf}$. Taking an inner product of (78) with W gives

$$\partial_t |W|^2 + u \cdot \nabla |W|^2 = 2(\nabla u \mathbf{w} \cdot \mathbf{w}) |W|^2, \quad \text{with } \mathbf{w} = \frac{W}{|W|}.$$

Along the flow,

$$\partial_t (\log |W \circ \Phi_{t,0}|) \leq |(\nabla u \mathbf{w} \cdot \mathbf{w}) \circ \Phi_{t,0}|.$$

Hence,

$$|\nabla \varphi|_{\inf}(t) \geq |\nabla \varphi_0|_{\inf} \exp \left(- \int_0^t \|(\nabla u \mathbf{w} \cdot \mathbf{w})\|_{L^\infty} ds \right).$$

By Lemma 4.6, we know that for any $\gamma \in (0, \mu]$,

$$\|\nabla u \mathbf{w} \cdot \mathbf{w}\|_{L^\infty} \leq C(m(\Delta_\gamma) + 1)(1 + \log \Delta_\gamma),$$

with $\Delta_\gamma = \frac{\|\nabla \varphi\|_{\dot{C}^\gamma}}{|\nabla \varphi|_{\inf}} + 1$. Collecting the above estimates yields that for each $\gamma \in (0, \mu]$,

$$|\nabla \varphi|_{\inf}(t) \geq |\nabla \varphi_0|_{\inf} \exp \left(- C \int_0^t (m(\Delta_\gamma(s)) + 1)(1 + \log \Delta_\gamma(s)) ds \right). \quad (81)$$

We now apply the method of losing estimates to derive Hölder estimates for W ; see [4, Section 3.3] for a general statement of this method for transport equations. Our novelty is that we avoid the use of Littlewood-Paley theory, which is essential when we extend the argument to domains with boundary.

Let $T > 0$ and $\mu \in (0, 1)$, $\varepsilon \in (0, \frac{\mu}{2})$ be fixed, and let $\gamma \in (0, \mu)$ be chosen later, then we define

$$V_\gamma(t) \triangleq \int_0^t (1 + \log \Delta_\gamma(s)) ds, \quad \eta_{\gamma,T} \triangleq \frac{\varepsilon}{V_\gamma(T)}, \quad \mu(t) \triangleq \mu - \eta_{\gamma,T} V_\gamma(t). \quad (82)$$

It is clear to see that $\mu(0) = \mu$, $\mu(T) = \mu - \varepsilon \geq \frac{\mu}{2}$, and $\mu(t)$ is decreasing for $t \in [0, T]$.

For any $x, y \in \mathbb{R}^2$, from (79) we have

$$\begin{aligned} |W(x, t) - W(y, t)| &\leq \|W_0\|_{\dot{C}^\mu} |\Phi_{0,t}(x) - \Phi_{0,t}(y)|^\mu \\ &\quad + \int_0^t \|\partial_W u(s)\|_{C_m^{\mu(s)}} |\Phi_{s,t}(x) - \Phi_{s,t}(y)|^{\mu(s)} (m(|\Phi_{s,t}(x) - \Phi_{s,t}(y)|^{-1}) + 1) ds. \end{aligned} \quad (83)$$

Recalling Lemma 4.1, for any $\gamma \in (0, 1)$, there exists $\overline{C} \geq 1$ such that

$$\|u(t)\|_{C_m^1(\mathbb{R}^2)} \leq \overline{C}(1 + \log \Delta_\gamma(t)), \quad t \in \mathbb{R}_+.$$

Applying (77) in Lemma 4.7, we deduce that for every $t \geq s \geq 0$,

$$\begin{aligned} |\Phi_{s,t}(x) - \Phi_{s,t}(y)| &\leq |x - y| \exp \left(\overline{C}(m(|x - y|^{-1}) + 1) \int_s^t (1 + \log \Delta_\gamma(\tau)) d\tau \right) \\ &= |x - y| e^{\mathfrak{M}_{s,t,\gamma}(|x-y|^{-1})}, \end{aligned} \quad (84)$$

where for convenience, we introduce the shortcut notation

$$\mathfrak{M}_{s,t,\gamma}(r) \triangleq \overline{C}(m(r) + 1) \int_s^t (1 + \log \Delta_\gamma(\tau)) d\tau. \quad (85)$$

It follows from (26), (84), and the monotonicity of m that

$$|\Phi_{s,t}(x) - \Phi_{s,t}(y)|^{\mu(s)} (m(|\Phi_{s,t}(x) - \Phi_{s,t}(y)|^{-1}) + 1)$$

$$\begin{aligned}
&\leq C|x-y|^{\mu(s)}e^{\mu(s)}\mathfrak{M}_{s,t,\gamma}(|x-y|^{-1})\left(m(|x-y|^{-1})e^{-\mathfrak{M}_{s,t,\gamma}(|x-y|^{-1})}+1\right) \\
&\leq C|x-y|^{\mu(s)}(m(|x-y|^{-1})+1)e^{\mu\mathfrak{M}_{s,t,\gamma}(|x-y|^{-1})},
\end{aligned} \tag{86}$$

where $C > 0$ depends only on μ (recalling that $\mu(s) \in [\mu - \varepsilon, \mu] \subset [\frac{\mu}{2}, \mu]$). Applying the above estimates to (83), we find

$$\begin{aligned}
\frac{|W(x,t) - W(y,t)|}{|x-y|^{\mu(t)}} &\leq \|W_0\|_{\dot{C}^\mu} |x-y|^{\mu-\mu(t)} e^{\mu\mathfrak{M}_{0,t,\gamma}(|x-y|^{-1})} \\
&\quad + C_1 \int_0^t (1 + \log \Delta_\gamma(s)) \|W(s)\|_{\dot{C}^{\mu(s)}} |x-y|^{\mu(s)-\mu(t)} (m(|x-y|^{-1}) + 1) e^{\mu\mathfrak{M}_{s,t,\gamma}(|x-y|^{-1})} ds \\
&\triangleq J_1 + J_2,
\end{aligned} \tag{87}$$

where $C_1 > 0$ depend only on μ and γ .

In the following, we divide into three cases to discuss the estimate of (87).

Case I: Suppose that $|x-y| \in (0, 1]$ is small enough such that

$$\frac{\log |x-y|^{-1}}{m(|x-y|^{-1}) + 1} \geq \frac{C'\overline{C}}{\eta_{\gamma,T}}, \tag{88}$$

where the constant $C' > \mu$ will be fixed later. By (88), we have that

$$\begin{aligned}
|x-y|^{\mu(s)-\mu(t)} &= \exp\left(\eta_{\gamma,T} \log |x-y| \int_s^t (1 + \log \Delta_\gamma(\tau)) d\tau\right) \\
&\leq \exp\left(-C'\overline{C}(m(|x-y|^{-1}) + 1) \int_s^t (1 + \log \Delta_\gamma(\tau)) d\tau\right) = e^{-C'\mathfrak{M}_{s,t,\gamma}(|x-y|^{-1})}.
\end{aligned}$$

Consequently, the terms J_1 and J_2 can be estimated as follows

$$J_1 \leq \|W_0\|_{\dot{C}^\mu} e^{-(C'-\mu)\mathfrak{M}_{0,t,\gamma}(|x-y|^{-1})},$$

and

$$\begin{aligned}
J_2 &\leq C_1 \sup_{s \in [0,t]} \|W(s)\|_{\dot{C}^{\mu(s)}} \int_0^t (1 + \log \Delta_\gamma(s)) (m(|x-y|^{-1}) + 1) e^{-(C'-\mu)\mathfrak{M}_{s,t,\gamma}(|x-y|^{-1})} ds \\
&= \frac{C_1}{\overline{C}} \sup_{s \in [0,t]} \|W(s)\|_{\dot{C}^{\mu(s)}} \int_0^t e^{-(C'-\mu)\mathfrak{M}_{s,t,\gamma}(|x-y|^{-1})} \frac{d}{ds} (-\mathfrak{M}_{s,t,\gamma}(|x-y|^{-1})) ds \\
&\leq \frac{1}{C' - \mu} \frac{C_1}{\overline{C}} \sup_{s \in [0,t]} \|W(s)\|_{\dot{C}^{\mu(s)}}.
\end{aligned}$$

Hence, choosing $C' = \mu + 2\frac{C_1}{\overline{C}}$, the estimates above imply

$$\frac{|W(x,t) - W(y,t)|}{|x-y|^{\mu(t)}} \leq \|W_0\|_{\dot{C}^\mu} + \frac{1}{2} \sup_{s \in [0,t]} \|W(s)\|_{\dot{C}^{\mu(s)}}. \tag{89}$$

Case II: Suppose $|x-y| \in (0, 1]$ and (88) is not satisfied, that is,

$$\frac{\log |x-y|^{-1}}{m(|x-y|^{-1}) + 1} \leq \frac{C'\overline{C}}{\eta_{\gamma,T}} = \frac{C'\overline{C}V_\gamma(T)}{\varepsilon}. \tag{90}$$

Thanks to (21), Lemma 2.1 and (90), for $|x-y| \leq \frac{1}{2}$, we have

$$\frac{\log^{\frac{1}{2}} |x-y|^{-1}}{C} \leq \frac{\log |x-y|^{-1}}{C(1 + \log^{\frac{1}{2}} |x-y|^{-1})} \leq \frac{\log |x-y|^{-1}}{m(|x-y|^{-1}) + 1} \leq \frac{C'\overline{C}V_\gamma(T)}{\varepsilon},$$

which implies that (for $|x - y| \in [\frac{1}{2}, 1)$, the following estimate is clear)

$$\log |x - y|^{-1} \leq \left(\frac{CC' \overline{C} V_\gamma(T)}{\varepsilon} \right)^2, \quad \text{and} \quad |x - y|^{-1} \leq e^{\left(\frac{CC' \overline{C} V_\gamma(T)}{\varepsilon} \right)^2}.$$

Consequently, we have

$$m(|x - y|^{-1}) \leq \tilde{m} \left(\left(\frac{CC' \overline{C} V_\gamma(T)}{\varepsilon} \right)^2 \right) \leq C \tilde{m} \left(\frac{CC' \overline{C} V_\gamma(T)}{\varepsilon} \right) \leq C(\tilde{m}(V_\gamma(T)) + 1) = C(m(e^{V_\gamma(T)}) + 1),$$

where we have used (H1), (23) and (25) in the three inequalities, respectively, and the constants C' , $\overline{C}$ and ε are absorbed in C . Together with (87), and the definition of $\mathfrak{M}_{s,t,\gamma}(\cdot)$ in (85), it follows that

$$J_1 \leq \|W_0\|_{\dot{C}^\mu} \exp \left(C(m(e^{V_\gamma(T)}) + 1) \int_0^t (1 + \log \Delta_\gamma(\tau)) d\tau \right) \leq \|W_0\|_{\dot{C}^\mu} e^{C \mathfrak{M}_{0,t,\gamma}(e^{V_\gamma(T)})}.$$

Similarly, since $\mu(s) \geq \mu(t)$ for any $t \geq s$, we also deduce that

$$J_2 \leq C(m(e^{V_\gamma(T)}) + 1) \int_0^t (1 + \log \Delta_\gamma(s)) \|W(s)\|_{\dot{C}^{\mu(s)}} e^{C \mathfrak{M}_{s,t,\gamma}(e^{V_\gamma(T)})} ds.$$

Hence, in this case, collecting the above estimates yields

$$\begin{aligned} \frac{|W(x,t) - W(y,t)|}{|x - y|^{\mu(t)}} &\leq \|W_0\|_{\dot{C}^\mu} e^{C \mathfrak{M}_{0,t,\gamma}(e^{V_\gamma(T)})} \\ &\quad + C(m(e^{V_\gamma(T)}) + 1) \int_0^t (1 + \log \Delta_\gamma(s)) \|W(s)\|_{\dot{C}^{\mu(s)}} e^{C \mathfrak{M}_{s,t,\gamma}(e^{V_\gamma(T)})} ds. \end{aligned} \quad (91)$$

Case III: When $|x - y| > 1$, we directly use the $\|W\|_{L^\infty}$ estimate (80) to obtain

$$\frac{|W(x,t) - W(y,t)|}{|x - y|^{\mu(t)}} \leq 2\|W\|_{L^\infty} \leq 2\|W_0\|_{L^\infty} + C \int_0^t (1 + \log \Delta_\gamma(s)) \|W(s)\|_{\dot{C}^{\mu(s)}} ds. \quad (92)$$

Based on the analysis of the three cases, it follows from (89), (91) and (92) that

$$\begin{aligned} \sup_{|x-y|>0} \frac{|W(x,t) - W(y,t)|}{|x - y|^{\mu(t)}} &\leq \frac{1}{2} \sup_{s \in [0,t]} \|W\|_{C^{\mu(s)}} + C\|W_0\|_{\dot{C}^\mu} e^{C \mathfrak{M}_{0,t,\gamma}(e^{V_\gamma(T)})} \\ &\quad + C(m(e^{V_\gamma(T)}) + 1) \int_0^t (1 + \log \Delta_\gamma(s)) \|W(s)\|_{\dot{C}^{\mu(s)}} e^{C \mathfrak{M}_{s,t,\gamma}(e^{V_\gamma(T)})} ds. \end{aligned}$$

Together with (80), we deduce that

$$\begin{aligned} \|W(t)\|_{C^{\mu(t)}} &\leq \frac{1}{2} \sup_{s \in [0,t]} \|W(s)\|_{C^{\mu(s)}} + C\|W_0\|_{\dot{C}^\mu} e^{C \mathfrak{M}_{0,t,\gamma}(e^{V_\gamma(T)})} \\ &\quad + C(m(e^{V_\gamma(T)}) + 1) \int_0^t (1 + \log \Delta_\gamma(s)) \|W(s)\|_{\dot{C}^{\mu(s)}} e^{C \mathfrak{M}_{s,t,\gamma}(e^{V_\gamma(T)})} ds. \end{aligned}$$

Taking the supremum over $s \in [0, t]$ leads to

$$\begin{aligned} \sup_{s \in [0,t]} \|W(s)\|_{C^{\mu(s)}} &\leq C\|W_0\|_{\dot{C}^\mu} e^{C \mathfrak{M}_{0,t,\gamma}(e^{V_\gamma(T)})} \\ &\quad + C(m(e^{V_\gamma(T)}) + 1) \int_0^t (1 + \log \Delta_\gamma(s)) \|W(s)\|_{\dot{C}^{\mu(s)}} e^{C \mathfrak{M}_{s,t,\gamma}(e^{V_\gamma(T)})} ds. \end{aligned} \quad (93)$$

Denote

$$\mathcal{E}(t) \triangleq \sup_{s \in [0,t]} \|W(s)\|_{C^{\mu(s)}} e^{-C \mathfrak{M}_{0,t,\gamma}(e^{V_\gamma(T)})}.$$

Then we find from (93) that

$$\mathcal{E}(t) \leq C\|W_0\|_{C^\mu} + C(m(e^{V_\gamma(T)}) + 1) \int_0^t (1 + \log \Delta_\gamma(s)) \mathcal{E}(s) ds.$$

A direct application of the Grönwall's inequality implies

$$\mathcal{E}(t) \leq C\|W_0\|_{C^\mu} \exp \left(\int_0^t C(m(e^{V_\gamma(T)}) + 1)(1 + \log \Delta_\gamma(s)) ds \right) = C\|W_0\|_{C^\mu} e^{C\mathfrak{M}_{0,t,\gamma}(e^{V_\gamma(T)})}.$$

Therefore, we conclude that

$$\begin{aligned} \|\nabla \varphi(T)\|_{C^{\mu-\varepsilon}} &= \|W(T)\|_{C^{\mu(T)}} \leq \mathcal{E}(T) e^{C\mathfrak{M}_{0,T,\gamma}(e^{V_\gamma(T)})} \leq C\|W_0\|_{C^\mu} e^{2C\mathfrak{M}_{0,T,\gamma}(e^{V_\gamma(T)})} \\ &\leq C\|\nabla \varphi_0\|_{C^\mu} \exp \left(C(m(e^{V_\gamma(T)}) + 1)V_\gamma(T) \right), \end{aligned} \quad (94)$$

for any $T > 0$ and $\gamma \in (0, 1)$, and the constant C is independent of T .

Now, choosing $\gamma = \mu - \varepsilon$, from (81) and (94), we find that (recalling that $\Delta_{\mu-\varepsilon}$ is defined in (58))

$$\Delta_{\mu-\varepsilon}(T) \leq C \exp \left(C(m(e^{V_{\mu-\varepsilon}(T)}) + 1)V_{\mu-\varepsilon}(T) + C \int_0^T m(\Delta_{\mu-\varepsilon}(s))(1 + \log \Delta_{\mu-\varepsilon}(s)) ds \right), \quad (95)$$

where $C = C(\mu, \varepsilon) > 0$ is independent of T , and $V_{\mu-\varepsilon}(T)$ is given by (82). By setting

$$f(t) \triangleq V'_{\mu-\varepsilon}(t) = 1 + \log \Delta_{\mu-\varepsilon}(t), \quad \mathcal{M}(r) \triangleq \begin{cases} r(\tilde{m}(r) + 1) = r(m(e^r) + 1), & \forall r \geq r_0, \\ r(m(e^{r_0}) + 1), & \forall 0 < r \leq r_0, \end{cases} \quad (96)$$

where $r_0 \geq 2$ is the constant appearing in (22), we rewrite the above inequality as

$$\begin{aligned} f(t) &\leq C + C \left(\tilde{m} \left(\int_0^t f(s) ds \right) + 1 \right) \int_0^t f(s) ds + C \int_0^t \tilde{m}(f(s)) f(s) ds \\ &\leq C + C \mathcal{M} \left(\int_0^t f(s) ds \right) + C \int_0^t \mathcal{M}(f(s)) ds, \end{aligned} \quad (97)$$

where $C > 0$ may depend on μ, ε , and the initial data. According to (22), (20) and (96), we have

$$r \mapsto \mathcal{M}(r) \text{ is strictly increasing and convex in } (0, +\infty), \quad (98)$$

and for any $\lambda > 0, r > 0, \epsilon > 0$

$$\mathcal{M}(\lambda r) \leq C\lambda r(\tilde{m}(\lambda r) + 1) \leq C\lambda(1 + \log_+^{\beta+\epsilon} \lambda)\mathcal{M}(r). \quad (99)$$

Applying Jensen's inequality, we get

$$\mathcal{M} \left(\int_0^t f(s) ds \right) = \mathcal{M} \left(\frac{1}{t} \int_0^t t f(s) ds \right) \leq \frac{1}{t} \int_0^t \mathcal{M}(t f(s)) ds \leq C \log^{\beta+\epsilon}(e+t) \int_0^t \mathcal{M}(f(s)) ds.$$

Consequently, it follows from (97) that

$$f(t) \leq C + C \log^{\beta+\epsilon}(e+t) \int_0^t \mathcal{M}(f(s)) ds. \quad (100)$$

Denote by

$$R(t) \triangleq \frac{r_0}{\log^{\beta+\epsilon}(e+t)} + \int_0^t \mathcal{M}(f(s)) ds,$$

so that $R(0) = r_0$ and $f(t) \leq C \log^{\beta+\epsilon}(e+t)R(t)$. Then, together with (99), we obtain

$$\begin{aligned} \frac{dR(t)}{dt} &= -\frac{\beta+\epsilon}{(e+t) \log^{1+\beta+\epsilon}(e+t)} + \mathcal{M}(f(t)) \\ &\leq \mathcal{M}(C \log^{\beta+\epsilon}(e+t)R(t)) \leq C \log^{\beta+2\epsilon}(e+t) \mathcal{M}(R(t)). \end{aligned}$$

Integrating the above differential inequality on $t \in [0, T]$ yields

$$\mathcal{H}(R(T)) = \mathcal{H}(R(T)) - \mathcal{H}(R(0)) \leq C(1+T) \log^{\beta+2\epsilon}(e+T),$$

where

$$\mathcal{H}(r) \triangleq \int_{r_0}^r \frac{d\tilde{r}}{\mathcal{M}(\tilde{r})} = \begin{cases} \int_{r_0}^r \frac{1}{\tilde{r}(\tilde{m}(\tilde{r})+1)} d\tilde{r}, & \forall r \geq r_0, \\ \frac{1}{m(e^{r_0})+1} \log \frac{r}{r_0}, & \forall 0 < r \leq r_0. \end{cases} \quad (101)$$

Note that $\mathcal{H}(R(0)) = \mathcal{H}(r_0) = 0$, $\lim_{r \rightarrow 0} \mathcal{H}(r) = -\infty$, and the Osgood condition (Osg) implies

$$\lim_{r \rightarrow +\infty} \mathcal{H}(r) = \int_{r_0}^{+\infty} \frac{1}{r(\tilde{m}(r)+1)} dr \geq \frac{1}{1 + \frac{1}{\tilde{m}(r_0)}} \int_{r_0}^{+\infty} \frac{dr}{r \tilde{m}(r)} = +\infty.$$

Together with (98), we infer that $r \mapsto \mathcal{H}(r)$ is strictly increasing in $[0, +\infty)$, and it has a unique inverse function $\mathcal{H}^{-1}(\cdot)$ in $(-\infty, +\infty)$. Hence, it follows that for any $T > 0$,

$$f(T) \leq \log^{\beta+\epsilon}(e+T)R(T) \leq \log^{\beta+\epsilon}(e+T) \mathcal{H}^{-1}(C(1+T) \log^{\beta+2\epsilon}(e+T)). \quad (102)$$

This implies the boundedness of $\Delta_{\mu-\epsilon}(T)$, which leads to the $C^{\mu-\epsilon}$ boundary regularity. Indeed, from (81), (94) and (102), we have that for any $T > 0$,

$$\begin{aligned} |\nabla \varphi(T)|_{\inf} &\geq |\nabla \varphi_0|_{\inf} \exp \left(-C \int_0^T \mathcal{M}(f(t)) dt \right) \geq |\nabla \varphi_0|_{\inf} e^{-CR(T)} \\ &\geq |\nabla \varphi_0|_{\inf} \exp \left(-C \mathcal{H}^{-1}(C(1+T) \log^{\beta+2\epsilon}(e+T)) \right), \end{aligned} \quad (103)$$

and

$$\begin{aligned} \|\nabla \varphi(T)\|_{C^{\mu-\epsilon}} &\leq C \|\nabla \varphi_0\|_{C^\mu} \exp \left(C \mathcal{M} \left(\int_0^T f(t) dt \right) \right) \leq C \|\nabla \varphi_0\|_{C^\mu} \exp \left(C \log^{\beta+\epsilon}(e+T)R(T) \right) \\ &\leq C \|\nabla \varphi_0\|_{C^\mu} \exp \left(C \log^{\beta+2\epsilon}(e+T) \mathcal{H}^{-1}(C(1+T) \log^{\beta+2\epsilon}(e+T)) \right). \end{aligned} \quad (104)$$

Since the preceding estimates are uniform with respect to the kernel regularization, the approximation argument in Subsection C.2 allows us to pass to the limit $\delta \downarrow 0$ and transfer these boundary estimates to the unique patch solution of Theorem 1.1.

This completes the proof of the following theorem.

Theorem 4.8. *Let $\mu \in (0, 1)$. Assume that $m(\xi) = m(|\xi|)$ is a radial function satisfying (H1)-(H2a)-(Osg). Consider the unique global patch solution (9) of the 2D Loglog-Euler type equation (1) associated with the patch data $\omega_0 = \mathbf{1}_{D_0}$ where D_0 is a simply connected and bounded domain with $\partial D_0 \in C^{1,\mu}$. Then for every $t > 0$ and every $\epsilon \in (0, \mu)$, the patch boundary $\partial D(t)$ retains $C^{1,\mu-\epsilon}$ regularity, namely, $\partial D(t) \in C^{1,\mu-\epsilon}$.*

More precisely, for any given $\epsilon > 0$ and any $\epsilon > 0$, there is some constant $C > 0$ depending on μ, ϵ, ϵ and the initial data, such that the estimates (103) and (104) hold, where φ is the solution to (51) and the map $\mathcal{H}$ is defined by (101).

In view of the definitions of $\mathcal{H}$ in (7) and $\mathcal{H}$ in (101), we have that for any $r \geq r_0$,

$$\begin{aligned} \mathcal{H}(r) &\geq \frac{1}{1 + \frac{1}{\tilde{m}(r_0)}} \int_{r_0}^r \frac{d\tilde{r}}{\tilde{r} \tilde{m}(\tilde{r})} = C \int_{e^{r_0}}^{e^r} \frac{d\tilde{r}}{\tilde{r}(\log \tilde{r})m(\tilde{r})} \\ &= C \left(\mathcal{H}(e^r) - \int_2^{e^{r_0}} \frac{d\tilde{r}}{\tilde{r}(\log \tilde{r})m(\tilde{r})} \right) = C(\mathcal{H}(e^r) - C). \end{aligned}$$

This implies

$$\mathcal{H}^{-1}(y) \leq \log(\mathcal{H}^{-1}(Cy + C)), \quad \forall y \geq 0. \quad (105)$$

Applying the bound to (103) and (104) yields the estimates (11) and (12). Then, a direct application of (54)-(55)-(56) gives (10), thereby completing the proof of Theorem 1.2 for $n = 1$.

5. GLOBAL REGULARITY OF A SINGLE $C^{n,\mu}$ VORTEX PATCH

In this section, we establish the global propagation of higher boundary regularity for patch solutions with initial data $\omega_0(x) = \mathbf{1}_{D_0}(x)$, where $\partial D_0 \in C^{n,\mu}$ for some $n \geq 2$ and $\mu \in (0, 1)$.

The main result of this section is the following.

Theorem 5.1. *Let $\mu \in (0, 1)$ and $n \geq 2$. Assume that $m(\xi) = m(|\xi|)$ is a radial function satisfying (H1)-(H2a)-(Osg). Let $\varphi_0 \in C^{n,\mu}(\mathbb{R}^2)$ be a level-set characterization of a bounded simply-connected domain D_0 . Then the corresponding global patch solution has the property that $\partial D(t)$ remains of class $C^{n,\mu'}$ for any $t > 0$ and every $\mu' \in (0, \mu)$.*

As a consequence of Theorem 5.1, C^∞ patch boundaries remain C^∞ for all positive time.

Corollary 5.2. *Assume that $m \in C^\infty(\mathbb{R}_+)$ satisfies (H1), (H2a), and (Osg), with the Mikhlin-Hörmander bounds in (H1) valid at every order. If $\partial D_0 \in C^\infty$, then the unique patch solution satisfies $\partial D(t) \in C^\infty$ for every $t > 0$.*

Proof. Fix $\mu = \frac{1}{2}$. By Theorem 5.1, $\partial D(t) \in C^{n,1/4}$ for every $n \geq 2$ and $t > 0$. Hence $\partial D(t) \in C^\infty$ for every $t > 0$. $\square$

According to the framework in Section 4.1, to prove Theorem 5.1, it is sufficient to control $\partial_W^{n-1}W$. More precisely, we will show that if $\varphi_0 \in C^{n,\mu}(\mathbb{R}^2)$, then for any $\mu' \in (0, \mu)$, there exists an increasing and positive function $R_{n,\mu,\mu'}(t)$ such that

$$\|\partial_W^{n-1}W(t)\|_{C^{\mu'}(\mathbb{R}^2)} \leq R_{n,\mu,\mu'}(t) < +\infty. \quad (106)$$

Indeed, given (106), for any $\gamma \in (0, 1)$, $\mu \in (0, 1)$ and $\mu' \in (0, \mu)$, we have

$$\sum_{j=1}^{n-1} \|\partial_W^{j-1}W(t)\|_{C^\gamma(\mathbb{R}^2)} + \|\partial_W^{n-1}W(t)\|_{C^{\mu'}(\mathbb{R}^2)} \leq \sum_{j=1}^{n-1} R_{j,\frac{\gamma+1}{2},\gamma}(t) + R_{n,\mu,\mu'}(t) < +\infty,$$

where we have used the fact that $C^{n,\mu} \subset C^{j,\frac{\gamma+1}{2}}$ when $j < n$. Consequently, we deduce $\partial D(t) \in C^{n,\mu'}$.

Moreover, the growth rate of $R_{n,\mu,\mu'}(t)$ matches with (104), namely

$$R_{n,\mu,\mu'}(T) \leq C\|\varphi_0\|_{C^{n,\mu}} \exp\left(C \log^{\beta+2\epsilon}(e+T) \mathcal{H}^{-1}(C(1+T) \log^{\beta+2\epsilon}(e+T))\right). \quad (107)$$

Applying the bound (105) to (107) yields the estimate (13). Then, a direct application of (54)-(55)-(56) gives (10) for $n \geq 2$, finishing the proof of Theorem 1.2.

5.1. The estimation of $\partial_W^k u$. Motivated by the argument of Radu [86, Prop. 6.2], we derive a desingularized integral representation for higher-order tangential derivatives $\partial_W^k u$, extending (69). For convenience, we use the notation

$$\delta_y^x[f] \triangleq f(x) - f(y), \quad \forall x, y \in \mathbb{R}^2.$$

Proposition 5.3 (Desingularized representation of $\partial_W^k u$). *Let $D \subset \mathbb{R}^2$ be a smooth bounded simply-connected domain, and let φ be a level-set characterization of D . Assume that m satisfies (H1)-(H2a). Let $W = (W_1, W_2) = \nabla^\perp \varphi$ be the tangential vector field associated with φ . Let $1 \leq k \leq n$,*

and suppose that $\partial_W^\gamma W \in C^\gamma(\mathbb{R}^2)$ for every $\gamma \in (0, 1)$ and $0 \leq l \leq k-2$ (the condition being vacuous when $k = 1$), and $\partial_W^{k-1} W \in C^\mu(\mathbb{R}^2)$, $\mu \in (0, 1)$. Let $u(x) = \nabla^\perp(-\Delta)^{-1}m(\Lambda)(\mathbf{1}_D)(x)$. Then

$$\partial_W^k u(x) = \sum_{j=1}^k \sum_{\substack{l_1 \geq \dots \geq l_j \geq 1 \\ l_1 + \dots + l_j = k}} c_{l_1, \dots, l_j}^k \text{p.v.} \int_D \left(\delta_y^x [\partial_W^{l_1-1} W] \otimes \dots \otimes \delta_y^x [\partial_W^{l_j-1} W] \right) \cdot \nabla^j K(x-y) dy, \quad (108)$$

where $c_{l_1, \dots, l_j}^k \in \mathbb{R}_+$, $c_k^k = 1$, “ $\otimes$ ” denotes the usual tensor product (e.g. $a \otimes b = (a_i b_j)_{2 \times 2}$ for two vectors $a, b \in \mathbb{R}^2$), and $K(x) = \frac{x^\perp}{|x|^2} G(|x|)$ is given by (27).

Remark 6. For example, when $k = 2, 3$, the representation becomes

$$\begin{aligned} \partial_W^2 u(x) &= \text{p.v.} \int_D (W(x) - W(y)) \otimes (W(x) - W(y)) \cdot \nabla^2 K(x-y) dy \\ &\quad + \text{p.v.} \int_D (\partial_W W(x) - \partial_W W(y)) \cdot \nabla K(x-y) dy, \end{aligned}$$

and

$$\begin{aligned} \partial_W^3 u(x) &= \text{p.v.} \int_D (W(x) - W(y)) \otimes (W(x) - W(y)) \otimes (W(x) - W(y)) \cdot \nabla^3 K(x-y) dy \\ &\quad + 3 \text{p.v.} \int_D (\partial_W W(x) - \partial_W W(y)) \otimes (W(x) - W(y)) \cdot \nabla^2 K(x-y) dy \\ &\quad + \text{p.v.} \int_D (\partial_W^2 W(x) - \partial_W^2 W(y)) \cdot \nabla K(x-y) dy. \end{aligned}$$

Proof of Proposition 5.3. We prove (108) by induction. For $k = 1$, (108) is a direct consequence of (69) with $c_1^1 = 1$. Suppose that (108) is true for k . We will show that it is also true for the $k+1$ case. The proof is analogous to Lemma 4.4.

Due to the roughness of the patch solution, we shall compute $\partial_W^{k+1} u$ in the distributional sense. Since $W = \nabla^\perp \varphi$ is divergence free, we have

$$\partial_W^{k+1} u = \text{div}(W \partial_W^k u).$$

Next, for any $\tilde{\chi} \in C_c^\infty(\mathbb{R}^2)$, we get

$$(\text{div}(W \partial_W^k u), \tilde{\chi}) = -(\partial_W^k u, W \cdot \nabla \tilde{\chi}) = - \int_{\mathbb{R}^2} \partial_W^k u(x) W(x) \cdot \nabla \tilde{\chi}(x) dx.$$

Denoting by

$$\mathcal{D}_W^{l_1, \dots, l_j}(x, y) \triangleq \delta_y^x [\partial_W^{l_1-1} W] \otimes \dots \otimes \delta_y^x [\partial_W^{l_j-1} W], \quad (109)$$

and through the induction hypotheses (108) and Fubini's theorem, we deduce that

$$\begin{aligned} &(\text{div}(W \partial_W^k u), \tilde{\chi}) \\ &= - \sum_{j=1}^k \sum_{\substack{l_1 \geq \dots \geq l_j \geq 1 \\ l_1 + \dots + l_j = k}} c_{l_1, \dots, l_j}^k \int_{\mathbb{R}^2} \left(\lim_{\varepsilon \rightarrow 0} \int_{D \cap \{|x-y| \geq \varepsilon\}} \mathcal{D}_W^{l_1, \dots, l_j}(x, y) \cdot \nabla^j K(x-y) dy \right) W(x) \cdot \nabla \tilde{\chi}(x) dx \\ &= - \sum_{j=1}^k \sum_{\substack{l_1 \geq l_2 \geq \dots \geq l_j \geq 1 \\ l_1 + \dots + l_j = k}} c_{l_1, \dots, l_j}^k \lim_{\varepsilon \rightarrow 0} \int_D \int_{|x-y| \geq \varepsilon} \mathcal{D}_W^{l_1, \dots, l_j}(x, y) \cdot \nabla^j K(x-y) W(x) \cdot \nabla \tilde{\chi}(x) dx dy \end{aligned}$$

$$\triangleq \sum_{j=1}^k \sum_{\substack{l_1 \geq l_2 \geq \dots \geq l_j \geq 1 \\ l_1 + \dots + l_j = k}} c_{l_1, \dots, l_j}^k \lim_{\varepsilon \rightarrow 0} \mathcal{A}_\varepsilon.$$

Since W is divergence free, the integration by parts gives us

$$\begin{aligned} \mathcal{A}_\varepsilon &= - \int_D \int_{|x-y| \geq \varepsilon} \mathcal{D}_W^{l_1, \dots, l_j}(x, y) \cdot \nabla^j K(x-y) W(x) \cdot \nabla \tilde{\chi}(x) dx dy \\ &= \int_D \int_{|x-y| \geq \varepsilon} W(x) \cdot \nabla_x \left(\mathcal{D}_W^{l_1, \dots, l_j}(x, y) \cdot \nabla^j K(x-y) \right) \tilde{\chi}(x) dx dy + I_\varepsilon \end{aligned}$$

where

$$I_\varepsilon \triangleq \int_D \int_{|x-y|=\varepsilon} \mathcal{D}_W^{l_1, \dots, l_j}(x, y) \cdot \nabla^j K(x-y) W(x) \cdot \frac{(x-y)}{|x-y|} \tilde{\chi}(x) dS(x) dy.$$

Define

$$J_\varepsilon \triangleq \int_D \int_{|x-y|=\varepsilon} \mathcal{D}_W^{l_1, \dots, l_j}(x, y) \cdot \nabla^j K(x-y) W(y) \cdot \frac{(x-y)}{|x-y|} \tilde{\chi}(x) dS(x) dy, \quad \text{and} \quad R_\varepsilon \triangleq I_\varepsilon - J_\varepsilon.$$

Noting that for $s = 1, \dots, j$,

$$|\delta_y^x [\partial_W^{l_s-1} W]| = |\partial_W^{l_s-1} W(x) - \partial_W^{l_s-1} W(y)| \leq \|\partial_W^{l_s-1} W\|_{C^\gamma} |x-y|^\gamma, \quad (110)$$

we control the difference R_ε by

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0} |R_\varepsilon| &= \lim_{\varepsilon \rightarrow 0} \left| \int_D \int_{|x-y|=\varepsilon} \mathcal{D}_W^{l_1, \dots, l_j}(x, y) \cdot \nabla^j K(x-y) (W(x) - W(y)) \cdot \frac{(x-y)}{|x-y|} \tilde{\chi}(x) dS(x) dy \right| \\ &\leq C \lim_{\varepsilon \rightarrow 0} \int_D \prod_{s=1}^j \left(\|\partial_W^{l_s-1} W\|_{C^\gamma} \varepsilon^\gamma \right) \cdot \frac{m(\varepsilon^{-1}) + 1}{\varepsilon^{j+1}} \cdot \|W\|_{C^\gamma} \varepsilon^\gamma \cdot 1 \cdot \|\tilde{\chi}\|_{L^\infty} \cdot 2\pi \varepsilon dy \\ &\leq C \lim_{\varepsilon \rightarrow 0} \varepsilon^{(j+1)\gamma-j} (m(\varepsilon^{-1}) + 1) = 0, \end{aligned}$$

by choosing $\gamma \in (\frac{k}{k+1}, 1)$ so that $\gamma > \frac{k}{k+1} \geq \frac{j}{j+1}$ and therefore $(j+1)\gamma - j > 0$. We have used (35) to estimate $\nabla^j K$.

For J_ε , a similar argument as (70) yields

$$\begin{aligned} J_\varepsilon &= \int_{\mathbb{R}^2} \int_{D \cap \{|x-y|=\varepsilon\}} \mathcal{D}_W^{l_1, \dots, l_j}(x, y) \cdot \nabla^j K(x-y) W(y) \cdot \frac{(x-y)}{|x-y|} \tilde{\chi}(x) dS(y) dx \\ &= \int_{\mathbb{R}^2} \int_{D \cap \{|x-y| \geq \varepsilon\}} W(y) \cdot \nabla_y \left(\mathcal{D}_W^{l_1, \dots, l_j}(x, y) \cdot \nabla^j K(x-y) \right) dy \tilde{\chi}(x) dx. \end{aligned}$$

Collecting the above computations yields that

$$\begin{aligned} \mathcal{A}_\varepsilon &= \int_{\mathbb{R}^2} \int_{D \cap \{|x-y| \geq \varepsilon\}} W(x) \cdot \nabla_x \left(\mathcal{D}_W^{l_1, \dots, l_j}(x, y) \cdot \nabla^j K(x-y) \right) dy \tilde{\chi}(x) dx \\ &\quad + \int_{\mathbb{R}^2} \int_{D \cap \{|x-y| \geq \varepsilon\}} W(y) \cdot \nabla_y \left(\mathcal{D}_W^{l_1, \dots, l_j}(x, y) \cdot \nabla^j K(x-y) \right) dy \tilde{\chi}(x) dx + R_\varepsilon \\ &= \int_{\mathbb{R}^2} \int_{D \cap \{|x-y| \geq \varepsilon\}} \mathcal{D}_W^{l_1, \dots, l_j}(x, y) \otimes (W(x) - W(y)) \cdot \nabla^{j+1} K(x-y) dy \tilde{\chi}(x) dx \\ &\quad + \int_{\mathbb{R}^2} \int_{D \cap \{|x-y| \geq \varepsilon\}} \sum_{s=1}^j \mathcal{D}_W^{l_1, \dots, l_{s+1}, \dots, l_j}(x, y) \cdot \nabla^j K(x-y) dy \tilde{\chi}(x) dx + R_\varepsilon. \end{aligned}$$

Observe that all the terms have the form $\mathcal{D}_W^{\tilde{l}_1, \dots, \tilde{l}_{\tilde{j}}}(x, y) \cdot \nabla^{\tilde{j}} K(x - y)$, with $\tilde{j} = j$ or $j + 1$ so that $\tilde{j} \in \{1, \dots, k + 1\}$, and $\tilde{l}_1 + \dots + \tilde{l}_{\tilde{j}} = k + 1$. Therefore, we obtain

$$\begin{aligned} (\operatorname{div}(W \partial_W^k u), \tilde{\chi}) &= \lim_{\varepsilon \rightarrow 0} \sum_{j=1}^k \sum_{\substack{l_1 \geq \dots \geq l_j \geq 1 \\ l_1 + \dots + l_j = k}} c_{l_1, \dots, l_j}^k \mathcal{A}_\varepsilon \\ &= \int_{\mathbb{R}^2} \left(\sum_{j=1}^{k+1} \sum_{\substack{l_1 \geq \dots \geq l_j \geq 1 \\ l_1 + \dots + l_j = k+1}} c_{l_1, \dots, l_j}^{k+1} \text{p.v.} \int_D \mathcal{D}_W^{l_1, \dots, l_j}(x, y) \cdot \nabla^j K(x - y) dy \right) \tilde{\chi}(x) dx, \end{aligned}$$

if we define the coefficients $c_{l_1, \dots, l_j}^{k+1}$ ($j = 1, \dots, k + 1$) properly by

$$c_{l_1, \dots, l_j}^{k+1} = \sum_{\substack{s=1 \\ l_s \geq 2}}^j c_{l_1, \dots, l_{s-1}, \dots, l_j}^k + \mathbf{1}_{\{l_j=1\}} c_{l_1, \dots, l_{j-1}}^k.$$

Recalling the notation (109), we conclude with (108) for $k + 1$, as desired. $\square$

Given the representation of $\partial_W^k u$ in (108), the leading term requiring the highest regularity of W corresponds to the case $j = 1$ and $l_1 = k$, which takes the form

$$\mathbf{G}_k(x) \triangleq \text{p.v.} \int_D (\partial_W^{k-1} W(x) - \partial_W^{k-1} W(y)) \cdot \nabla K(x - y) dy.$$

The next lemma provides estimates for the principal term $\mathbf{G}_k$ and the remaining lower-order terms.

Lemma 5.4. *Let $2 \leq k \leq n$. Under the assumptions of Proposition 5.3, we have*

$$\|\mathbf{G}_k\|_{C_m^\mu(\mathbb{R}^2)} \leq C(1 + \log \Delta_\gamma) \|\partial_W^{k-1} W\|_{\dot{C}^\mu(\mathbb{R}^2)}, \quad (111)$$

and

$$\|\partial_W^k u - \mathbf{G}_k\|_{C^\mu(\mathbb{R}^2)} \leq C \sum_{j=2}^k \sum_{\substack{l_1 \geq \dots \geq l_j \geq 1 \\ l_1 + \dots + l_j = k}} \left(\prod_{s=1}^j \|\partial_W^{l_s-1} W\|_{\dot{C}^{\gamma_k}} \right), \quad (112)$$

for every $\mu \in (0, 1)$, $\gamma \in (0, 1)$, and $\gamma_k \in (1 - \frac{1-\mu}{k}, 1)$.

Proof. The proof of (111) is analogous to that of Lemma 4.5, replacing W on the right-hand side of (71) by $\partial_W^{k-1} W$.

To prove (112), we also follow a similar procedure as in Lemma 4.5. Recall that

$$\partial_W^k u(x) - \mathbf{G}_k(x) = \sum_{j=2}^k \sum_{\substack{l_1 \geq \dots \geq l_j \geq 1 \\ l_1 + \dots + l_j = k}} c_{l_1, \dots, l_j}^k \Psi_{l_1, \dots, l_j}(x),$$

where we denote by

$$\begin{aligned} \Psi_{l_1, \dots, l_j}(x) &\triangleq \text{p.v.} \int_D \mathcal{D}_W^{l_1, \dots, l_j}(x, y) \cdot \nabla^j K(x - y) dy \\ &= \text{p.v.} \int_D \left(\delta_y^x [\partial_W^{l_1-1} W] \otimes \dots \otimes \delta_y^x [\partial_W^{l_j-1} W] \right) \cdot \nabla^j K(x - y) dy. \end{aligned}$$

By virtue of (110), (35), and (21), we have the L^∞ bound:

$$\begin{aligned}
\|\partial_W^k u - \mathbf{G}_k\|_{L^\infty} &\leq C \sum_{j=2}^k \sum_{\substack{l_1 \geq \dots \geq l_j \geq 1 \\ l_1 + \dots + l_j = k}} \left(\prod_{s=1}^j \|\partial_W^{l_s-1} W\|_{\dot{C}^{\gamma_k}} \right) \int_D |x-y|^{-j-1+j\gamma_k} (m(|x-y|^{-1}) + 1) dy \\
&\leq C \sum_{j=2}^k \sum_{\substack{l_1 \geq \dots \geq l_j \geq 1 \\ l_1 + \dots + l_j = k}} \left(\prod_{s=1}^j \|\partial_W^{l_s-1} W\|_{\dot{C}^{\gamma_k}} \right) \int_0^L r^{-j+j\gamma_k} (m(r^{-1}) + 1) dy \\
&\leq C \sum_{j=2}^k \sum_{\substack{l_1 \geq \dots \geq l_j \geq 1 \\ l_1 + \dots + l_j = k}} \left(\prod_{s=1}^j \|\partial_W^{l_s-1} W\|_{\dot{C}^{\gamma_k}} \right),
\end{aligned}$$

Choosing $x, h \in \mathbb{R}^2$ such that $|h| \leq \frac{\bar{c}_0}{3}$ (recalling that $\bar{c}_0 > 0$ is the constant appearing in Lemma 2.3), we write

$$\Psi_{l_1, \dots, l_j}(x) - \Psi_{l_1, \dots, l_j}(x+h) = \mathfrak{J}_1 + \mathfrak{J}_2 + \mathfrak{J}_3 + \mathfrak{J}_4,$$

where

$$\begin{aligned}
\mathfrak{J}_1 &\triangleq \text{p.v.} \int_{D \cap \{|x-y| < 2|h|\}} \mathcal{D}_W^{l_1, \dots, l_j}(x, y) \cdot \nabla^j K(x-y) dy, \\
\mathfrak{J}_2 &\triangleq -\text{p.v.} \int_{D \cap \{|x-y| < 2|h|\}} \mathcal{D}_W^{l_1, \dots, l_j}(x+h, y) \cdot \nabla^j K(x+h-y) dy, \\
\mathfrak{J}_3 &\triangleq \int_{D \cap \{|x-y| \geq 2|h|\}} \left(\mathcal{D}_W^{l_1, \dots, l_j}(x, y) - \mathcal{D}_W^{l_1, \dots, l_j}(x+h, y) \right) \cdot \nabla^j K(x-y) dy, \\
\mathfrak{J}_4 &\triangleq \int_{D \cap \{|x-y| \geq 2|h|\}} \mathcal{D}_W^{l_1, \dots, l_j}(x+h, y) \cdot \left(\nabla^j K(x-y) - \nabla^j K(x+h-y) \right) dy.
\end{aligned}$$

For $\mathfrak{J}_1$, from (110), (35) and (72), we get

$$\begin{aligned}
|\mathfrak{J}_1| &\leq C \int_{|x-y| \leq 2|h|} \left(\prod_{s=1}^j \|\partial_W^{l_s-1} W\|_{\dot{C}^{\gamma_k}} |x-y|^{\gamma_k} \right) \frac{m(|x-y|^{-1})}{|x-y|^{j+1}} dy \\
&\leq C \left(\prod_{s=1}^j \|\partial_W^{l_s-1} W\|_{\dot{C}^{\gamma_k}} \right) \int_0^{2|h|} \rho^{j\gamma_k-j} m(\rho^{-1}) d\rho \leq C \left(\prod_{s=1}^j \|\partial_W^{l_s-1} W\|_{\dot{C}^{\gamma_k}} \right) |h|^{j\gamma_k-j+1} m(|h|^{-1}) \\
&\leq C \left(\prod_{s=1}^j \|\partial_W^{l_s-1} W\|_{\dot{C}^{\gamma_k}} \right) |h|^\mu,
\end{aligned}$$

where for the last inequality, we have used the fact that $j\gamma_k - j + 1 > \mu$, which follows from assumptions $\gamma_k > 1 - \frac{1-\mu}{k}$ and $j \leq k$. A similar argument yields

$$|\mathfrak{J}_2| \leq C \left(\prod_{s=1}^j \|\partial_W^{l_s-1} W\|_{\dot{C}^{\gamma_k}} \right) \int_0^{3|h|} \rho^{j\gamma_k-j} m(\rho^{-1}) d\rho \leq C \left(\prod_{s=1}^j \|\partial_W^{l_s-1} W\|_{\dot{C}^{\gamma_k}} \right) |h|^\mu.$$

For the term $\mathfrak{J}_3$, since $|x-y| \geq 2h$, we have $|x+h-y| \leq |x-y| + |h| \leq \frac{3}{2}|x-y|$. We apply (110) and obtain

$$\begin{aligned}
&|\mathcal{D}_W^{l_1, \dots, l_j}(x, y) - \mathcal{D}_W^{l_1, \dots, l_j}(x+h, y)| \\
&= \left| \sum_{s=1}^j \left(\delta_y^{x+h} [\partial_W^{l_s-1} W] \otimes \dots \otimes (\delta_y^x [\partial_W^{l_s-1} W] - \delta_y^{x+h} [\partial_W^{l_s-1} W]) \otimes \dots \otimes \delta_y^x [\partial_W^{l_j-1} W] \right) \right|
\end{aligned}$$

$$\leq C \left(\prod_{s=1}^j \|\partial_W^{l_s-1} W\|_{\dot{C}^{\gamma_k}} \right) |h|^{\gamma_k} |x-y|^{(j-1)\gamma_k}.$$

Then, it follows from (35) and analogous estimates for $\mathfrak{J}_1$ that

$$\begin{aligned} |\mathfrak{J}_3| &\leq C \left(\prod_{s=1}^j \|\partial_W^{l_s-1} W\|_{\dot{C}^{\gamma_k}} \right) |h|^{\gamma_k} \text{p.v.} \int_{D \cap \{|x-y| \geq 2|h|\}} |x-y|^{(j-1)\gamma_k} \frac{m(|x-y|^{-1}) + 1}{|x-y|^{j+1}} dy \\ &\leq C \left(\prod_{s=1}^j \|\partial_W^{l_s-1} W\|_{\dot{C}^{\gamma_k}} \right) |h|^{\gamma_k} \int_{2|h|}^L \rho^{(j-1)\gamma_k-j} (m(\rho^{-1}) + 1) d\rho \\ &\leq C \left(\prod_{s=1}^j \|\partial_W^{l_s-1} W\|_{\dot{C}^{\gamma_k}} \right) |h|^{j\gamma_k-j+1} (m(|h|^{-1}) + 1) \leq C \left(\prod_{s=1}^j \|\partial_W^{l_s-1} W\|_{\dot{C}^{\gamma_k}} \right) |h|^\mu. \end{aligned}$$

Using the mean value theorem and (35), the term $\mathfrak{J}_4$ can be bounded as

$$\begin{aligned} |\mathfrak{J}_4| &\leq C \int_{D \cap \{|x-y| \geq 2|h|\}} \left(\prod_{s=1}^j \|\partial_W^{l_s-1} W\|_{\dot{C}^{\gamma_k}} \left(\frac{3}{2} |x-y| \right)^{\gamma_k} \right) |h| \frac{m((\frac{1}{2}|x-y|)^{-1}) + 1}{(\frac{1}{2}|x-y|)^{j+2}} dy \\ &\leq C \left(\prod_{s=1}^j \|\partial_W^{l_s-1} W\|_{\dot{C}^{\gamma_k}} \right) |h| \int_{2|h|}^L \rho^{j\gamma_k-j-1} (m(\rho^{-1}) + 1) d\rho \\ &\leq C \left(\prod_{s=1}^j \|\partial_W^{l_s-1} W\|_{\dot{C}^{\gamma_k}} \right) |h|^{j\gamma_k-j+1} (m(|h|^{-1}) + 1) \leq C \left(\prod_{s=1}^j \|\partial_W^{l_s-1} W\|_{\dot{C}^{\gamma_k}} \right) |h|^\mu. \end{aligned}$$

Gathering the above estimates yields

$$\|\Psi_{l_1, \dots, l_j}\|_{\dot{C}^\mu(\mathbb{R}^2)} \leq C \prod_{s=1}^j \left(\|\partial_W^{l_s-1} W\|_{\dot{C}^{\gamma_k}} \right),$$

and the desired estimate (112) follows. $\square$

5.2. Proof of Theorem 5.1. We prove (106)-(107) by induction. The case $n = 1$ has been proved in Theorem 4.8, where the bound is given by (104) with $\mu' = \mu - \varepsilon$.

Assume that (106)-(107) holds for any $1 \leq n \leq k$. We will show that it holds for $n = k + 1$, that is, if $\|\partial_{W_0}^k W_0\|_{C^\mu(\mathbb{R}^2)} < +\infty$, then

$$\|\partial_W^k W(t)\|_{C^{\mu'}(\mathbb{R}^2)} \leq R_{k+1, \mu, \mu'}(t) < +\infty, \quad (113)$$

where $R_{k+1, \mu, \mu'}(t)$ satisfies (107).

Recalling that $\Phi_{t,s}(x)$ is the flow map given by (5), and from the equation (57), we have

$$\partial_W^k W(x, t) = \partial_{W_0}^k W_0(\Phi_{0,t}(x)) + \int_0^t \partial_W^{k+1} u(s, \Phi_{s,t}(x)) ds.$$

Let $\varepsilon = \mu - \mu'$, $\gamma = \mu'$. Fix $T > 0$, and define $V_\gamma(t)$, $\eta_{\gamma,T}$ and $\mu(t)$ as in (82). Consequently,

$$\begin{aligned} |\partial_W^k W(x, t) - \partial_W^k W(y, t)| &\leq \|\partial_{W_0}^k W_0\|_{\dot{C}^\mu} |\Phi_{0,t}(x) - \Phi_{0,t}(y)|^\mu \\ &\quad + \int_0^t \left| \partial_W^{k+1} u(\Phi_{s,t}(x), s) - \partial_W^{k+1} u(\Phi_{s,t}(y), s) \right| ds. \end{aligned}$$

It follows from Proposition 5.3 and Lemma 5.4 that for every $\gamma_k \in (1 - \frac{1-\mu}{k+1}, 1)$,

$$\left| \partial_W^{k+1} u(x, s) - \partial_W^{k+1} u(y, s) \right| \leq C(1 + \log \Delta_\gamma(s)) \|\partial_W^k W(s)\|_{\dot{C}^{\mu(s)}} |x-y|^{\mu(s)} (m(|x-y|^{-1}) + 1)$$

$$+ C \sum_{j=2}^{k+1} \sum_{\substack{l_1 \geq \dots \geq l_j \geq 1 \\ l_1 + \dots + l_j = k+1}} \left(\prod_{i=1}^j \|\partial_W^{l_i-1} W(s)\|_{\dot{C}^{\gamma_k}} \right) |x-y|^\mu. \quad (114)$$

Note that the first term can be handled analogously to the argument in Theorem 4.8 (in the case $n = 1$), while the second term is controlled by the induction hypothesis (113):

$$\sum_{j=2}^{k+1} \sum_{\substack{l_1 \geq \dots \geq l_j \geq 1 \\ l_1 + \dots + l_j = k+1}} \left(\prod_{i=1}^j \|\partial_W^{l_i-1} W(t)\|_{\dot{C}^{\gamma_k}} \right) \leq \sum_{j=2}^{k+1} \sum_{\substack{l_1 \geq \dots \geq l_j \geq 1 \\ l_1 + \dots + l_j = k+1}} \left(\prod_{i=1}^j R_{l_i, \frac{\gamma_k+1}{2}, \gamma_k}(t) \right) \triangleq R_{k+1, \mu}(t) < +\infty.$$

Indeed, $R_{l_i, \frac{\gamma_k+1}{2}, \gamma_k}(t) < +\infty$ due to the fact $l_i \leq k$ (since $l_i \geq 1$, $l_1 + \dots + l_j = k+1$, and $j \geq 2$), and consequently $C^{k+1, \mu} \subset C^{l_i, \frac{\gamma_k+1}{2}}$. Applying (84) and (86), we obtain the following bound analogously to (87):

$$\begin{aligned} \frac{|\partial_W^k W(x, t) - \partial_W^k W(y, t)|}{|x-y|^{\mu(t)}} &\leq C \|\partial_{W_0}^k W_0\|_{\dot{C}^\mu} |x-y|^{\mu-\mu(t)} e^{\mu \mathfrak{M}_{0,t,\gamma}(|x-y|^{-1})} \\ &\quad + C_1 \int_0^t (1 + \log \Delta_\gamma(s)) \|\partial_W^k W(s)\|_{\dot{C}^\mu(s)} |x-y|^{\mu(s)-\mu(t)} (m(|x-y|^{-1}) + 1) e^{\mu \mathfrak{M}_{s,t,\gamma}(|x-y|^{-1})} ds \\ &\quad + C |x-y|^{\mu-\mu(t)} \int_0^t R_{k+1,\mu}(s) e^{\mu \mathfrak{M}_{s,t,\gamma}(|x-y|^{-1})} ds \\ &\leq C \left(\|\partial_{W_0}^k W_0\|_{\dot{C}^\mu} + \int_0^T R_{k+1,\mu}(t) dt \right) |x-y|^{\mu-\mu(t)} e^{\mu \mathfrak{M}_{0,t,\gamma}(|x-y|^{-1})} \\ &\quad + C_1 \int_0^t (1 + \log \Delta_\gamma(s)) \|\partial_W^k W(s)\|_{\dot{C}^\mu(s)} |x-y|^{\mu(s)-\mu(t)} (m(|x-y|^{-1}) + 1) e^{\mu \mathfrak{M}_{s,t,\gamma}(|x-y|^{-1})} ds. \end{aligned}$$

Then, through the same argument as in the proof of Theorem 4.8, we obtain a bound analogously to (104):

$$\begin{aligned} \|\partial_W^k W(T)\|_{C^{\mu'}} &= \|\partial_W^k W(T)\|_{C^{\mu-\varepsilon}} \\ &\leq C \left(\|\partial_{W_0}^k W_0\|_{C^\mu} + \int_0^T R_{k+1,\mu}(t) dt \right) \exp \left(C \log^{\beta+2\epsilon}(e+T) \mathcal{H}^{-1}(C(1+T) \log^{\beta+2\epsilon}(e+T)) \right) \\ &\triangleq R_{k+1,\mu,\mu'}(T) < +\infty. \end{aligned}$$

This finishes the proof of (113). By induction, (106) holds.

We are left to verify that $R_{k+1,\mu,\mu'}(t)$ satisfies (107). By the induction hypothesis, we have

$$R_{k+1,\mu}(t) \leq C \|\varphi_0\|_{C^{k+1,\mu}} \exp \left(C \log^{\beta+2\epsilon}(e+t) \mathcal{H}^{-1}(C(1+t) \log^{\beta+2\epsilon}(e+t)) \right).$$

Denote

$$g(t) \triangleq C \log^{\beta+2\epsilon}(e+t) \mathcal{H}^{-1}(C(1+t) \log^{\beta+2\epsilon}(e+t)).$$

It is easy to check that g is a strictly increasing function in $[0, +\infty)$ with $\min_{t \geq 0} g'(t) = c > 0$. Then,

$$\int_0^T R_{k+1,\mu}(t) dt \leq \int_0^T C \|\varphi_0\|_{C^{k+1,\mu}} e^{g(t)} dt \leq \frac{C \|\varphi_0\|_{C^{k+1,\mu}}}{c} \int_0^T g'(t) e^{g(t)} dt \leq C \|\varphi_0\|_{C^{k+1,\mu}} e^{g(T)}.$$

Therefore, we find that

$$\begin{aligned} R_{k+1,\mu,\mu'}(T) &\leq C (\|\varphi_0\|_{C^{k+1,\mu}} + C \|\varphi_0\|_{C^{k+1,\mu}} e^{g(T)}) e^{g(T)} \\ &\leq C \|\varphi_0\|_{C^{k+1,\mu}} \exp \left(C \log^{\beta+2\epsilon}(e+T) \mathcal{H}^{-1}(C(1+T) \log^{\beta+2\epsilon}(e+T)) \right). \end{aligned}$$

Applying the same regularization and compactness argument at order n in Subsection C.2 transfers the preceding uniform estimates to the unique patch solution and completes the proof.

6. GLOBAL REGULARITY OF MULTIPLE VORTEX PATCHES

In this section, we extend the previous analysis to vortex patch solutions consisting of multiple patches.

Suppose that $\omega_0(x)$ takes the form of (4), where $N > 1$ and $D_i(0) \subset \mathbb{R}^2$ for $i \in \{1, \dots, N\}$ are simply connected disjoint bounded domains with

$$d_0 \triangleq \min_{i \neq j} \text{dist}(D_i(0), D_j(0)) > 0.$$

Then, according to Theorem 1.1, the 2D Loglog-Euler type equation (1) admits a unique global vortex patch solution

$$\omega(x, t) = \sum_{j=1}^N a_j \mathbf{1}_{D_j(t)}(x), \quad a_j \in \mathbb{R}, \quad D_j(t) \triangleq \Phi_t(D_j(0)), \quad (115)$$

where $\Phi_t = \Phi_{t,0}$ denotes the flow map defined by (5).

Suppose that $\partial D_i(0) \in C^{n,\mu}$, $i \in \{1, \dots, N\}$. Let $\varphi_{i,0} \in C^{n,\mu}$ be a level-set characterization of $D_i(0)$ with a compact support $\Omega_i \triangleq \{x \in \mathbb{R}^2 : \varphi_{i,0}(x) \neq 0\}$ slightly larger than $D_i(0)$, such that

$$D_i(0) \subset \Omega_i, \quad \text{dist}(\partial \Omega_i, \partial D_i(0)) \geq \frac{d_0}{3}, \quad \text{and} \quad d_i(0) \triangleq \min_{j \neq i} \text{dist}(\Omega_i, D_j(0)) \geq \frac{d_0}{2}.$$

Denote by

$$\varphi_i(x, t) = \varphi_{i,0}(\Phi_{0,t}(x)), \quad \text{and} \quad W_i(x, t) \triangleq \nabla^\perp \varphi_i(x, t).$$

From the framework discussed in Section 4.1, we know that $\varphi_i(\cdot, t)$ is a level-set characterization of the domain $D_i(t)$, and that the boundary regularity $\partial D_i(t) \in C^{n,\mu}$ follows from the control of $\|\partial_{W_i}^{n-1} W_i(t)\|_{C^\mu(\mathbb{R}^2)}$. Furthermore, we have

$$\text{supp } \varphi_i(\cdot, t) = \Omega_i(t) \triangleq \Phi_t(\Omega_i).$$

and

$$\|\partial_{W_i}^l u(t)\|_{C^{\mu-\varepsilon}(\mathbb{R}^2)} = \|\partial_{W_i}^l u(t)\|_{C^{\mu-\varepsilon}(\Omega_i(t))}, \quad \|\partial_{W_i}^{l-1} W_i(t)\|_{C^{\mu-\varepsilon}(\mathbb{R}^2)} = \|\partial_{W_i}^{l-1} W_i(t)\|_{C^{\mu-\varepsilon}(\Omega_i(t))}.$$

Define the distance

$$d_i(t) \triangleq \min_{j \neq i} \text{dist}(\Omega_i(t), D_j(t)).$$

Applying the estimate (6) on the flow map, we obtain

$$d_i(t) \geq \frac{1}{H^{-1}(H(d_i(0)^{-1}) + Ct)} \geq \frac{1}{H^{-1}(H(\frac{2}{d_0}) + Ct)} > 0, \quad \forall t \geq 0. \quad (116)$$

Hence, different patches remain separated for all time.

Let us also comment on the velocity field u , which takes the form

$$u(x, t) = \sum_{j=1}^N u_j(x, t) \triangleq \sum_{j=1}^N \nabla^\perp (-\Delta)^{-1} m(\Lambda)(a_j \mathbf{1}_{D_j(t)}(x)). \quad (117)$$

Since the patches are separated, the main contribution to $u(x, t)$ for $x \in \Omega_i(t)$ comes from the i -th patch $u_i(x, t)$. Indeed, for $j \neq i$, we apply (35) and control $u_j(x, t)$ by

$$\sum_{j \neq i} \|\nabla^l u_j(t)\|_{L^\infty(\Omega_i(t))} \leq \sum_{j \neq i} |a_j| \left\| \int_{D_j(t)} \nabla^l K(x - y) dy \right\|_{L_x^\infty(\Omega_i(t))}$$

$$\leq C \frac{m(\mathbf{d}_i(t)^{-1}) + 1}{\mathbf{d}_i(t)^{l+1}} \leq C(\mathbf{d}_i(t)^{-l-2} + 1), \quad (118)$$

for $l \in \{0, 1, \dots, n+1\}$ where we do not intend to find the optimized exponent in the last inequality. Therefore, the global regularity results for a single vortex patch extend to the case of multiple vortex patches.

Theorem 6.1. *Let $\mu \in (0, 1)$ and $n \in \mathbb{N}^*$. Assume that $m(\xi) = m(|\xi|)$ is a radial function satisfying (H1)-(H2a)-(Osg). For each patch solution ω given by (115), if $\partial D_i(0) \in C^{n, \mu}$ for all $i \in \{1, \dots, N\}$, then the patch boundaries $\partial D_i(t) \in C^{n, \mu-\varepsilon}$ for every $\varepsilon \in (0, \mu)$ and $i \in \{1, \dots, N\}$.*

Moreover, for every $\varepsilon > 0$ and every $\epsilon > 0$, there are some constant $C > 0$ depending on initial data, ε and ϵ such that for all $i = \{1, 2, \dots, N\}$, the following estimates hold:

$$|\nabla \varphi_i(t)|_{\inf} \geq |\nabla \varphi_{i,0}|_{\inf} \exp \left(-C\mathcal{H}^{-1}(C(1+t) \log^{\beta+\epsilon}(e+t) + C\mathcal{H}(\mathbf{g}(t))) \right), \quad (119)$$

and

$$\|\partial_{W_i}^{n-1} W_i(t)\|_{C^{\mu-\varepsilon}} \leq C\|\varphi_{i,0}\|_{C^{n,\mu}} \exp \left(C \log^{\beta+\epsilon}(e+t) \mathcal{H}^{-1}(C(1+t) \log^{\beta+\epsilon}(e+t) + C\mathcal{H}(\mathbf{g}(t))) \right), \quad (120)$$

where φ_i is a level-set characterization of $D_i(t)$, $W_i \triangleq \nabla^\perp \varphi_i$, $\mathcal{H}$ is defined by (101), $\mathbf{g}(t)$ is defined by

$$\mathbf{g}(t) \triangleq \left(\mathcal{H}^{-1}(\mathcal{H}(\frac{2}{d_0}) + Ct) \right)^3. \quad (121)$$

Remark 7. The main difference of the bounds (119)-(120) compared to the single-patch case (103)-(104) is the appearance of the term $\mathcal{H}(\mathbf{g}(t))$, which grows exponentially (or faster) in time. This growth arises because patches may approach one another as time evolves. As a consequence, the regularity bounds grow triple-exponentially for the 2D Euler equation (i.e. $\beta = \varepsilon = \epsilon = 0$), and potentially faster in the general case.

Proof of Theorem 6.1. We start by proving this result for the case $n = 1$. Define

$$\Delta_{i,\gamma}(t) \triangleq \frac{\|\nabla \varphi_i(t)\|_{\dot{C}^\gamma(\Omega_i(t))}}{\|\nabla \varphi_i(t)\|_{\inf(\partial D_i(t))}} + 1, \quad \gamma \in (0, 1).$$

To estimate u , we use the decomposition (117). The control of $u_i(t)$ on $\Omega_i(t)$ is the same as the single patch case, see (60), (71) and (74); while the control of $u_j(t)$ on $\Omega_i(t)$ follows from (118). We deduce

$$\begin{aligned} \|u(t)\|_{C_m^1(\Omega_i(t))} &\leq C(1 + \log \Delta_{i,\gamma}(t) + \mathbf{d}_i(t)^{-3}), \\ \|\partial_{W_i} u\|_{C_m^\sigma(\Omega_i(t))} &\leq C(1 + \log \Delta_{i,\gamma}(t) + \mathbf{d}_i(t)^{-3}) \|W_i(t)\|_{C^\sigma(\Omega_i(t))}, \quad \sigma \in (0, 1), \\ \left\| \nabla u \frac{W_i}{|W_i|} \cdot \frac{W_i}{|W_i|} \right\|_{L^\infty(\Omega_i(t))} &\leq C(m(\Delta_{i,\gamma}(t)) + 1)(1 + \log \Delta_{i,\gamma}(t) + \mathbf{d}_i(t)^{-3}). \end{aligned}$$

The rest of the proof follows the same argument as in Section 4.5, with $1 + \log \Delta_\gamma(\cdot)$ replaced by $1 + \log_+ \Delta_{i,\gamma}(\cdot) + \mathbf{d}_i(\cdot)^{-3}$. In particular, we deduce the bound on $\Delta_{i,\gamma}(t)$ analogous to (95):

$$\Delta_{i,\mu-\varepsilon}(t) \leq C \exp \left(C(m(e^{V_{i,\mu-\varepsilon}(t)}) + 1)V_{i,\mu-\varepsilon}(t) + C \int_0^t (m(\Delta_{i,\mu-\varepsilon}(s)) + 1)V'_{i,\mu-\varepsilon}(s) ds \right),$$

where

$$V_{i,\gamma}(t) \triangleq \int_0^t \left(1 + \log \Delta_{i,\gamma}(s) + \mathbf{d}(s)^{-3} \right) ds, \quad \gamma \in (0, 1).$$

Define

$$f_i(t) \triangleq V'_{i,\gamma}(t) = 1 + \log \Delta_{i,\mu-\varepsilon}(t) + \mathbf{d}_i(t)^{-3},$$

and $\mathcal{M}$ as in (96). It follows from (116) and the inequality above that for every $t \in [0, T]$ and $\epsilon > 0$,

$$f_i(t) \leq C \left(\mathbf{g}(T) + \log^{\beta+\epsilon}(e+t) \int_0^t \mathcal{M}(f_i(s)) ds \right),$$

where $\mathbf{g}$ is defined by (121). The function

$$R_i(t) \triangleq \frac{\mathbf{g}(T)}{\log^{\beta+\epsilon}(e+t)} + \int_0^t \mathcal{M}(f_i(s)) ds$$

satisfies

$$\begin{aligned} \frac{dR_i(t)}{dt} &= -\frac{(\beta+\epsilon)\mathbf{g}(T)}{(e+t)\log^{\beta+\epsilon}(e+t)} + \mathcal{M}(f_i(t)) \\ &\leq \mathcal{M}(C \log^{\beta+\epsilon}(e+t) R_i(t)) \leq C \log^{\beta+2\epsilon}(e+t) \mathcal{M}(R_i(t)), \end{aligned}$$

which implies

$$\mathcal{H}(R_i(T)) - \mathcal{H}(R_i(0)) = \int_0^T \frac{R'_i(t) dt}{\mathcal{M}(R_i(t))} \leq C(1+T) \log^{\beta+2\epsilon}(e+T),$$

where $\mathcal{H}$ is defined in (101). Therefore, we conclude that

$$f_i(T) \leq C \log^{\beta+\epsilon}(e+T) R_i(T) \leq \log^{\beta+2\epsilon}(e+T) \mathcal{H}^{-1}(C(1+T) \log^{\beta+2\epsilon}(e+T) + \mathcal{H}(\mathbf{g}(T))).$$

This leads to (119) and (120), and hence we complete the proof of Theorem 6.1 for the case $n = 1$.

Next, we address the case of higher regularity ($n \geq 2$) and aim to prove that $\partial D_i(t) \in C^{n, \mu-\epsilon}$ for each $i \in \{1, \dots, N\}$. The proof follows closely the single-patch case (Theorem 5.1), with additional arguments to account for interactions among patches.

To estimate $\partial_{W_i}^{k+1} u$, we use the decomposition (117)

$$\partial_{W_i}^{k+1} u(x, t) = \partial_{W_i}^{k+1} u_i(x, t) + \sum_{\ell \neq i} \partial_{W_i}^{k+1} u_\ell(x, t).$$

The first term can be treated like the single-patch case. For the second term, since $D_\ell(t) \cap \Omega_i(t) = \emptyset$, we are still able to apply the expression formula (108) and obtain

$$\begin{aligned} \partial_{W_i}^{k+1} u_\ell(x) &= a_\ell \text{p.v.} \int_{D_\ell(t)} (\partial_{W_i}^k W_i(x) - \partial_{W_i}^k W_i(y)) \cdot \nabla K(x-y) dy \\ &\quad + a_\ell \sum_{j=2}^{k+1} \sum_{\substack{l_1 \geq \dots \geq l_j \geq 1 \\ l_1 + \dots + l_j = k+1}} c_{l_1, \dots, l_j}^{k+1} \int_{D_\ell(t)} \mathcal{D}_{W_i}^{l_1, \dots, l_j}(x, y) \cdot \nabla^j K(x-y) dy \\ &\triangleq \tilde{\mathbf{G}}_{k+1}(x) + \left(\partial_{W_i}^{k+1} u_\ell(x, t) - \tilde{\mathbf{G}}_{k+1}(x) \right), \end{aligned}$$

with

$$\tilde{\mathbf{G}}_{k+1}(x) = a_\ell \text{p.v.} \int_{D_\ell(t)} (\partial_{W_i}^k W_i(x) - \partial_{W_i}^k W_i(y)) \cdot \nabla K(x-y) dy.$$

For $\tilde{\mathbf{G}}_{k+1}$, we proceed similarly as in Lemma 5.4, and apply (108) to obtain

$$\|\tilde{\mathbf{G}}_{k+1}(t)\|_{C^\mu(\Omega_i(t))} \leq \|\partial_{W_i}^k W_i\|_{\dot{C}^\mu} (\mathbf{d}_i(t)^{-3} + 1).$$

For $\partial_{W_i}^{k+1} u_\ell(x, t) - \tilde{\mathbf{G}}_{k+1}(x)$, the same proof of (112) yields

$$\|\partial_{W_i}^{k+1} u_\ell(x, t) - \tilde{\mathbf{G}}_{k+1}(x)\|_{C^\mu(\Omega_i(t))} \leq C \sum_{j=2}^{k+1} \sum_{\substack{l_1 \geq \dots \geq l_j \geq 1 \\ l_1 + \dots + l_j = k+1}} \left(\prod_{s=1}^j \|\partial_{W_i}^{l_s-1} W_i\|_{\dot{C}^{\gamma_k}} \right),$$

Combining the above estimates, we obtain the analogue of (114), where the term $1 + \log \Delta_\gamma(\cdot)$ is replaced with $1 + \log \Delta_{i,\gamma}(\cdot) + d_i(\cdot)^{-3}$.

The rest of the proof follows the single-patch case, with the additional $d_i(\cdot)^{-3}$ term treated as in the case $n = 1$ of Theorem 6.1. We thus obtain (120).

Finally, the uniform separation estimate prevents distinct patches from touching in the limit. The compactness and kernel-convergence argument developed in Appendix C.2 therefore applies to each component $D_i^\delta(t) = \Phi_t^\delta(D_i(0))$. The resulting limiting multi-patch solution coincides with the unique solution of Theorem 1.1, thereby completing the proof of Theorem 6.1. $\square$

APPENDIX A. PROOF OF LEMMA 2.3

Properties (32) and (33) with $l = 1, 2$ have been proved in [83, Lemma 2]. For (33) with general l and (34) can be proved analogously. Here we sketch the proof.

Starting from the explicit expression (28) on G , we claim that for every $l \in \{1, 2, \dots, n+1\}$, the l -th order derivative $G^{(l)}$ can be expressed by

$$G^{(l)}(\rho) = \mathcal{G}_l(\rho) + \frac{(-1)^l}{2\pi\rho^l} \int_0^\infty J_0(\rho r) M_l(r) dr, \quad (122)$$

where $\mathcal{G}_l(\rho)$ and $M_l(r)$ are iteratively defined as

$$\mathcal{G}_l(\rho) = \mathcal{G}'_{l-1}(\rho) - \frac{l-1}{\rho} (G^{(l-1)}(\rho) - \mathcal{G}_{l-1}(\rho)), \quad \mathcal{G}_0(\rho) = \mathcal{G}_1(\rho) = 0,$$

$$M_l(r) = M_{l-1}(r) + r M'_{l-1}(r), \quad M_0(r) = m'(r);$$

or equivalently,

$$\forall l \in \{2, 3, \dots, n+1\}, \quad \mathcal{G}_l(\rho) = \sum_{j=1}^{l-1} a_{j,l} \frac{G^{(j)}(\rho)}{\rho^{l-j}}, \quad \{a_{j,l}\}_{1 \leq j \leq l-1} \text{ are iteratively defined,}$$

$$\forall l \in \{1, 2, \dots, n+1\}, \quad M_l(r) = \sum_{j=1}^{l+1} b_{j,l} r^{j-1} m^{(j)}(r), \quad b_{1,l} = b_{l+1,l} = 1,$$

and $b_{j,l} = b_{j,l-1} + (j-1)b_{j,l-1} + b_{j-1,l-1}$ for every $j \in \{2, \dots, l\}$. Indeed, we calculate $G^{(l)}$ by differentiating $G^{(l-1)}$ and using the following integration by parts

$$\begin{aligned} \int_0^\infty J'_0(\rho r) r M_{l-1}(r) dr &= \left. \frac{J_0(\rho r) r M_{l-1}(r)}{\rho} \right|_{r=0}^{+\infty} - \frac{1}{\rho} \int_0^\infty J_0(\rho r) (M_{l-1}(r) + r M'_{l-1}(r)) dr \\ &= -\frac{1}{\rho} \int_0^\infty J_0(\rho r) M_l(r) dr, \end{aligned}$$

where we have used the fact that for every $l \in \{1, \dots, n+1\}$,

$$\lim_{r \rightarrow +\infty} |J_0(\rho r) r M_{l-1}(r)| \leq C \lim_{r \rightarrow +\infty} r^{-\frac{1}{2}} \cdot r \cdot m'(r) = C\beta_1 \lim_{r \rightarrow +\infty} \frac{m(r)}{r^{1/2} \log r} = 0,$$

and

$$\lim_{r \rightarrow 0^+} r^l m^{(l)}(r) = 0, \quad \lim_{r \rightarrow 0^+} r M_{l-1}(r) = 0,$$

applying the Mikhlin-Hörmander condition in (H1), (H2a), (21) and (19).

Next, we prove (33) and (34) by induction. Note that (33)-(34) with $l = 1$ have already been proven in [83, Lemma 2] (although it was only stated that $|G'(\rho)| \leq C$ in $[\bar{c}_0, +\infty)$, it can easily be extended to (34) with $l = 1$; see also below). Let $k \in \{1, \dots, N\}$. Suppose that (33) and (34)

with each $l \in \{1, \dots, k\}$ hold, we intend to show that they also hold for the $l = k + 1$ case. Let $\chi(\xi) = \chi(|\xi|) \in C_c^\infty(\mathbb{R}^2)$ be a smooth radial function such that

$$\chi \equiv 1, \text{ in } \{|\xi| \leq 1\}, \quad \chi \equiv 0, \text{ in } \{|\xi| \geq 2\}, \quad 0 \leq \chi \leq 1.$$

Thanks to identity (28) with $|x| = \rho$ and formulas (122), we have

$$\begin{aligned} G^{(k+1)}(\rho) &= \mathcal{G}_{k+1}(\rho) + \frac{(-1)^{k+1}}{2\pi\rho^{k+1}} \int_0^\infty J_0(\rho r) \chi(\rho r) M_{k+1}(r) dr \\ &\quad + \frac{(-1)^{k+1}}{(2\pi)^2 \rho^{k+1}} \int_{\mathbb{R}^2} e^{ix \cdot \xi} (1 - \chi(\rho|\xi|)) \frac{M_{k+1}(|\xi|)}{|\xi|} d\xi \\ &\triangleq \mathbf{I}_1 + \mathbf{I}_2 + \mathbf{I}_3. \end{aligned}$$

For the first term, noting that $\mathbf{I}_1 = \sum_{j=1}^k a_{j,k+1} \frac{G^{(j)}(\rho)}{\rho^{k+1-j}}$, we immediately use (H1) and the induction assumptions to deduce that

$$|\mathbf{I}_1| \leq C \frac{1 + m(\rho^{-1})}{\rho^{k+1}}.$$

For the term $\mathbf{I}_2$, since the Bessel function $J_0(\cdot)$ satisfies that $0 < J_0(r) \leq 1$ for $r \in [0, 2]$, and using (H1), we derive

$$\begin{aligned} |\mathbf{I}_2| &\leq \frac{1}{2\pi\rho^{k+1}} \int_0^\infty J_0(\rho r) \chi(\rho r) \left(m'(r) + |b_{2,k+1}| r |m''(r)| + \dots + |b_{k+2,k+1}| r^{k+1} |m^{(k+2)}(r)| \right) dr \\ &\leq \frac{C}{\rho^{k+1}} \int_0^{2\rho^{-1}} J_0(\rho r) m'(r) dr \leq \frac{C}{\rho^{k+1}} \left(m(2\rho^{-1}) - m(0^+) \right). \end{aligned}$$

For the last term $\mathbf{I}_3$, through the integration by parts and (H1), observe that

$$\begin{aligned} |\mathbf{I}_3| &= \frac{1}{(2\pi)^2 \rho^{k+3}} \left| \sum_{j=1}^{k+2} b_{j,k+1} \int_{\mathbb{R}^2} e^{ix \cdot \xi} \Delta_\xi \left((1 - \chi(\rho|\xi|)) \frac{|\xi|^{j-1} m^{(j)}(|\xi|)}{|\xi|} \right) d\xi \right| \\ &\leq \frac{C}{\rho^{k+3}} \sum_{j=1}^{k+2} \left(\int_{\rho^{-1} \leq |\xi| \leq 2\rho^{-1}} (\rho^2 + \rho|\xi|^{-1}) \frac{|\xi|^{j-1} m^{(j)}(|\xi|)}{|\xi|} d\xi + \int_{|\xi| \geq \rho^{-1}} \Delta_\xi \left(\frac{|\xi|^{j-1} m^{(j)}(|\xi|)}{|\xi|} \right) d\xi \right) \\ &\leq \frac{C}{\rho^{k+3}} \left(\int_{\rho^{-1} \leq |\xi| \leq 2\rho^{-1}} (\rho^2 + \rho|\xi|^{-1}) \frac{m'(|\xi|)}{|\xi|} d\xi + \int_{|\xi| \geq \rho^{-1}} \frac{m'(|\xi|)}{|\xi|^3} d\xi \right). \end{aligned}$$

By virtue of the fact that $r \mapsto m'(r)$ is non-increasing and $r m'(r) \leq m(r)$ for $r > 0$ large (from (H2a)-(Osg)), we infer that for $\rho > 0$ small enough, i.e., $\rho \in (0, \bar{c}_0]$,

$$|\mathbf{I}_3| \leq \frac{C}{\rho^{k+2}} m'(\rho^{-1}) \leq \frac{C}{\rho^{k+1}} m(\rho^{-1});$$

while for every $\rho \in [\bar{c}_0, \infty)$,

$$\begin{aligned} |\mathbf{I}_3| &\leq \frac{C}{\rho^{k+3}} \left(\rho^2 \int_{\rho^{-1}}^{2\rho^{-1}} m'(r) dr + \int_{\rho^{-1}}^{\bar{c}_0^{-1}} \frac{m'(r)}{r^2} dr + \int_{\bar{c}_0^{-1}}^\infty \frac{m'(r)}{r^2} dr \right) \\ &\leq \frac{C}{\rho^{k+3}} \left(\rho^2 m(2\rho^{-1}) + \bar{c}_0^2 m(\bar{c}_0^{-1}) + m(\bar{c}_0^{-1}) \int_{\rho^{-1}}^{\bar{c}_0^{-1}} \frac{1}{r^3} dr + m'(\bar{c}_0^{-1}) \int_{\bar{c}_0^{-1}}^\infty \frac{1}{r^2} dr \right) \leq \frac{C}{\rho^{k+1}}. \end{aligned}$$

Gathering the above estimates leads to the inequalities (33) and (34) with $l = k + 1$, as desired. Hence, we complete the proof of (33) and (34).

APPENDIX B. PROOF OF PROPOSITION 4.3: THE CIRCULAR PATCH

This appendix section is devoted to the proof of Proposition 4.3. Before we discuss the main proof, we first present two auxiliary lemmas.

Lemma B.1 (Abelian estimate for Fourier sine transform). *Let $Q \in L^1_{\text{loc}}([1, \infty))$. Assume that there exists $R_0 > 1$ such that Q is positive and C^1 on $[R_0, \infty)$ and*

$$q_Q(\rho) \triangleq \frac{\rho Q'(\rho)}{Q(\rho)} \rightarrow 0, \quad \text{as } \rho \rightarrow \infty. \quad (123)$$

Then, for every $\delta > 0$, the integral

$$I_Q(\delta) \triangleq \int_1^\infty \frac{Q(\rho)}{\rho} \sin(\delta \rho) d\rho$$

converges in the improper oscillatory sense. Moreover,

$$I_Q(\delta) = \left(\frac{\pi}{2} + o(1) \right) Q(\delta^{-1}), \quad \text{as } \delta \downarrow 0. \quad (124)$$

Proof. By (123), we may choose $R \geq R_0$ such that

$$|q_Q(\rho)| \leq \frac{1}{2}, \quad \forall \rho \geq R.$$

For $R \leq r \leq s$, integration of the logarithmic-derivative bound gives

$$\left(\frac{s}{r} \right)^{-1/2} \leq \frac{Q(s)}{Q(r)} \leq \left(\frac{s}{r} \right)^{1/2}. \quad (125)$$

Furthermore, for $\rho \geq R$,

$$\frac{d}{d\rho} \log \frac{Q(\rho)}{\rho} = \frac{q_Q(\rho) - 1}{\rho} \leq -\frac{1}{2\rho}.$$

Thus, $Q(\rho)/\rho$ eventually decreases to zero, and Dirichlet's test shows that $I_Q(\delta)$ is well defined for every $\delta > 0$.

Set

$$\lambda \triangleq \delta^{-1}, \quad Q_\lambda(s) \triangleq \frac{Q(\lambda s)}{Q(\lambda)}.$$

Splitting the integral at R and changing variables $s = \rho/\lambda$ on $[R, \infty)$, we obtain

$$\frac{I_Q(\delta)}{Q(\delta^{-1})} = \frac{I_Q(\lambda^{-1})}{Q(\lambda)} = E_\lambda + \int_{R/\lambda}^\infty Q_\lambda(s) \frac{\sin s}{s} ds, \quad (126)$$

where

$$|E_\lambda| \leq \frac{1}{\lambda Q(\lambda)} \int_1^R |Q(\rho)| d\rho \rightarrow 0, \quad \text{as } \lambda \rightarrow \infty.$$

Here we have used $\lambda Q(\lambda) \rightarrow \infty$ (by (125)).

Fix $0 < \eta < 1 < A$. For sufficiently large λ , the integration of $(\log Q(\rho))' = q_Q(\rho)/\rho$ gives as $\lambda \rightarrow \infty$,

$$\sup_{s \in [\eta, A]} |\log Q_\lambda(s)| = \sup_{s \in [\eta, A]} \left| \int_\lambda^{\lambda s} \frac{q_Q(\rho)}{\rho} d\rho \right| \leq \sup_{\rho \in [\eta\lambda, A\lambda]} |q_Q(\rho)| \max\{|\log \eta|, \log A\} \rightarrow 0.$$

Consequently,

$$\int_\eta^A Q_\lambda(s) \frac{\sin s}{s} ds \rightarrow \int_\eta^A \frac{\sin s}{s} ds, \quad \text{as } \lambda \rightarrow \infty.$$

For $R/\lambda \leq s \leq 1$, (125) yields

$$\left| \int_{R/\lambda}^{\eta} Q_{\lambda}(s) \frac{\sin s}{s} ds \right| \leq \int_0^{\eta} s^{-\frac{1}{2}} ds = 2\sqrt{\eta}.$$

It remains to estimate the oscillatory tail. For $s \geq 1$, (125) implies $Q_{\lambda}(s) \leq \sqrt{s}$. Moreover, the function $\frac{Q_{\lambda}(s)}{s^{3/4}}$ decreases on $[1, \infty)$ for sufficiently large λ , since

$$\frac{d}{ds} \log \frac{Q_{\lambda}(s)}{s^{3/4}} = \frac{qQ(\lambda s) - \frac{3}{4}}{s} \leq -\frac{1}{4s}.$$

The bound $\frac{Q_{\lambda}(s)}{s^{3/4}} \leq s^{-\frac{1}{4}}$ also shows that $\frac{Q_{\lambda}(s)}{s} \rightarrow 0$ as $s \rightarrow \infty$. Integration by parts therefore gives

$$\left| \int_A^{\infty} Q_{\lambda}(s) \frac{\sin s}{s} ds \right| \leq \left| \frac{Q_{\lambda}(A)}{A} \right| + \left| \int_A^{\infty} \frac{Q_{\lambda}(s)(qQ(\lambda s) - 1)}{s^2} \cos s ds \right| \leq 3 \left| \frac{Q_{\lambda}(A)}{A} \right| \leq 3A^{-\frac{1}{2}}.$$

To conclude, we find

$$\limsup_{\lambda \rightarrow \infty} \left| \frac{I_Q(\lambda^{-1})}{Q(\lambda)} - \int_{\eta}^A \frac{\sin s}{s} ds \right| \leq 2\sqrt{\eta} + 3A^{-\frac{1}{2}}.$$

Letting first $\eta \downarrow 0$ and then $A \rightarrow \infty$, and using the fact $\int_0^{\infty} \frac{\sin s}{s} ds = \frac{\pi}{2}$, we obtain (124). $\square$

Abel interpretation of oscillatory integrals. Let $a \geq 0$ and let $F \in L^1_{\text{loc}}([a, \infty))$ be complex-valued. Assume that

$$e^{-\varepsilon \rho} F(\rho) \in L^1([a, \infty)) \quad \text{for every } \varepsilon > 0.$$

Whenever the following limit exists, we define the Abel value of oscillatory integral as

$$\text{Abel} \int_a^{\infty} F(\rho) d\rho \triangleq \lim_{\varepsilon \downarrow 0} \int_a^{\infty} e^{-\varepsilon \rho} F(\rho) d\rho.$$

This summability method is regular: if the ordinary improper integral $\int_a^{\infty} F(\rho) d\rho$ converges, then its Abel value exists and agrees with the improper integral; see [97, Chapter 4, §1, Lemma 1]. All Fourier-Bessel integrands appearing below are locally integrable and have at most polynomial growth. Thus, for every $\varepsilon > 0$, the exponentially regularized integrals are absolutely convergent. Differentiation and integration by parts are always performed first at fixed ε , and the limit $\varepsilon \downarrow 0$ is taken afterwards. We refer to [97, Chapter 4] for the Abel summability of Fourier integrals and the use of exponential convergence factors.

Lemma B.2 (Integration by parts for Abel integrals). *Let $M \in C^1([1, \infty))$ have at most polynomial growth, and assume that M' is eventually nonnegative and decreases to zero. Then, for every $a \neq 0$, the Abel integral*

$$\text{Abel} \int_1^{\infty} M(\rho) e^{ia\rho} d\rho \triangleq \lim_{\varepsilon \downarrow 0} \int_1^{\infty} e^{-\varepsilon \rho} M(\rho) e^{ia\rho} d\rho$$

exists and satisfies

$$\text{Abel} \int_1^{\infty} M(\rho) e^{ia\rho} d\rho = \frac{i}{a} \left[M(1) e^{ia} + \int_1^{\infty} M'(\rho) e^{ia\rho} d\rho \right]. \quad (127)$$

The convergence is locally uniform for $a \in \mathbb{R} \setminus \{0\}$. In particular,

$$\text{Abel} \int_1^{\infty} M(\rho) \cos(a\rho) d\rho = -\frac{M(1) \sin a}{a} - \frac{1}{a} \int_1^{\infty} M'(\rho) \sin(a\rho) d\rho, \quad (128)$$

$$\text{Abel} \int_1^{\infty} M(\rho) \sin(a\rho) d\rho = \frac{M(1) \cos a}{a} + \frac{1}{a} \int_1^{\infty} M'(\rho) \cos(a\rho) d\rho. \quad (129)$$

Proof. For $\varepsilon > 0$, set $z_\varepsilon \triangleq \varepsilon - ia$. Ordinary integration by parts gives

$$\int_1^\infty e^{-z_\varepsilon \rho} M(\rho) d\rho = \frac{e^{-z_\varepsilon} M(1)}{z_\varepsilon} + \frac{1}{z_\varepsilon} \int_1^\infty e^{-z_\varepsilon \rho} M'(\rho) d\rho.$$

The improper integral $\int_1^\infty M'(\rho) e^{ia\rho} d\rho$ converges locally uniformly for $a \neq 0$ by Dirichlet's test, and its Abel regularizations converge to the same value. Letting $\varepsilon \downarrow 0$ proves (127). Taking real and imaginary parts gives (128)-(129). $\square$

Now we turn to the proof of Proposition 4.3.

Proof of Proposition 4.3. With the convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}^2} e^{-ix \cdot \xi} f(x) dx, \quad f(x) = \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} e^{ix \cdot \xi} \widehat{f}(\xi) d\xi,$$

the polar coordinates and standard Bessel identities (see [85, (10.6.6) and (10.9.1)]) give

$$\widehat{\mathbf{1}}_B(\xi) = \int_0^1 \int_0^{2\pi} e^{-i|\xi|s \sin \theta} d\theta ds = 2\pi \int_0^1 J_0(|\xi|s) s ds = 2\pi \frac{J_1(|\xi|)}{|\xi|},$$

where J_n , $n \geq 0$, denotes the Bessel functions of the first kind. Therefore, for $r \neq 1$,

$$g_0(r) = \frac{1}{2\pi} \int_{\mathbb{R}^2} e^{ir\xi_2} m(|\xi|) \frac{J_1(|\xi|)}{|\xi|} d\xi = \int_0^\infty m(\rho) J_1(\rho) J_0(r\rho) d\rho. \quad (130)$$

The asymptotic expansion of Bessel functions (see [85, (10.17.2)–(10.17.3)])

$$J_\nu(s) = \left(\frac{2}{\pi s}\right)^{1/2} \cos\left(s - \frac{\nu\pi}{2} - \frac{\pi}{4}\right) + O(s^{-3/2}) \quad \text{as } s \rightarrow \infty \quad (131)$$

gives, uniformly for $r \in [1/2, 3/2]$,

$$J_1(\rho) J_0(r\rho) = \frac{\sin((1-r)\rho) - \cos((1+r)\rho)}{\pi\sqrt{r}\rho} + O(\rho^{-2}), \quad \text{as } \rho \rightarrow \infty.$$

Set

$$\kappa(\rho) \triangleq \frac{\rho m'(\rho)}{m(\rho)}.$$

Since $\tilde{m}(t) = m(e^t)$, assumption (H2a) gives

$$\kappa(\rho) = \frac{1}{\log \rho} \frac{(\log \rho) \tilde{m}'(\log \rho)}{\tilde{m}(\log \rho)} \rightarrow 0, \quad \text{as } \rho \rightarrow \infty. \quad (132)$$

Thus, m slowly varies at infinity and $m(\rho)/\rho$ eventually decreases to zero. Moreover, for all sufficiently large ρ , $\frac{m(\lambda\rho)}{m(\rho)} \leq \sqrt{\lambda}$, $\forall \lambda \geq 1$. In particular, $m(\rho) \leq C(1+\rho)^{1/2}$ and hence $\int_1^\infty \frac{m(\rho)}{\rho^2} d\rho < \infty$. The cosine term is therefore uniformly bounded by Dirichlet's test $\frac{1}{\pi\sqrt{r}} \left| \int_1^\infty \frac{m(\rho) \cos((1+r)\rho)}{\rho} d\rho \right| < \infty$, and the remainder is absolutely integrable. Consequently,

$$g_0(r) = \frac{1}{\pi\sqrt{r}} \int_1^\infty \frac{m(\rho)}{\rho} \sin((1-r)\rho) d\rho + O(1), \quad \text{as } r \rightarrow 1.$$

It remains to determine the sine-transform asymptotics. Since (132) holds, Lemma B.1 applied with $Q = m$ gives

$$\int_1^\infty \frac{m(\rho)}{\rho} \sin(\delta\rho) d\rho = \left(\frac{\pi}{2} + o(1)\right) m(\delta^{-1}), \quad \text{as } \delta \downarrow 0.$$

Since the sine transform is odd and $m(|1-r|^{-1}) \rightarrow \infty$ as $r \rightarrow 1$, the bounded remainder above is negligible. This proves part (i).

Next, we consider part (ii). By rotational symmetry,

$$u(x) = V(r)e_\theta, \quad r = |x|,$$

with $e_\theta = (-\sin \theta, \cos \theta) = (-\frac{x_2}{r}, \frac{x_1}{r})$. Since $\nabla^\perp = (\partial_{x_2}, -\partial_{x_1})$, the identity $\operatorname{curl} u = \partial_{x_1} u_2 - \partial_{x_2} u_1 = g_0$ becomes

$$V'(r) + \frac{V(r)}{r} = g_0(r), \quad (133)$$

which implies $(rV(r))' = rg_0(r)$. The preceding asymptotics and the boundedness of V near $r = 1$ therefore give

$$(1 \pm h)V(1 \pm h) - V(1) = \int_1^{1 \pm h} r \left(\frac{1}{2} + o(1) \right) \operatorname{sgn}(1 - r) m(|1 - r|^{-1}) dr,$$

and

$$|V(1 \pm h) - V(1)| = \left(\frac{1}{2} + o(1) \right) \int_0^h m(t^{-1}) dt + O(h), \quad \text{as } h \downarrow 0.$$

Arguing as (38),

$$\frac{1}{h m(h^{-1})} \int_0^h m(t^{-1}) dt \rightarrow 1, \quad \text{as } h \downarrow 0.$$

Since $m(h^{-1}) \rightarrow \infty$ by (H2a), the term $O(h)$ is negligible, which demonstrates the velocity estimate (67). The identity $u \cdot \mathbf{n} = 0$ on ∂B follows immediately from the rotational symmetry.

We finally prove part (iii). Set

$$L(\rho) \triangleq \rho m'(\rho).$$

Assumption (H2a) and (16) give

$$\frac{\rho L'(\rho)}{L(\rho)} = 1 + \frac{\rho m''(\rho)}{m'(\rho)} \rightarrow 0, \quad \text{as } \rho \rightarrow \infty, \quad (134)$$

which yields $m''(\rho) < 0$ for sufficiently large ρ . Similarly as (125), integrating the logarithmic-derivative bound shows that, $L(\rho) \leq C\sqrt{\rho}$ for all sufficiently large ρ . Hence $m'(\rho) = L(\rho)/\rho$ eventually decreases to zero. We shall also use the facts established above that $m(\rho)/\rho$ eventually decreases to zero and $\int_1^\infty \frac{m(\rho)}{\rho^2} d\rho < \infty$.

To justify differentiation of the Fourier-Bessel representation (130), set

$$g_{0,\varepsilon}(r) \triangleq \int_0^\infty e^{-\varepsilon\rho} m(\rho) J_1(\rho) J_0(r\rho) d\rho.$$

For $\varepsilon > 0$, ordinary differentiation gives

$$g'_{0,\varepsilon}(r) = \int_0^\infty e^{-\varepsilon\rho} \rho m(\rho) J_1(\rho) J'_0(r\rho) d\rho = - \int_0^\infty e^{-\varepsilon\rho} \rho m(\rho) J_1(\rho) J_1(r\rho) d\rho.$$

On every compact subset of $[1/2, 3/2] \setminus \{1\}$, the asymptotic expansion (131), Lemma B.2, and Dirichlet's test imply that $g_{0,\varepsilon} \rightarrow g_0$ and $g'_{0,\varepsilon}$ converge locally uniformly. Therefore,

$$g'_0(r) = - \operatorname{Abel} \int_0^\infty \rho m(\rho) J_1(\rho) J_1(r\rho) d\rho, \quad r \neq 1.$$

Using the first correction term in the asymptotic expansion of Bessel functions [85, (10.17.3)], we obtain, uniformly for $r \in [1/2, 3/2]$,

$$\rho J_1(\rho) J_1(r\rho) = \frac{\cos((1-r)\rho) - \sin((1+r)\rho)}{\pi\sqrt{r}}$$

$$- \frac{3}{8\pi\sqrt{r}\rho} [(1+r^{-1})\cos((1+r)\rho) + (1-r^{-1})\sin((1-r)\rho)] + O(\rho^{-2}).$$

For $r = 1 \pm h$, the fixed-frequency term with amplitude m is uniformly bounded by Lemma B.2; the fixed-frequency term with amplitude m/ρ is bounded by Dirichlet's test; and the $O(m/\rho^2)$ remainder is absolutely integrable. Moreover, Lemma B.1 applied with $Q = m$ gives

$$\left| (1-r^{-1}) \int_1^\infty \frac{m(\rho)}{\rho} \sin((1-r)\rho) d\rho \right| \leq Ch m(h^{-1}).$$

It follows that

$$g'_0(1 \pm h) = -\frac{1}{\pi\sqrt{1 \pm h}} \text{Abel} \int_1^\infty m(\rho) \cos(h\rho) d\rho + O(1 + h m(h^{-1})). \quad (135)$$

Under (134), applying Lemma B.1 with $Q = L$ gives

$$\int_1^\infty m'(\rho) \sin(h\rho) d\rho = \left(\frac{\pi}{2} + o(1)\right) h^{-1} m'(h^{-1}).$$

Hence, (128) with $M = m$ yields

$$\text{Abel} \int_1^\infty m(\rho) \cos(h\rho) d\rho = -\left(\frac{\pi}{2} + o(1)\right) h^{-2} m'(h^{-1}), \quad \text{as } h \downarrow 0. \quad (136)$$

Here, the boundary term in (128) is $O(1)$ and is negligible because $h^{-2} m'(h^{-1}) = h^{-1} L(h^{-1}) \rightarrow \infty$ as $h \downarrow 0$.

Finally, recalling (134) and $\kappa(\rho) = \frac{\rho m'(\rho)}{m(\rho)}$ satisfies (132), we have

$$\frac{\rho \kappa'(\rho)}{\kappa(\rho)} = \frac{\rho L'(\rho)}{L(\rho)} - \kappa(\rho) \rightarrow 0, \quad \text{as } \rho \rightarrow \infty.$$

The logarithmic-derivative comparison applied to $\kappa(\rho)$ gives $\rho^{1-\vartheta} \kappa(\rho) \rightarrow \infty$ for all fixed $\vartheta \in [0, 1)$. Thus, for all fixed $\vartheta \in [0, 1)$,

$$\frac{h^{-\vartheta}(1 + m(h^{-1}))}{h^{-2} m'(h^{-1})} = \frac{1}{h^{-(1-\vartheta)} L(h^{-1})} + \frac{1}{h^{-(1-\vartheta)} \kappa(h^{-1})} \rightarrow 0, \quad \text{as } h \downarrow 0. \quad (137)$$

Combining this with (135)-(136) gives

$$g'_0(1 \pm h) = \left(\frac{1}{2} + o(1)\right) h^{-2} m'(h^{-1}), \quad \text{as } h \downarrow 0.$$

With $\tau_\nu \triangleq \nu^\perp$, rotational symmetry gives $u(r\nu) = V(r)\tau_\nu$ and $V(r)$ satisfies the equality (133). Therefore,

$$V''(r) = g'_0(r) - \frac{V'(r)}{r} + \frac{V(r)}{r^2}.$$

Since V is bounded near $r = 1$ and

$$|V'(1 \pm h)| = \left(\frac{1}{2} + o(1)\right) m(h^{-1}), \quad \text{as } h \downarrow 0,$$

we obtain

$$V''(1 \pm h) = \left(\frac{1}{2} + o(1)\right) h^{-2} m'(h^{-1}), \quad \text{as } h \downarrow 0.$$

Writing $u(x) = \frac{V(r)}{r} x^\perp$ with $x^\perp = (-x_2, x_1)$, and differentiating one and two times give

$$\nabla u(r\nu) = V'(r)\tau_\nu \otimes \nu + \mathcal{R}_1(r, \nu), \quad \text{with } |\mathcal{R}_1(r, \nu)| \leq C|V(r)|, \quad (138)$$

and

$$\nabla^2 u(r\nu) = V''(r)\tau_\nu \otimes \nu \otimes \nu + \mathcal{R}_2(r, \nu), \quad \text{with} \quad |\mathcal{R}_2(r, \nu)| \leq C(|V'(r)| + |V(r)|).$$

Hence, as $h \downarrow 0$,

$$\nabla^2 u((1 \pm h)\nu) = \left(\frac{1}{2} + o(1)\right) h^{-2} m'(h^{-1}) \tau_\nu \otimes \nu \otimes \nu + O(1 + m(h^{-1})),$$

which combined with (137) proves part (iii). $\square$

APPENDIX C. JUSTIFICATION OF THE *a priori* ESTIMATES VIA KERNEL REGULARIZATION

C.1. Construction of Yudovich-type solutions. By the kernel regularization process, we shall rigorously construct a weak solution to the 2D Loglog-Euler type equation (1) that satisfies the *a priori* estimates in the proof of Theorem 1.1.

Let $\zeta \in C^\infty([0, \infty))$ satisfy $\zeta = 0$ on $[0, 1]$ and $\zeta = 1$ on $[2, \infty)$, and set

$$K_\delta(x) \triangleq \zeta\left(\frac{|x|}{\delta}\right) K(x), \quad \delta > 0,$$

where $K(x)$ is the kernel function given by (27). Since K is odd, divergence free, and tangent to circles centered at the origin, the same properties hold for K_δ . Moreover, estimate (35) and the fact that $|x| \approx \delta$ on the support of the derivatives of the cutoff give, uniformly in δ ,

$$|\nabla^\ell K_\delta(x)| \leq \frac{C_\ell(m(|x|^{-1}) + 1)}{|x|^{\ell+1}}, \quad 0 \leq \ell \leq n+1. \quad (139)$$

Consider

$$\partial_t \Phi_t^\delta(a) = \int_{\mathbb{R}^2} K_\delta(\Phi_t^\delta(a) - \Phi_t^\delta(b)) \omega_0(b) db, \quad \Phi_0^\delta(a) = a. \quad (140)$$

For fixed $\delta > 0$, write $\Phi_t^\delta = \text{Id} + q^\delta$ and work in $X = C_b(\mathbb{R}^2; \mathbb{R}^2)$. The right-hand side of (140), regarded as a map $\mathcal{F}_\delta : X \rightarrow X$, satisfies

$$\|\mathcal{F}_\delta(q)\|_X \leq \|K_\delta\|_{L^\infty} \|\omega_0\|_{L^1}, \quad \|\mathcal{F}_\delta(q) - \mathcal{F}_\delta(\tilde{q})\|_X \leq 2\|\nabla K_\delta\|_{L^\infty} \|\omega_0\|_{L^1} \|q - \tilde{q}\|_X.$$

The Picard-Lindelöf theorem in Banach spaces and the usual continuation criterion therefore give a unique global solution of (140). Defining

$$u^\delta(x, t) \triangleq \int_{\mathbb{R}^2} K_\delta(x - \Phi_t^\delta(b)) \omega_0(b) db,$$

we have $u^\delta \in C([0, t]; C_b^1)$ and $\partial_t \Phi_t^\delta = u^\delta \circ \Phi_t^\delta$. Let $\Phi_{t,s}^\delta$ denote the two-parameter flow generated by u^δ (similar to (5)), so that $\Phi_t^\delta = \Phi_{t,0}^\delta$. Since the radial cutoff function ensures $\text{div } K_\delta = 0$, the Jacobi-Liouville formula gives

$$\det \nabla \Phi_t^\delta = 1.$$

Together with uniqueness of the forward and backward trajectories, this shows that Φ_t^δ is a global volume-preserving diffeomorphism. Consequently, setting

$$\omega^\delta(\cdot, t) = \omega_0 \circ \Phi_{0,t}^\delta(\cdot),$$

a change of variables yields

$$u^\delta(x, t) = K_\delta * \omega^\delta(x, t), \quad \|\omega^\delta(t)\|_{L^1 \cap L^\infty} = \|\omega_0\|_{L^1 \cap L^\infty}. \quad (141)$$

The proof of Lemma 3.1 applies to K_δ with constants independent of δ . Indeed, the radial cutoff preserves the kernel bounds used there, including on the annulus $|x| \approx \delta$. It follows from (141) that

$$\|u^\delta(t)\|_{L^\infty} + \sup_{x \neq y} \frac{|u^\delta(x, t) - u^\delta(y, t)|}{\nu(|x - y|)} \leq C\|\omega_0\|_{L^1 \cap L^\infty}$$

uniformly in δ and $t \geq 0$. Hence, on every finite interval $[0, T]$, the flows $\Phi_{t,0}^\delta$ and their inverses $\Phi_{0,t}^\delta$ have uniform spatial and temporal moduli of continuity and satisfy

$$|\Phi_{t,s}^\delta(x) - x| \leq C \|\omega_0\|_{L^1 \cap L^\infty} |t - s|. \quad (142)$$

The local Arzelà-Ascoli theorem therefore gives, as $\delta \rightarrow 0$, along a subsequence,

$$\Phi_{t,0}^\delta \rightarrow \Phi_{t,0}, \quad \Phi_{0,t}^\delta \rightarrow \Phi_{0,t}$$

locally uniformly on $[0, T] \times \mathbb{R}^2$. Passing to the limit in the identities for the regularized flows shows that

$$\Phi_{0,t} = \Phi_{t,0}^{-1}.$$

Using the bound (142) to localize the supports, the local uniform convergence permits passage to the limit in the volume-preserving identities for both the flows and their inverses:

$$\int_{\mathbb{R}^2} \eta(\Phi_{t,0}(a)) \, da = \int_{\mathbb{R}^2} \eta(a) \, da = \int_{\mathbb{R}^2} \eta(\Phi_{0,t}(a)) \, da, \quad \forall \eta \in C_c(\mathbb{R}^2).$$

Since compactly supported continuous functions determine Radon measures, these identities extend to every nonnegative Borel function. We may therefore set

$$\omega(x, t) = \omega_0(\Phi_{0,t}(x)).$$

Taking respectively $\eta = |\omega_0|$ and $\eta = \mathbf{1}_{\{|\omega_0| > \lambda\}}$ gives

$$\|\omega(t)\|_{L^1} = \|\omega_0\|_{L^1}, \quad \|\omega(t)\|_{L^\infty} = \|\omega_0\|_{L^\infty}.$$

Finally, a near-far decomposition gives the convergence of the velocities. The only point requiring attention is the singular region, where volume preservation and the kernel bound (139) lead to

$$\int_{\{|x - \Phi_t^\delta(a)| < r\}} |K_\delta(x - \Phi_t^\delta(a))| |\omega_0(a)| \, da \leq C \|\omega_0\|_{L^\infty} \int_0^r (m(\rho^{-1}) + 1) \, d\rho \rightarrow 0 \quad \text{as } r \downarrow 0,$$

uniformly in δ . Away from the singularity, we first truncate the a -integration to a large ball. On this ball, $K_\delta \rightarrow K$ and the local uniform convergence of the flows apply, while the remaining tail is bounded, uniformly in δ , by $C_r \int_{\{|a| > R\}} |\omega_0(a)| \, da$, which tends to zero as $R \rightarrow \infty$. Consequently,

$$u^\delta \rightarrow u \quad \text{locally uniformly}, \quad u(x, t) = \int_{\mathbb{R}^2} K(x - \Phi_{t,0}(a)) \omega_0(a) \, da = K * \omega(x, t).$$

Passing to the limit in the integral form of the flow equation gives

$$\partial_t \Phi_{t,0}(a) = u(\Phi_{t,0}(a), t).$$

The Lagrangian identity

$$\int_{\mathbb{R}^2} \omega(x, t) \tilde{\chi}(x, t) \, dx = \int_{\mathbb{R}^2} \omega_0(a) \tilde{\chi}(\Phi_{t,0}(a), t) \, da$$

then gives the weak formulation in Definition 3.1. Since $T > 0$ is arbitrary, a diagonal argument produces a global weak solution. This completes the proof of existence part of Theorem 1.1.

C.2. Justification of the propagation of boundary regularity. We briefly justify the level-set formulation of the evolved patch and the *a priori* arguments used in Sections 4-6.

We keep $D_0 \subset \mathbb{R}^2$ fixed as in Theorem 1.2 and consider the regularized Lagrangian equation (140) with $\omega_0 = \mathbf{1}_{D_0}$,

$$\partial_t \Phi_t^\delta(a) = \int_{D_0} K_\delta(\Phi_t^\delta(a) - \Phi_t^\delta(b)) db, \quad \Phi_0^\delta(a) = a.$$

As we did for (140) with fixed $\delta > 0$, the mapping Φ_t^δ is a global volume-preserving diffeomorphism. The bounded higher derivatives of K_δ give the corresponding higher spatial regularity of the flow and hence preserve the $C^{n,\mu}$ regularity of the regularized patch $D^\delta(t) \triangleq \Phi_t^\delta(D_0)$. In particular,

$$u^\delta(x, t) \triangleq \int_{D_0} K_\delta(x - \Phi_t^\delta(b)) db = K_\delta * \mathbf{1}_{D^\delta(t)}(x).$$

Define

$$\varphi^\delta(x, t) \triangleq \varphi_0(\Phi_{0,t}^\delta(x)), \quad W^\delta \triangleq \nabla^\perp \varphi^\delta,$$

with φ_0 a $C^{n,\mu}$ level-set characterization of D_0 . Differentiating $\varphi^\delta(\Phi_t^\delta(a), t) = \varphi_0(a)$ and using $\det \nabla \Phi_t^\delta = 1$, we obtain

$$W^\delta(\Phi_t^\delta(a), t) = \nabla \Phi_t^\delta(a) W_0(a).$$

This is the primary formulation of the evolution of W^δ at the Hölder regularity level. In particular,

$$W^\delta(x, t) = W_0(\Phi_{0,t}^\delta(x)) + \int_0^t \partial_{W^\delta} u^\delta(\Phi_{s,t}^\delta(x), s) ds.$$

Thus the corresponding transport equation (78) is understood through this Lagrangian formula, or equivalently, in the distributional sense, when W^δ is only Hölder continuous. More generally, for $1 \leq k \leq n$, the highest-order equation in (57) is understood through

$$\partial_{W^\delta}^{k-1} W^\delta(x, t) = \partial_{W_0}^{k-1} W_0(\Phi_{0,t}^\delta(x)) + \int_0^t \partial_{W^\delta}^k u^\delta(\Phi_{s,t}^\delta(x), s) ds.$$

The radial cutoff function preserves the cancellation identities used in the preceding sections. For instance,

$$\partial_{W^\delta} u^\delta(x, t) = \int_{D^\delta(t)} (W^\delta(x, t) - W^\delta(y, t)) \cdot \nabla K_\delta(x - y) dy.$$

The regularized counterpart of (108) holds as well. Together with the above uniform kernel bounds (139), this shows that all the *a priori* estimates derived in Sections 4-6 hold with constants independent of δ .

Define $z^\delta(\xi, t) \triangleq \Phi_t^\delta(z_0(\xi))$, with $z_0 : \mathbb{T} \rightarrow \partial D_0$ a $C^{n,\mu}$ -parameterization of ∂D_0 . Fix $\mu' < \bar{\mu} < \mu$. Applying the uniform estimates with regularity loss $\mu - \bar{\mu}$, and using the preceding Lagrangian identities for time equicontinuity, the compact embedding $C^{n,\bar{\mu}}(\mathbb{T}) \subset\subset C^{n,\mu'}(\mathbb{T})$ and the Arzelà–Ascoli theorem give, as $\delta \rightarrow 0$, after extraction,

$$z^\delta \rightarrow z \quad \text{in } C([0, T]; C^{n,\mu'}(\mathbb{T})).$$

The full-flow compactness and kernel-convergence argument in Appendix C.1 identifies the limit with its unique Yudovich patch solution. Hence $z = \Phi_t \circ z_0$, and the uniform boundary estimates pass to the limit by lower semicontinuity. The same compactness argument applied to φ^δ and W^δ identifies their limits with $\varphi = \varphi_0 \circ \Phi_{0,t}$ and $W = \nabla^\perp \varphi$, respectively; hence the uniform level-set estimates also pass to the limit by lower semicontinuity.

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